FINAL WELL REPORT

Drilling

License: NO PL046BS

Well: NO 15/9-F-1 C

Statoil

Title:		
FINAL WELL REPORT **Drilling** **License: NO PL046BS** **Well: NO 15/9-F-1 C**		
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1 General well data

Licence number:	NO PL046BS
Well name:	NO 15/9-F-1 C
Slot:	F-1
Type of well:	Production
Water depth / airgap:	91,0 m MSL / 54, 9 m RT
Primary objective:	Producer to the Volve Northwest segment
Completion type:	7” monobore perforated liner

Section	Start time	End time	Rig name
NO 15/9-F-1 B Preparations	25.Jan.2014 11:00	31.Jan.2014 07:45	MÆRSK INSPIRER
NO 15/9-F-1 B Pull Completion	31.Jan.2014 07:45	03.Feb.2014 20:30	MÆRSK INSPIRER
NO 15/9-F-1 B Permanent P&A (DP) w/ RIG	03.Feb.2014 20:30	19.Feb.2014 21:00	MÆRSK INSPIRER
NO 15/9-F-1 B Prepare Sidetrack (DP) w/ RIG	19.Feb.2014 21:00	21.Feb.2014 14:30	MÆRSK INSPIRER
NO 15/9-F-1 C 17 1/2"	21.Feb.2014 14:30	03.Mar.2014 18:15	MÆRSK INSPIRER
NO 15/9-F-1 C 12 1/4"	03.Mar.2014 18:15	11.Mar.2014 04:30	MÆRSK INSPIRER
NO 15/9-F-1 C 8 1/2"	11.Mar.2014 04:30	15.Mar.2014 16:30	MÆRSK INSPIRER
NO 15/9-F-1 C Liner	15.Mar.2014 16:30	18.Mar.2014 16:15	MÆRSK INSPIRER

Structure centre coordinates :

Geographic:	Lat.	58° 26' 29.807" N	Long.	1° 53' 14.929" E
UTM:		6 478 563,53 m N		435 050,02 m E

Wellhead coordinates :

Geographic:	Lat.	58° 26' 29.706" N	Long.	1° 53' 15.149" E
UTM:		6 478 560,36 m N		435 053,55 m E
Rectangular:		-3,17 m N		3,53 m E (from structure centre)

2 Dispensations

Disp no	Title	Doc no	Status
85553	Unntak fra krav om standard survey intervall på 30 m.	TR3513	Approved

3 Health, environment, safety and quality (HES&Q)

3.1 RUH

Type (colour code)	Synergi code	Number of
Green	4	8
Yellow	3	2
Sum		10

3.1.1 RUH details

WELLBORE: NO 15/9-F-1 B

Synergi no	Hazard	Description
1394527	3	Synergi 1394527 (Closed) - 31.Jan.2014 - Non-conformities - TDS spin/drill mode malfunction. While POOH with the THRT on 5 1/2" DP it was not possible to use the spin function of the TDS to connect to the stand at connection height in the derrick. Erratic torque readings were observed without actual movement of the shaft. The error was intermittent and was observed earlier during similar operations with DP stands at height.
1394407	4	Synergi 1394407 (Closed) - 31.Jan.2014 - Non-conformities - Unable to open and unlatch BX elevator. While troubleshooting a TDS spin function problem in the derrick it was not possible to unlatch the elevator from a stand of 5 1/2" DP including a THRT.
1394525	4	Synergi 1394525 (In process) - 05.Feb.2014 - Non-conformities - Stuck DrilQuip adjustable sub setting tool. Unable to release adjustable sub setting tool from 10 3/4" surface tieback string after engaging it and releasing tension from tieback.Had to lay out the tool together with the surface casing hanger and one DP single. The same incident occured on F-15 C during P&A operations in October 2013. (see superior case below)

WELLBORE: NO 15/9-F-1 C

Synergi no	Hazard	Description
1398615	3	Synergi 1398615 (In process) - 11.Mar.2014 - Non-conformities - Lost communication with AutoTrak RSS. After drilling the 9 5/8" shoe track/rat hole and 3 m new formation a FIT was performed. When drilling was about to commence again, communication with the AutoTrak steering head was not achieved. Diagnostic downlinking and troubleshooting was performed without success. It was decided to POOH and change tools.
1397650	4	Synergi 1397650 (Closed) - 04.Mar.2014 - Non-conformities - Mud leakage through weep hole on annular Annular preventer leaking through weep. BOP tested on 2nd of March to 345 bar
1398010	4	Synergi 1398010 (Closed) - 05.Mar.2014 - Non-conformities - Malfunction on drop gyro tool. Prior to P/U the drop gyro tool it was not possible to set-up tool and establish communication between the computer and the gyro.
1398013	4	Synergi 1398013 (Closed) - 07.Mar.2014 - Non-conformities - Experienced difficulties to make up X-over joint with casing tong. The lower body of the 10 3/4" x 9 5/8" x-over joint had an 11 3/4" OD. No jaws for this OD were available, resulting in difficulties making up the joint and getting correct torque/turn graphs. The tong had to grip on the 9 5/8" coupling and the 10 3/4" part of the x/o body. In addition to the above described circumstances causing downtime, it is necessary to repair the not working stabbing arm on the tong frame.
1398087	4	Synergi 1398087 (Closed) - 07.Mar.2014 - Non-conformities - Leaking solenoid valve on TDS. A leaking solenoid valve was discovered on the TDS. The leakage was stopped and the valve will be replaced.
1398776	4	Synergi 1398776 (Closed) - 12.Mar.2014 - Non-conformities - Decoding problems while drilling 8 1/2" hole. Encountered decoding problems from downhole tools. Pulled into previous casing shoe and performed downlinking/calibration to change decoding mode.
1399165	4	Synergi 1399165 (Closed) - 15.Mar.2014 - Non-conformities - Drawworks shut-down due to cooling water alarm While RIH with 7" liner at 71 m the drawworks shut down due to a cooling water alarm. Cooling water pump had tripped.

3.2 Experience listing

Subject:	THRT not fully engaged while pulling tubing hanger free.		
Section:	NO 15/9-F-1 B, Pull Completion		
Rep date:	03.Feb.2014		
Keywords:	TUBING HANGER		
Downtime:		**Pot time improvements:**	
Comp inv:	GE Vetco		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
RIH with THRT and engaged TH as per GE Vetco procedure. Set 2 MT weight on hanger, marked pipe. Rotated tool RH and observed tool drop. Turned tool LH. Did not observe positive stop in J-slot profile. Suspected lock ring rotating. Applied 5 MT overpull and confirmed tool engaged.
Pulled ring free with 40 MT overpull. Continued to pull, observed weight droping to string weight at 140 MT hookload. POOH and inspected retrieval tool. No damage observed.

RIH and set 2 MT on TH. Observed mark at same spot. Rotated tool RH and observed tool drop. Rotated LH and did not observed positive stop. Confimed tool holding with 140 MT overpull. Set down 4 MT weight on hanger and turned LH to force tool fully into rotating lock ring. Repeated process. Picked up and pulled TH and completion free at 190 MT hookload. POOH and landed TH on c-plate.

Tool was fully engaged when inspecting same on surface. Required several attempts to release THRT from TH. Had to rotate tool using chain tongs, suspected allignment/hung-up on communication collar. Inspected TH and observed 1/3 lock ring J-slot was bent upwards due to THRT not fully engaged during first pull.

Immediate solution:
POOH and inspected retrieval tool. No damage observed.
Future recommended solution:
THRT with hydraulic line should be used. Then the THRT can be locked in properly with use of hydraulics. This take some more time, but is a safer solution.

Subject:	2 7/8" tubing equipment not fit for purpose		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	13.Feb.2014		
Keywords:	TUBING		
Downtime:		**Pot time improvements:**	
Comp inv:	Statoil		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Prior to run 2 7/8" tubing it was discovered that it had only been supplied manual slips.
2 7/8" tubing and handling equipment had been ordered by Statoil.
This resulted in a lot of time spent both on SJA/TBT/TOFS and also while tripping in hole.

While picking up the first joint various means of stabbing remotely was attemped with no success. Rigged up tong tailing arm and function tested same. Continued running in with the stinger.
Immediate solution:
Performed a TOFS, SJA and TBT prior to use of manual slips.
Future recommended solution:
Non-standard tubular and related equipment should be specified by Statoil, but ordered by the rig contracor. That way we will get the correct equipment.

Subject:	2 7/8" tubing equipment not fit for purpose		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	13.Feb.2014		
Keywords:	TUBING		
Downtime:		**Pot time improvements:**	
Comp inv:	Statoil		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Attempted to P/U first 2 7/8" joint from HTS with the air operated elevator. Unable to tilt and close center latch elevator, bail rotator not designed for this type of elevator. Evaluated different options. Attempted
elevator using tugger. Unable to latch elevator.
Postponed the operation to a later stage.
2 7/8" tubing and handling equipment had been ordered by Statoil.

Immediate solution:
Decided to use the BX elevator, but discovered that correct inserts were missing. Had some problems with the TDS at the same time, and decided together with town to go ahead and repair the TDS. Rigged down the Weatherford equipment.

Future recommended solution:
Non-standard tubular and related equipment should be specified by Statoil, but ordered by the rig contracor. That way we will get the correct equipment.

Subject:	Make up torque for 2 7/8" tubing		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	14.Feb.2014		
Keywords:	CEMENT PLUG		
Downtime:		**Pot time improvements:**	
Comp inv:	Maersk Drilling		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
2 7/8" P110 8,7 # TPS TS-6 tubing was used as cement stinger. 2 7/8" drill pipe was not used due setup of Iron Roughneck. The optimum make up torque for this tubing is 5339 Nm with maximum makeup of 5935 Nm.

A 300 m plug was to be set inside 7" liner and up inside 9 5/8" casing.

When conditioned mud prior to cement an RPM of 20 was used giving 8 kN. This is above the makeup torque for 2 7/8". Due to this rotation was stopped.
When displacing cement and rotating pipe with 30 RPM 8 kN was seen. However it was necessary for a good cement job.

Immediate solution:

Future recommended solution:
Evaluate to use tubing/pipe with higher makeup torque.

Subject:	No rotator sent out for Weatherford BX elevator		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	16.Feb.2014		
Keywords:	CASING/LINER		
Downtime:		**Pot time improvements:**	
Comp inv:	Weatherford		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
No rotator was sent out with the Weatherford BX elevator for pulling the 10 3/4" x 9 5/8" casing and 13 3/8" casing. Weatherford said that they did not have any available rotators to send out to the rig.
Immediate solution:
Used Maersk elevator
Future recommended solution:
Ensure that rotator is shipped out together with the casing handling equipment. Attempt getting one though a different vendor if none available with casing contractor.

Subject:	Lockdown assembly retrieval tool overfilled with hydraulic oil and air		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	16.Feb.2014		
Keywords:	WELLHEAD/ASA		
Downtime:		**Pot time improvements:**	
Comp inv:	GE Vetco		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
The lockdown assembly/seal assembly retrieval tool (SRT) is set in retrieval position on surface by pushing up a sleeve on the outer body of the tool past a ledge, locking the sleeve and tool in position for being able to engage and retrieve the lockdown assembly. The SRT hydraulic oil chamber was overfilled with oil and air, causing the sleeve never to pass the ledge that locks the SRT in retrieving position. As a result the sleeve slid back down again once the force that pushed the sleeve upwards was removed. It was not noticed that the sleeve slipped back down again priro to running the SRT in hole for retrieving the lockdown assembly.
After the 1st mis-run the tool operator noticed hydraulic oil coming out of the chamber when pushing up the sleeve to set the tool in retrieval position. The operator bleed of oil and air from the hydraulic oil chamber, but did not notice that the sleeve once again did not pass the locking ledge, and slid back down again prior to running in hole.
When re-setting the tool in retrieving position after the 2nd mis-run, the operator noticed that the sleeve slid back down, and bleed of more oil and air from the hydraulic chamber ¿ enabling the sleeve to be porperly locked in retrieving position.

Immediate solution:
Bleed of oil and air from the tool hydraulic chamber, allowing the tool to be properly set in retrieving position.
Future recommended solution:
Verify that the retrieval tool is in retrival mode priror to running the tool in hole.
Ensure that the tool has proper amount of hydraulic oil in chamber prior to shipping the tool offshore.

Subject:	Oil leakage in seal assembly retrieval tool		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	16.Feb.2014		
Keywords:	WELLHEAD/ASA		
Downtime:		**Pot time improvements:**	
Comp inv:	GE Vetco		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
After breaking out the seal assy retrieving tool (SRT) from the jet sub stand, the lock down assy was removed. When spinning the SRT back into the jet sub stand the outer body continued to rotate after the make-up was done. This was caused by the weight of the outer body and spinnig in to fast with the iron roughneck spinner.

As the outer body continued to rotate, the lower inner piston that keeps the SRT hydraulic oil in place was spunn out (left hand threads), resulting in the hydraulic chamber being emptied.

Immediate solution:
Screwed the sleeve back into place and re-filled the tool with hydraulic oil. Used chain tongs instead of IRN for next ops.

Future recommended solution:
Ensure that caution is taken when spinning the SRT tool back into the jetting sub stand. Evaluate to use chain tongs instead of IRN in future ops.

Subject:	Excessive amount of slops created by not circulating over Hi-Vis pill prior to cement job		
Section:	NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG		
Rep date:	23.Feb.2014		
Keywords:	CEMENT PLUG		
Downtime:		**Pot time improvements:**	
Comp inv:	Statoil		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Excessive amount of slops was created when setting the KO plug on the F-1 B P&A.

The cement stinger was run in hole to the cut 10 3/4" casing, and the well circulated and the mud conditioned prior to setting a Hi-Vis pill and settting the KO cement plug.

The Hi-Vis pill was set from the top of the cut 10 3/4" casing at 1457 m MD and up to the cut 13 3/8" casing at 1399 m MD (where the cement KO plug would be set from). After spotting the Hi-Vis pill the stinger was pulled out to top of the cut 13 3/8" casing at 1399 m MD, and the KO cement plug set before pulling 80 m above theoretical top of cement and circulating out the spacer and Hi-Vis pill.

The result from the above was that excessive slops was created from the Hi-Vis pill mixing with mud and spacer as the top of the Hi-Vis pill had not been circulated out prior to setting the cement plug.

Immediate solution:
N/A

Future recommended solution:
It is recommedned for future KO plugs to spot the Hi-Vis pill, then pull out to the top of the pill with the stinger, and then circulate out the top of the pill while conditioning the well for the cement job.

Subject:	Excessive time spent on circulating hole clean and weighting up mud.		
Section:	NO 15/9-F-1 C, 17 1/2"		
Rep date:	18.Mar.2014		
Keywords:	DRILLING FLUID		
Downtime:		**Pot time improvements:**	
Comp inv:	Baroid		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Circulated 1 x BU, pumped a high-vis pill and circulated out of hole with little cutting over the shakers. Pumped a 2nd high-vis pill and circulated out of hole before starting to weight up mud from 1.40 sg to 1.41 sg.

Immediate solution:
NA
Future recommended solution:
If little cuttings are seen over the shakers with the 1st hi-vis pill, then evaluate to wegiht up mud while circulating the 2nd hi-vis pill.

Subject:	Weight up mud from 1.40 sg to 1.41 sg while circulating 17 1/2" hole clean at TD		
Section:	NO 15/9-F-1 C, 17 1/2"		
Rep date:	18.Mar.2014		
Keywords:	DRILLING FLUID		
Downtime:		**Pot time improvements:**	
Comp inv:	Baroid		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Increased MW from 1.40 sg to 1.41 sg prior to POOH with 17 1/2" BHA to compensate for weight of cuttings in well to ensure good hole conditions for running 13 3/8" casing.

The 13 3/8" casing was RIH without experiencing any drag. When running the 13 3/82 casing on well F-15 D, the casing had to be washed down the last stand to get past a restriction.
Immediate solution:
NA
Future recommended solution:
Evaluate to increase MW while circulating hole clean to compensate for weight of cuttings in the well, to ensure good hole conditions for running of casing.

Subject:	Increased ROP throguth out 17 1/2" section using Baker Hughes Kymera bit		
Section:	NO 15/9-F-1 C, 17 1/2"		
Rep date:	27.Feb.2014		
Keywords:	REF BIT RECORD		
Downtime:		**Pot time improvements:**	
Comp inv:	Baker Hughes		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Drilling the 17 1/2" section using a 17 1/2" Baker Hughes HP633F Kymera bit improved Net ROP compared to ROP in 17 1/2" sections on the previous well in the Volve field were PDC bits were used.

Net ROP F-1 C 17 1/2" seksjon: Kymera bit - 29,3 m/hr. Bit grading: 0-1-WT-S-E-I-NO-TD

Net ROP F-15 D 17 1/2" seksjon: PDC bit - 16,9 m/hr. Bit grading: 1-2-WT-S-X-I-LT-TD
Net ROP F-1 17 1/2" seksjon: PDC bit - 18,1 m/hr. Bit grading: 4-3-CT-N-X-I-LT-TD
Net ROP F-11 T2 17 1/2" seksjon: PDC bit - 21,5 m/hr. Bit grading: 1-3-WR-G-X-I-BT-TD

Immediate solution:
N/A
Future recommended solution:
Same bit recommended for future 17 1/2" sections

Subject:	Problem while making up the casing crossover (10 3/4" x 9 5/8")		
Section:	NO 15/9-F-1 C, 12 1/4"		
Rep date:	07.Mar.2014		
Keywords:	CASING/LINER		
Downtime:		**Pot time improvements:**	
Comp inv:	Weatherford		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
After picking the 10 3/4" x 9 5/8" crossover; the crossover body was found too big to be grabbed and made up by Weather ford tong. Some adjustment was done to fit the crossover in the tong but it leaded to have some compromise on make-up curve. The torque curve showed the base and the optimum torque but curve continuity could not be verified due to the adjustment.
Immediate solution:
Changing the jaws and lifting the the Weatherford tong to grab the top 10 3/4" casing instead of crossover.
Future recommended solution:
It is suggested to have pre-made 10 3/4" and 9 5/8" pups to the Cross over.

Subject:	Leaking cement head swivel		
Section:	NO 15/9-F-1 C, Liner		
Rep date:	19.Mar.2014		
Keywords:	CEMENTING		
Downtime:		**Pot time improvements:**	
Comp inv:	Weatherford Norge AS		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
when circulating the hole clean with the 7" liner on setting depth rotation was estbalished with 20 rpm.
It was observed that the swivel on the Weatherford cement head started leaking.

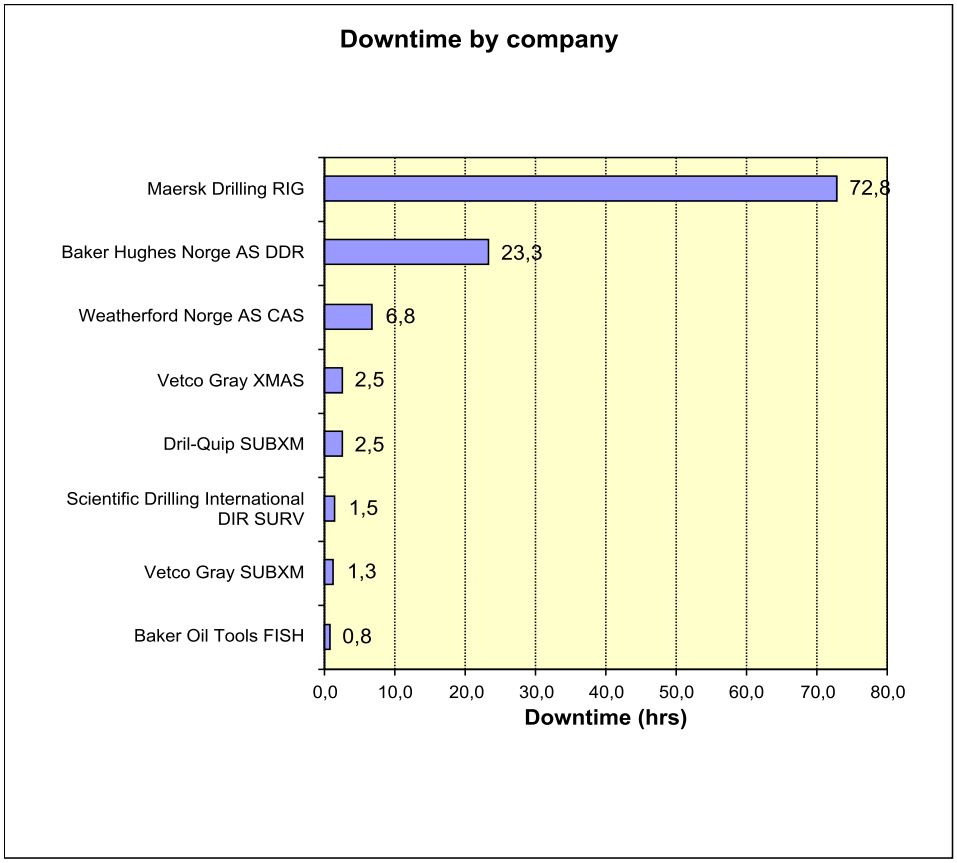
Immediate solution:
It was decided to stop rotation to prevent further damage to the swivel and maintain the option to rotate at a later stage.
During the cement job rotation was established and maintained while cement was displaced into the annulus.
Future recommended solution:
Improved quality of tools sent to the rig

Subject:	Encountered tight hole when running 7" liner		
Section:	NO 15/9-F-1 C, Liner		
Rep date:	25.Mar.2014		
Keywords:	CASING/LINER		
Downtime:		**Pot time improvements:**	
Comp inv:	Weatherford		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Tight hole was encountered when RIH with the 7" liner that was managed to slide through with 10-15 MT WOB. Also experienced pack-offs that had to be worked and reamed through with 20 rpm.

The above resulted in the liner being set a few meters shallow. The liner was RIH with a reamer shoe that had an OD of 8,25".
Immediate solution:
Reamed and worked though the restrictions and pack-offs.
Future recommended solution:
Evaluate to not run a reamer shoe with the liner as the OD of the reamer shoe is 8.25", while the OD of a regular guide shoe is 7.75 ". It is thought that the large OD of the reamer shoe could be what is casusing the problems with tight hole. By running a regular shoe with a smaller OD,one might be able to avoid any problems with tight hole.

3.3 Incidents by service and company



SERVICE COMPANY: Casing CAS Weatherford Norge AS
WELLBORE: NO 15/9-F-1 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
16.feb.2014	CAS-E03 Hydraulic operated tongs	1395647	Down time for 10 3/4" x 9 5/8" casing handling equipment.	460000	2007273	5,5	100	5,5

WELLBORE: NO 15/9-F-1 C

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
07.mar.2014	CAS-E03 Hydraulic operated tongs	1398013	Experienced difficulties to make up X-over joint with casing tong.	103075	1979040	1,3	100	1,3
			Total	563075				6,8

SERVICE COMPANY: Directional Drilling DDR Baker Hughes Norge AS
WELLBORE: NO 15/9-F-1 C

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
11.mar.2014	DDR-E32 Electrical	1398615	Lost communication with AutoTrak RSS.	1793505	1979040	21,8	100	21,8
12.mar.2014	DDR-E36 Other	1398776	Decoding problems while drilling 8 1/2" hole.	123690	1979040	1,5	100	1,5
			Total	1917195				23,3

SERVICE COMPANY: Directional survey DIR SURV Scientific Drilling International
WELLBORE: NO 15/9-F-1 C

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
05.mar.2014	DIR SURV-E01 Electrical logging equipment	1398010	Malfunction on drop gyro tool.	123690	1979040	1,5	100	1,5
			Total	123690				1,5

SERVICE COMPANY: Fishing service FISH Baker Oil Tools
WELLBORE: NO 15/9-F-1 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
14.feb.2014	FISH-E30 Other Equipment	1395495	Float in 8 1/2" pre-made Clean-out assy not working	60000	1920000	0,8	100	0,8
			Total	60000				0,8

SERVICE COMPANY: Rig Operations RIG Maersk Drilling
WELLBORE: NO 15/9-F-1 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
31.jan.2014	RIG-E07 Top drive	1394527	TDS spin/drill mode malfunction.	4638375	1979040	56,3	100	56,3
31.jan.2014	RIG-E345 Elevator	1394407	Unable to open and unlatch BX elevator.	350455	1979040	4,3	100	4,3
12.feb.2014	RIG-E02 BOP control system	1395172	BOP UPS alarm	85000	2040000	1,0	100	1,0
13.feb.2014	RIG-E348 Slips	1395364	Equipment downtime during tripping.	106000	2035200	1,3	100	1,3
14.feb.2014	RIG-E341 Vert pipe handling system	1395480	DP came out of fingerboard			3,5	100	3,5
14.feb.2014	RIG-E341 Vert pipe handling system	1395494	leaking hose on PRS	147000	2016000	1,8	100	1,8
16.feb.2014	RIG-E341 Vert pipe handling system	1395650	PRS failures during pulling casing 16-Feb 2014	21000	2016000	0,3	100	0,3

WELLBORE: NO 15/9-F-1 C

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
27.feb.2014	RIG-E341 Vert pipe handling system	1396957	Drillpipe stand "fell out" in derrick durring tripping. Operation was stopped.			3,3	100	3,3
07.mar.2014	RIG-E07 Top drive	1398087	Leaking solenoid valve on TDS.	41230	1979040	0,5	100	0,5
15.mar.2014	RIG-E10 Drawwork	1399165	Drawworks shut-down due to cooling water alarm	61845	1979040	0,8	100	0,8
			Total	5450905				72,8

SERVICE COMPANY: Subsea WH/X-mas tree SUBXM Dril-Quip
WELLBORE: NO 15/9-F-1 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
05.feb.2014	SUBXM-E26 Other Equipment (serv. or perm)	1394525	Stuck DrilQuip adjustable sub setting tool.	206150	1979040	2,5	100	2,5
			Total	206150				2,5

SERVICE COMPANY: Subsea WH/X-mas tree SUBXM Vetco Gray
WELLBORE: NO 15/9-F-1 B

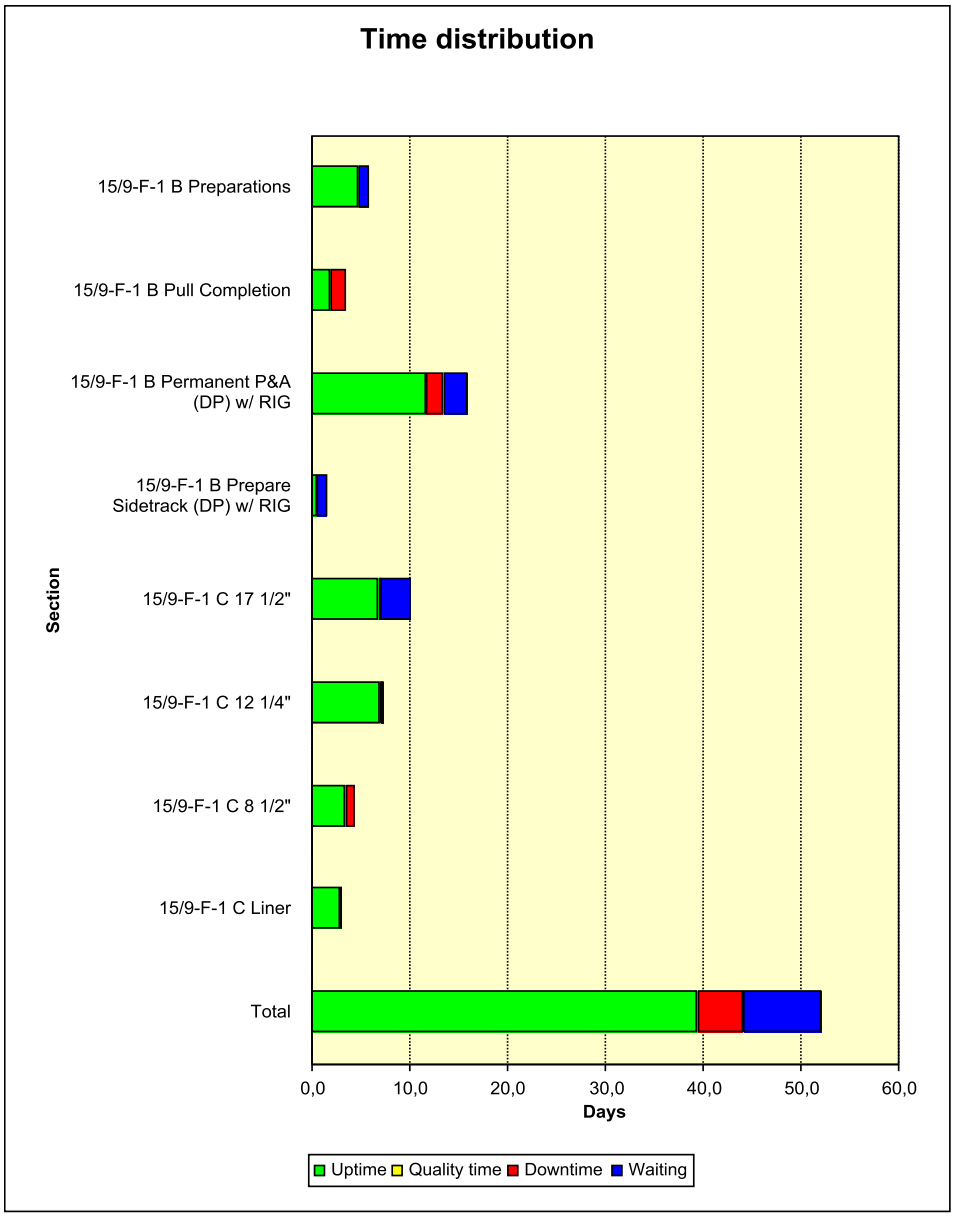
Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
15.feb.2014	SUBXM-E23 Wellhead equipment/Running tool					1,3	100	1,3
			Total	0				1,3

SERVICE COMPANY: Surf. WH/X-mas tree XMAS Vetco Gray
WELLBORE: NO 15/9-F-1 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
15.feb.2014	XMAS-E20 Other Equipment	1395581	2 misrun when attempting to pull 10 3/4" Lock Down assy.	273000	2620800	2,5	100	2,5
			Total	273000				2,5

3.4 Time distribution

		Length	Budget		TL		Actual		Ops
Section	Start time	m	hrs	days	hrs	days	hrs	days	(f)
NO 15/9-F-1 B Preparations	25.Jan.2014 11:00	0,0	126,0	5,3	104,9	4,4	140,8	5,9	100,0
NO 15/9-F-1 B Pull Completion	31.Jan.2014 07:45	318,0	59,8	2,5	49,7	2,1	84,5	3,5	55,9
NO 15/9-F-1 B Permanent P&A (DP) w/ RIG	03.Feb.2014 20:30	1,0	513,7	21,4	491,8	20,5	384,5	16,0	86,6
NO 15/9-F-1 B Prepare Sidetrack (DP) w/ RIG	19.Feb.2014 21:00	66,0	38,5	1,6	43,1	1,8	41,5	1,7	100,0
NO 15/9-F-1 C 17 1/2"	21.Feb.2014 14:30	1153,0	273,0	11,4	309,6	12,9	243,8	10,2	98,1
NO 15/9-F-1 C 12 1/4"	03.Mar.2014 18:15	542,0	196,5	8,2	176,1	7,3	178,3	7,4	98,2
NO 15/9-F-1 C 8 1/2"	11.Mar.2014 04:30	1038,0	123,0	5,1	106,0	4,4	108,0	4,5	78,5
NO 15/9-F-1 C Liner	15.Mar.2014 16:30	-55,0	90,0	3,8	86,0	3,6	71,8	3,0	99,0
Sum			1420,5	59,3	1367,2	57,0	1253,2	52,2	


Time distribution
15/9-F-1 B Preparations
15/9-F-1 B Pull Completion
15/9-F-1 B Permanent P&A (DP) w/ RIG
15/9-F-1 B Prepare Sidetrack (DP) w/ RIG
15/9-F-1 C 17 1/2"
15/9-F-1 C 12 1/4"
15/9-F-1 C 8 1/2"
15/9-F-1 C Liner
Total
Section
0,0
10,0
20,0
30,0
40,0
50,0
60,0
Days
Uptime
Quality time
Downtime
Waiting

The graph above is based on the following details:

Section	Downtime	Uptime	Quality time	Waiting time	Total time
15/9-F-1 B Preparations	0,0	4,8	0,0	1,0	5,9
15/9-F-1 B Pull Completion	1,6	2,0	0,0	0,0	3,5
15/9-F-1 B Permanent P&A (DP) w/ RIG	1,8	11,7	0,0	2,5	16,0
15/9-F-1 B Prepare Sidetrack (DP) w/ RIG	0,0	0,6	0,0	1,2	1,7
15/9-F-1 C 17 1/2"	0,1	6,9	0,0	3,1	10,2
15/9-F-1 C 12 1/4"	0,1	7,0	0,0	0,3	7,4
15/9-F-1 C 8 1/2"	1,0	3,5	0,0	0,0	4,5
15/9-F-1 C Liner	0,0	3,0	0,0	0,0	3,0
Total	4,6	39,5	0,0	8,0	52,2

4 Activity highlights

4.1 NO 15/9-F-1 B, Preparations (25.Jan.2014 11:00)

START: 25.Jan.2014 11:00 3465 mMD
END : 31.Jan.2014 07:45 3465 mMD

OBJECTIVE
Kill the well and prepare the tubing for pulling.

SUMMARY
The well was handed over from production at 11:00 hrs 25.01.14.

Rigged up W/L BOP and riser. Removed xmas tree cap and installed short riser and W/L BOP. Schlumberger built and tested the W/L toolstring for setting the EV0 plug. Filled the riser with 40% MEG and 60% water with the production MEG manifold. Pressure tested the wireline rig up and made up BHA for running the EV0 plug. The EV0 plug was succesfully sat at 3147m MD.

Rigged up for W/L run # 2 (Tubing punch) and RIH and punched at 2942-2945m MD.

Pumped 70m3 of 1,03sg packer fluid followed by 10m3 of 1,50 sg CaBa brine pill followed by 42m3 of packer fluid, waited for 30 minutes to allowed the brine pill to settle into the 7" tubing/liner. Pumped 100m3 1,37sg OBM at 600-900 lpm until OBM in returns. Circulated to condition mud, in total 4 x B/U.

M/U BHA for run # 3 (tubing cutter) and RIH. Positioned cutter at 2948,5m MD and performed cut.

Removed flowline from F-1 tree, drained xmas tree and removed same.
Nippled up BOP, installed diverter and R/D the equipment.

4.2 NO 15/9-F-1 B, Pull Completion (31.Jan.2014 07:45)

START: 31.Jan.2014 07:45 3147 mMD
END : 03.Feb.2014 20:30 3465 mMD

OBJECTIVE
Clear the borehole for tubing to make access for slot recovery

SUMMARY
P/U THRT and landed same in rotary. Installed bushings. M/U 5 1/2" DP stand to THRT. RIH w/ THRT on 5 1/2" DP stand from surface to 21 m.
Engaged TH as per GE Vetco instructions. Pulled to 110 t (40t overpull), observed actuating ring release. Continued to pick up, weight dropped of at 140 t hookload. POOH and inspected RT, no damage observed. RIH and engaged TH as per GE Vetco instructions, no clear indication of tool beeing fully engaged. Suspected actuator ring rotating. Picked up to 140 t, set back weight to 3 t on TH. Re-engaged retrieving tool. Picked up to 190 t
hookload (120 t overpull) and pulled TH free. POOH w/ tubing and tubing hanger on THRT and 5 1/2" DP stand from 21 to surface and landed TH in c-plate. Attempted to break out TDS, observed high torque on motor. Confirmed IBOPs not breaking out. Broke out from stand, 20 - 30 kNm torque.
Released THRT from tubing hanger. Required several attempts to come free. Inspected hanger, observed first quarter of actuator ring shoulder bent upwards. R/B stand and THRT in aft PRS. Removed communication collar and BPV from hanger. R/U 7" tubing handling equipment and Weatherford power tong. Installed TH handling tool. P/U string and installed Weatherford compact spider slips. POOH and L/O 7" tubing from 2926 m to 1860 m, removed cable clamps and spooled control line/cable. Continued to POOH with 7" tubing.

4.3 NO 15/9-F-1 B, Permanent P&A (DP) w/ RIG (03.Feb.2014 20:30)

START: 03.Feb.2014 20:30 3147 mMD
END : 19.Feb.2014 21:00 3148 mMD

SUMMARY

Attempted to P/U first 2 7/8" joint from HTS. Unable to tilt and close center latch elevator, bail rotator not designed for this type of elevator. Evaluated different options. Attempted to tilt elevator using tugger. Unable to latch elevator.. Decided to go ahead with TDS repair/change out of torque encoder.

RIH and sat asafelock plug at 619,9 m. Pulled casing hanger and 10 3/4" Tieback. Removed diverter.
Pulled overshot and removed BOP tensioners. Nippled down BOP. Disconnected surface riser and pulled same.
Rigged down cylinders, tensioner ring and centralizer.
Removed hatches and installed sheaves for guidewires. Ran guidewires and minireel. Disconnected production riser and pulled same.

Had to wait due to weather window for running the HPDR.

Ran drilling riser joints, tensioning joint and BOP joint. Ran down with riser until just above guide funnel. Lowered connector down inside guide funnel without any issues Landed drilling riser connector and sat down 20 T. Installed hot stab.
Pressured up by minireel to lock the connector. Performed OP test.
Installed hatches and build scaffolding. Tested riser and connector to 100 bar. Installed centralizers and tensioning system. Moved BOP and installed diverter. Finalized nippling of BOP. Performed connector test to 232 bar.

P/U Safe Lock retrieving tool and RIH, stopped at 170 m and displaced riser to 1,37 OBM.
Continued RIH to above plug. Established circulation and latched onto plug. Released same. POOH and L/D plug and pulling tool.
P/U and M/U cement stinger with a muleshoe. RIH 350 m 2 7/8" tubing and continued on 5 1/2" DP. Washed down and tagged EVO plug at 3148 m. Pulled up and conditioned mud.
Placed a 300 m cement plug. Pulled out of cement and circulated btm up before POOH. M/U and RIH with 8 1/2" Clean-out assy.
Washed down and drilled cement to confirmed hard cement at 2878 m. Confimed same by setting down 10 T.
Circulated btm up and POOH to 1457 m. Circulated to take down MW from 1,38 to 1,28 SG.
Continued POOH with 8 1/2" Clean-out assy.

M/U and RIH with a 9 5/8" cutting assembly to 1457 m MD. Cut the 9 5/8" casing with 800 lpm, and POOH and layed down assembly. P/U a jet sub and a lockdown assembly retrieval tool and RIH to retrieve the 10 3/4" lockdown assembly. Experienced 2 mis-runs due to problems with the retrieval tool before retrieving the lockdown assembly on the 3rd attempt, with 20 ton overpull.

After retrieving the lockdown assembly, the extension sleeve was removed from the retrieval tool, and the tool RIH again to to retrieve the 10 3/4" seal assembly. The assembly was retireved on the 1st attempt, with 12 ton overpull, with no trapped pressure below. A 10 3/4" fishing spear was then P/U and RIH, engaging the 10 3/4" casing right below the hanger. The cut 10 3/4" x 9 5/8" casing was pulled free without any overpull, and the well circulated to 1.40 sg OBM before pulling and retrieving the casing at surface.

The 13 3/8" wearbushing was installed, and a 13 3/4" cutting BHA RIH and the 13 3/8" casing cut at 1399 m MD with 1000 lpm and 80 rpm. The BHA was POOH before retrieving the 13 3/8" wearbushing and seal assembly, no trapped pressure below. A 13 3/8" spear assembly was P/U and RIH, engaging the 13 3/8" casing at 140 m, and the cut 13 3/8" casing pulled free and retrieved at surface with no overpull.

The nominal bore protector was installed, and a 5 1/2" cement stinger RIH to top of the cut 10 3/4" casing at 1457 m MD. The well was circulated and the mud conditioned for settitng a Hi-Vis bentonite pill and the KO cement plug. The Hi-Vis pill was spotted from top of the cut 10 3/4" casing to the top of the cut 13 3/8" casing before pulling above and setting a balanced cement kick-off plug to 50 m inside the 20" casing, at 1298 m MD.

4.4 NO 15/9-F-1 B, Prepare Sidetrack (DP) w/ RIG (19.Feb.2014 21:00)

START: 19.Feb.2014 21:00 1298 mMD
END : 21.Feb.2014 14:30 1364 mMD

OBJECTIVE
Drill out KO plug, and sidetrack well to wellbore F-1 C below 20" shoe.

SUMMARY
A 17 1/2" drilling BHA was MU and RIH, tagging the top of cement at 1293 m MD, and soft cement drilled down to 1302 m MD before WOC.
After WOC, the wellbore was sidetracked at 1355 m MD, with 50% formation returns at 1364 m MD confirming a successful kick-off.

4.5 NO 15/9-F-1 C, 17 1/2" (21.Feb.2014 14:30)

START: 21.Feb.2014 14:30 1364 mMD
END : 03.Mar.2014 18:15 2517 mMD

OBJECTIVE
Drill 17 1/2" hole to TD at 2514 m MD and set 13 3/8" casing in the shales of the lower Lista formation to seal off the lower Hordaland Group and the Balder formation.

SUMMARY
50% formation seen at 1364 m MD. Continued drilling to 1482 m MD, circulating BU prior to performing a check trip into the 20" shoe. Ran back down to 1482 m MD again, and continued drilling to 1540 m MD at a controlled ROP of 12 m/hr due to WOW to transfer cuttings to ISO-tank vessel. No drag was observed pulling into the shoe or when running back through again.

WOW to transfer cuttings to ISO-tank vessel for 14 hrs. Meanwhile circulated B/U and pulled into 20" shoe.

Ran back down to 1520 m MD again. Drilled down to 1684 m MD before WOW to transfer cuttings to ISO-tank vessel.

WOW to transfer cuttings to ISO-tank vessel for 48 hrs.

Drilled from 1684 m MD to 1864 m MD at 30 - 35 m/hr before WOW to transfer cuttings to ISO-tank vessel.

WOW to transfer cuttings to ISO-tank vessel for 6 hrs.

Drilled from 1864 m MD to 2088 m MD at 40 m/hr before WOW to transfer cuttings to ISO-tank vessel.

WOW to transfer cuttings to ISO-tank vessel for 4.25 hrs

Drilled from 2088 m MD to TD at 2514 m MD at 30-40 m/hr. Frequent stringer were encountered while drilling from bottom Skade to the bottom of the Lista formation.

BIT RUNS
A 17 ½" HP633F type bit and a Baker Hughes MWD/LWD string was run in hole on 5 ½" HWDP and 5 ½" drill pipe (DP) and drilled the section in one run.

Parameters: 3750 - 4300 lpm, 210-285 bar, 1150 m, 40,0 drill hrs, 87,8 circ hrs, 28,8 m/hr, 2 - 12 tons, 60 - 140 rpm, 3 - 28 kNm

Bit grading: 0 - 1 - WT - S - E - I - NO - TD

DRILLING FLUID
1.40 sg Enviromul Yellow mud system was selected for drilling the 17 1/2" section. At the end of mud weight was increased to 1.41 sg.

CASING
A shoe joint, intermediate joint and landing collar was run before 190 joints 13 3/8", 72lbs/ft, Q125, Vam Top casing was run to casing depth, at 2507,65 m MD RKB on a 5 ½" HWDP landing string.

The casing was tested to 270 bars for 10 minutes after plug bumped during cement/displacement job.

MWD/BHA
A 17 1/2" Baker Hughes Kymera, HP633F bit was selected by Baker Hughes for this section.

A Baker Hughes 9 1/2" AutoTrak steering unit was run together with a Baker Hughes MWD/LWD for formation evaluation and directional control.
8 1/4" drill collars (DC), 5 ½" heavy weight drill pipe (HWDP) and 5 ½" drill pipe were used above the MWD/LWD string.

CEMENTING
Circulated 1 x casing 195 m3 with 2500 lpm, 35 bar. No losses observed.

Pumped 25 m3 of 1,6 SG spacer with 2100 lpm, 23 bar through TDS with rig pumps.
Released dart for bottom wiper blug.Mixed and pumped 30,9 m3 of 1,92 SG cement slurry with 800 lpm. Released top dart. Displaced the cement to the rig floor with 1,5 m3 of drill water. Observed bottom dart landed in bottom wiper plug after 970 ltr. Shear pressure 117 bar. Observed top plug shared after 700 ltr, 64 bar. Released top dart and displaced cement using rig pumps with 2500 lpm, 8 bar. Decreased to 2000 lpm prior to shear ,Sheared btm plug at 135 bar, 6686 stroke. Top plug landed at 8250 strokes. FCP 10 bar

4.6 NO 15/9-F-1 C, 12 1/4" (03.Mar.2014 18:15)

START: 03.Mar.2014 18:15 2517 mMD
END : 11.Mar.2014 04:30 3059 mMD

OBJECTIVE
Drill the 12 ¼" section down to 3506 mMD. Set 10 3/4" x 9 5/8" casing into the Shetland Group to ensure good cement above the production packer setting depth.

SUMMARY
M/U 12 1/4" drilling assembly and RIH with 12 1/4" BHA to TOC at 2473,8 m MD. Performed choke drill and drilled out plug and cement to 2475 m. Displaced the well to 1,28 SG OBM. Drilled out shoe track and rat hole to 2514 m. Drilled 3 m new formation to 2517 m. Circulated well clean and condition mud. prior to the FIT. Performed FIT to 1,55 SG EMW. The 12 1/4" section was drilled in one run to TD @ 3056 by average 22.1 ROP.

BIT RUNS
A 12 1/4" Hughes Christensen QT406 type bit and Baker Hughes MWD/LWD was run in hole on 5 ½" HWDP and 5 ½" drill pipe (DP) and drilled to section TD at 3056 m MD in one run.

Parameters: 3850 - 3880lpm, 250 bar, 542 m, 27,4 drill hrs, 41 circ hrs, 23,5 m/hr, 6 - 12 tons, 120 rpm, 12 - 16 kNm

Bit grading: 1 - 1 - WT - A - X - I - NO - TD

Drilled 40 m cement for 4,3 hrs

DRILLING FLUID
The section was drilled with 1.28 sg Enviromul Yellow and LCM was bled into the mud due to a possible loss zone in Ty Fm. At TD circulated the hole clean with 3880 lpm, 242 bar, 140 RPM, 11-16 kNm, ECD 1,31 sg.

CASING
A shoe joint, an intermediate joint and a landing collar were run before 24 joints of centralized 9 5/8", 53,5 lb/ft, P100, VamTop. Then it was followed by 169 joints of normal 9 5/8", 53,5 lb/ft, P100, VamTop, a cross over and 39 joints of 10 3/4" 65,7lb/ft P110 VamTop casing. Casing setting depth at 3047 m MD. All the 10 3/4" joints and 72 joints of 9 5/8" had been racked in the derrick while drilling the 12 1/4" hole, and was RIH in stands. The average rate of casing running was 8 joints/hr.
During making up the crossover joint caused problem due to large OD size. Tight spots were experienced while running last stands, 500 l/min circulation was established and casing was washed down until the casing hanger landed on 5 1/2" landing string (Shoe at 3047 mMD).
Casing was tested to 345 bar. Casing was cemented with 1.90 sg slurry. After 6 hrs and prior to setting the seal assambly the annulus was bullheaded by pumping 95 m3 1.1 sg brine with 1000 l/min,100 bar. Brine was displaced with 20 m3 sea water. Seal asembely installed and tested to 345 bar.

MWD/BHA
An 12 1/4" Hughes Christensen, QT406 bit, IADC M333 was selected by Baker for this section. A Baker Hughes 9 1/2" AutoTrak steering unit was run to together with a Baker Hughes MWD and LWD (logging while drilling) for formation evaluation, directional control and vibration monitoring purposes.

CEMENTING
Circulated 1 x casing 115 m3 with 1500 lpm, 37 bar. No losses observed.

Pumped 20 m3 of 1,6 SG spacer with 1500 lpm, 38 bar through TDS with rig pumps.
Released dart for bottom wiper blug.Mixed and pumped 15 m3 of 1,92 SG cement slurry with 900 lpm. Released top dart.
Displaced the cement to the rig floor with 1,5 m3 of drill water. Observed bottom dart landed in bottom wiper plug after 900 ltr. Shear pressure 119 bar.
Observed top plug shared after 920 ltr, 106 bar. Started to displace cement using rig pumps with 1500 lpm, 53 bar.
Bumped top plug after 112 m3 displacement (5170 strokes with 94,9 %eff). Lost 8 m3 during displacemen.

The casing was pressure tested to 20/345 bar for 5/10 minutes after the cement job.

4.7 NO 15/9-F-1 C, 8 1/2" (11.Mar.2014 04:30)

START: 11.Mar.2014 04:30 3056 mMD
END : 15.Mar.2014 16:30 4094 mMD

OBJECTIVE
Drilling the 8 ½" with the planned trajectory down to section depth to 4039 mMD. Setting the liner 7in liner at TD. Cementing the liner and ensure that there is sufficient cement bond behind the liner.

SUMMARY
A 8 1/2" drilling assembly was made and RIH with 6 3/4" BHA to TOC at 3015 m MD. A choke drill was performed and the plug and cement were drilled out to 3047 m. The well was displaced to 1,34 SG OBM. The shoe track and rat hole were drilled out to 3056m, and 3 m of new formation was drilled to 3059 m. The well was circulated clean and mud conditioned, prior to the FIT. The FIT was performed to 1,60 SG EMW. Communication to the Auto track steering unit was lost and after several attempts decided to pull out of hole. Gyro dropped at shoe. Changed the steering unint (Baker Autotrak) and run in hole.
Drilled from 3059 to 3290 m. Avg. ROP: 19 m/hr. WOB 3 - 9 MT. 2100-2400 l/min/
180-220 bar. 100 rpm / 12 - 15 kNm. ECD 1,50-1,53 sg.

Commenced drilling and drilled to TD at 4094 m MD without any problems. Stopped rotation and re-logged. Circulated the hole clean with 3 x B/U while reciprocating the string. Performed sticky test across the reservoir and took 13 pressure points before POOH.

BIT RUNS
8 1/2" Baker Hughes TD606X Talon bit (IADC code: M223) was selected for section. Bit drilled shoe track and 3m of new formation

Parameters: 2200- 2400lpm,180 - 220 bar, 3m formation/ 32m cement, 7 drill hrs (5 hrs cement), 1circ hrs, 19 m/hr, 3 - 9 tons, 100 rpm, 12 - 15 kNm, ECD 1,50-1,53 sg. Due to problem with steering unit the BHA was pulled out.

Bit graded as 0 - 0 - NO - A X - I - NO - DTF and was re-ran for the second run.

Bit run # 2

Parameters: 2150- 2400lpm, 190 - 230 bar, 1035m, 44,9 drill hrs, 65,8circ hrs, 23,1 m/hr, 3 - 9 tons, 120-150 rpm, 12 - 22 kNm ECD 1,49-1,52 sg.

Bit grading: 1 - 1 - WT - A - X - I - NO - TD

DRILLING FLUID
While drilling the cement at 9 5/8" shoe; the mud (Enviromul Yellow) was conditioned and weight up from 1.28 to 1,34.
At TD the hole was circulated clean with 3 x B/U.

CASING

Discribed in Liner section below.

MWD/BHA
A Hughes Christensen TD606X type bit, IADC M233 was selected by Baker for this section. The bit was part of the misrun BHA and was re-run on run # 2. A Baker Hughes 6 3/4" AutoTrak steering unit was run to together with a Baker Hughes MWD and LWD (logging while drilling) including radioactive sources for formation evaluation, directional control and vibration monitoring purposes.

5 ½" heavy weight drill pipe (HWDP) and 5 ½" drill pipe were used above the MWD/LWD string.

CEMENTING
Pumped 15 m3 of 1,56 sg spacer w/700 l/min/ 83 - 58 bar. Isolated rig pumps from cement line, lined up and applied 30 bar against IBOP with rig pumps. Lined up through cement hose and verified line-up. Dropped dart # 1 for bottom plug. Mixed and pumped 18,4 m3 1,90 sg cement slurry w/ 650 l/min. Pumped 890 l drill water after cement job. Rotated liner while pumping cement, but had to stop rotation due to leaking swivel.

Released dart # 2 for top plug. Displaced cement with 900 l/min. Slowed down to 700 l/min prior to first dart releasing bottom WP at 570 strokes, 61 bar SPP. Continued pumping with 900 l/min/ 35 - 96 bar. Reduced pump rate at 1350 strokes to 700 l/min/ 67 bar. Observed second dart releasing top WP. Continued displacement with 1000 l/min. Established liner rotation with 20 rpm prior cement entering the annulus. Lowered rate to 800 l/min after experiencing loss tendencies. Slowed down to 500 l/min on the last 2 m3. Bumped the plug after 2430 strokes / 52,7 m3, 95 %pump efficiency, 120 bar. Increased pressure to 190 bar with rig pumps. Equalized with cement unit and pressure tested the liner to 300 bar/ 10 min.

4.8 NO 15/9-F-1 C, Liner (15.Mar.2014 16:30)

START: 15.Mar.2014 16:30 4094 mMD
END : 18.Mar.2014 16:15 4087 mMD

OBJECTIVE
Run and cement the 7" liner

SUMMARY
P/U 7" reamer shoe joint. M/U shoe joint, 3 x intermediate joints and float/landing collar joint and RIH to 59 m. Tested shoe float holding, filled the liner and checked for flow through. Installed swedge and verified circulation before it was layed out.

92 standard joints was run, together with 3 pups for later correlation.

The liner was RIH on a 5 1/2" HWD and 5 1/2" DP landingstring, broke circulation at the 9 5/8" casing shoe and circulated one annulus volume.
Continued to RIH until 4036 m MD. Established circulation and washed and worked the liner down to 4075m MD w/ 250-450 LPM and 30-60 bar. Took weight and experienced pack-offs. Pulled above restrictions to clear pack-off and worked the string down. Continue to wash and work the string to 4081m MD, reamed down w/ 20 RPM to 4084m MD and was able to get the string to 4090 m MD.

Decided to set the liner at 4087m MD (6m shallower than planned depth).

5 Formation evaluation

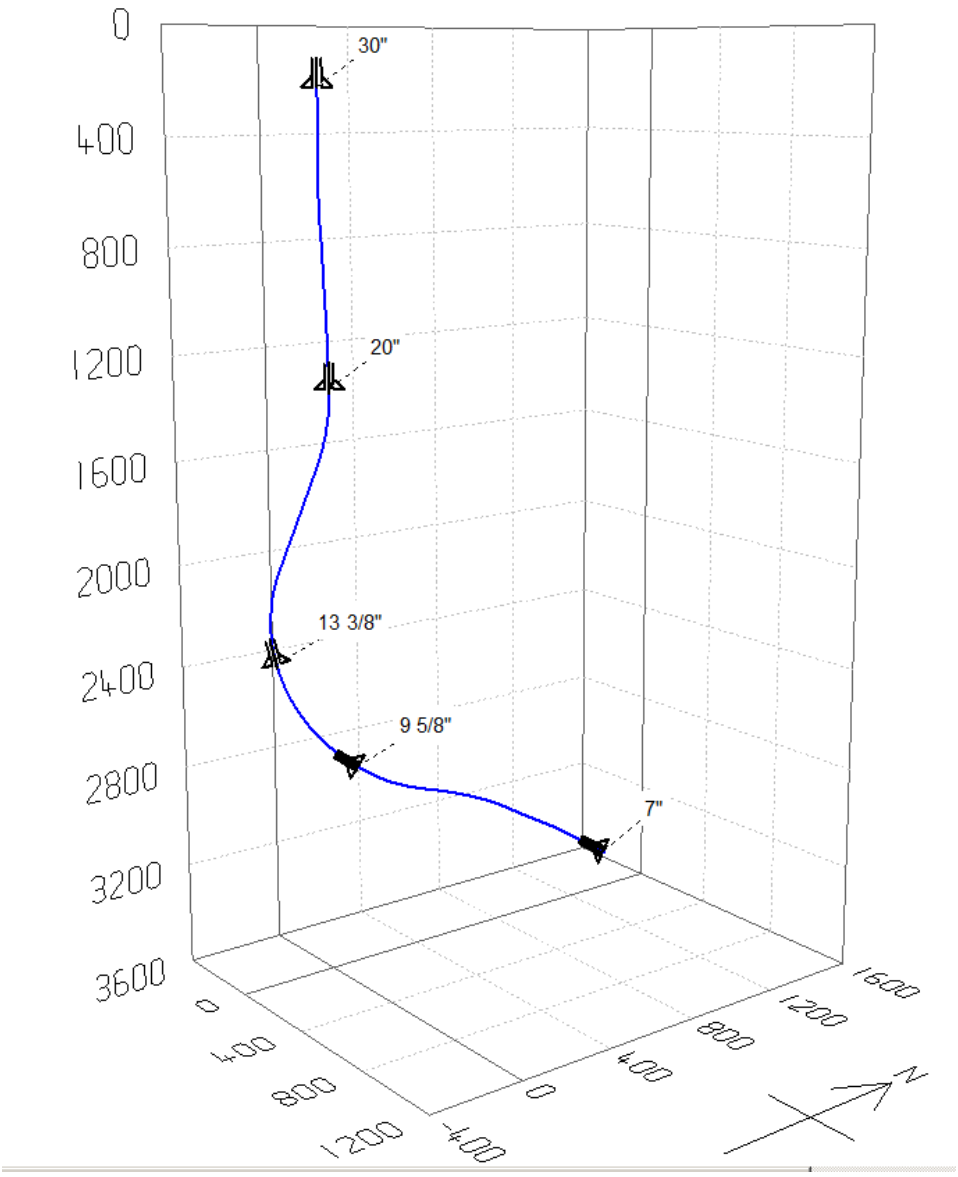
5.1 Pressure points

Test no	Depth m MD RKB	Depth m TVD RKB	Form. pressure BAR	Hydrostatic pressure Before BAR	Hydrostatic pressure After BAR	Formation	Comments
1	3977,0	3138,7	354,58	446,10	446,95	Hugin	mobility 148.7
2	3925,1	3120,2	353,17	444,30	445,00	Hugin	mobility 84.2
3	3858,0	3096,5	352,20	440,57	441,19	Hugin	mobility 35.4
4	3823,1	3084,9	350,95	438,83	439,33	Hugin	mobility 8.0
5	3768,1	3067,0	349,50	435,89	436,39	Hugin	mobility 270.6
6	3739,9	3057,6	348,79	434,85	435,41	Hugin	mobility 610.6
7	3720,1	3050,6	348,27	433,52	434,09	Hugin	mobility 323.0
8	3620,2	3012,3	344,35	428,72	429,12	Hugin	mobility 550.5
9	3603,0	3006,7	345,38	426,57	426,67	Hugin	mobility 77.7
10	3551,1	2992,9	341,51	422,84	426,67	Hugin	mobility 95.9
11	3523,2	2986,7	340,98	421,42	423,10	Hugin	mobility 53.8
12	3493,0	2980,2	340,45	421,67	421,85	Hugin	mobility 458.9
13	3273,1	2948,3	338,18	415,41	415,76	Hugin	mobility 1152.9

5.2 Geological formation tops

PROGNOSIS					ACTUAL		
Formation	depth m MD RKB	depth m TVD RKB	m MSL	Incertaint ± m TVD	depth m MD RKB	depth m TVD RKB	difference m TVD RKB
KOP							
Grid Fm Top	2134	2078	2023	50	2112	2058	-20
Grid Fm Base	2204	2145	2090	50	2163	2106	-39
Balder Fm Top	2319	2258	2204	30	2319	2259	1
Sele Fm Top	2377	2316	2261	30	2384	2324	8
Lista Fm Top	2444	2383	2328	30	2454	2393	10
Ty Fm Top	2537	2473	2419	20	2541	2478	5
Ekofisk Fm Top	2674	2601	2546	25	2668	2596	-5
Tor Fm Top	2684	2610	2555	25	2685	2611	1
Hod Fm Top	2868	2757	2702	20	2877	2764	7
Blodøks_Fm._Top					2979	2830	
Hidra Fm Top	3112	2896	2842	30	3067	2879	-18
Rødby Fm Top	3122	2901	2846	40	3089	2889	-12
Åsgard Fm Top	3171	2919	2864	40	3142	2912	-7
Draupne Fm Top (BCU)	3240	2938	2883	20	3205	2933	-5
Heather Fm Top	3247	2940	2885	20	3223	2938	-2
Heather 'SST' Top	3256	2942	2887	20			
Hugin Fm Top	3259	2943	2888	15	3230	2939	-3
Hugin Fm Base	3940	3152	3097	15	4004	3148	-3
TD	4039	3216	3161		4094	3180	

6 Appendix 1: Directional data



Norway

SLEIPNER
Volve F
F-1

F-1 C

Design: F-1 C

Standard Survey Report

22 mars, 2014

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54,90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54,90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
Design:	F-1 C	Database:	Production EDM P246N

Project SLEIPNER, Norway

Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site Volve F, 15/9

Site Position:		Northing:	6 478 563,53 m	Latitude:	58° 26' 29,807 N
From:	Map	Easting:	435 050,02 m	Longitude:	1° 53' 14,929 E
Position Uncertainty:	0,00 m	Slot Radius:	13,200 in	Grid Convergence:	-0,95 °

Well F-1

Well Position	+N/-S	3,17 m	Northing:	6 478 566,69 m	Latitude:	58° 26' 29,907 N
	+E/-W	-3,53 m	Easting:	435 046,49 m	Longitude:	1° 53' 14,707 E
Position Uncertainty		0,00 m	Wellhead Depth:	91,00 m	Water Depth:	91,00 m

Wellbore F-1 C

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	GNETICREFERENCE	31.10.2013	-1,09	71,62	50 525

Design F-1 C

Audit Notes:

Version:	1.0	Phase:	ACTUAL	Tie On Depth:	1 335,60

Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145,90	3,17	-3,53	27,98

Survey Program Date 18.03.2014

From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
150,15	210,70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)
237,60	358,50	15/9 F1 26" Gyro Scientific Modular (F-1)	MWD gyro, Baker Hughes	Baker Hughes' MWD gyro service
378,00	737,40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools
756,80	1 335,60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)
1 388,70	2 501,10	15/9 F-1 C 17 1/2" AutoTrak G3 (F-1 C)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
2 548,10	3 032,10	15/9 F-1 C 12 1/4" AutoTrak G3 (F-1 C)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3 032,10	4 079,20	15/9 F-1 C 8 1/2" AutoTrak G3 (F-1 C)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
145,90	0,00	0,00	145,90	3,17	-3,53	0,00	0,000	0,000	0,000
150,15	0,00	0,00	150,15	3,17	-3,53	0,00	0,000	0,000	0,000
160,45	0,00	0,00	160,45	3,17	-3,53	0,00	0,000	0,000	0,000
170,04	0,00	0,00	170,04	3,17	-3,53	0,00	0,000	0,000	0,000
179,79	0,00	0,00	179,79	3,17	-3,53	0,00	0,000	0,000	0,000

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54,90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54,90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
Design:	F-1 C	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
189,74	0,00	0,00	189,74	3,17	-3,53	0,00	0,000	0,000	0,000
198,50	0,00	0,00	198,50	3,17	-3,53	0,00	0,000	0,000	0,000
210,70	0,00	0,00	210,70	3,17	-3,53	0,00	0,000	0,000	0,000
237,60	1,84	302,24	237,60	3,40	-3,90	0,03	2,052	2,052	0,000
251,10	1,91	304,70	251,09	3,64	-4,27	0,07	0,237	0,156	5,467
264,90	2,47	305,76	264,88	3,94	-4,70	0,14	1,221	1,217	2,304
279,20	2,05	306,81	279,17	4,28	-5,15	0,22	0,885	-0,881	2,203
292,70	1,80	304,00	292,66	4,54	-5,52	0,28	0,594	-0,556	-6,244
302,80	1,13	330,72	302,76	4,72	-5,70	0,35	2,791	-1,990	79,366
315,50	1,41	333,18	315,45	4,96	-5,83	0,51	0,674	0,661	5,811
332,20	2,40	314,19	332,14	5,39	-6,18	0,73	2,086	1,778	-34,114
342,20	2,78	301,18	342,13	5,66	-6,54	0,80	2,093	1,140	-39,030
358,50	3,30	293,10	358,41	6,05	-7,30	0,78	1,238	0,957	-14,871
378,00	3,07	298,60	377,88	6,52	-8,28	0,74	0,588	-0,354	8,462
414,80	3,30	293,64	414,62	7,42	-10,11	0,67	0,292	0,187	-4,043
455,40	3,07	292,72	455,16	8,31	-12,19	0,48	0,174	-0,170	-0,680
495,40	2,84	290,41	495,11	9,07	-14,10	0,25	0,194	-0,172	-1,732
535,60	2,77	293,07	535,26	9,79	-15,93	0,04	0,110	-0,052	1,985
575,80	3,06	287,04	575,41	10,49	-17,85	-0,25	0,315	0,216	-4,500
616,50	4,01	305,32	616,03	11,63	-20,05	-0,27	1,078	0,700	13,474
656,80	6,13	317,80	656,17	14,04	-22,65	0,64	1,770	1,578	9,290
697,10	8,24	324,71	696,15	17,99	-25,76	2,67	1,695	1,571	5,144
720,00	9,57	322,37	718,78	20,84	-27,87	4,19	1,805	1,742	-3,066
737,40	10,71	320,81	735,91	23,24	-29,78	5,41	2,022	1,966	-2,690
756,80	10,39	320,18	754,98	25,98	-32,04	6,77	0,526	-0,495	-0,974
798,90	9,88	320,50	796,42	31,68	-36,76	9,59	0,366	-0,363	0,228
838,30	10,04	315,72	835,23	36,75	-41,31	11,93	0,641	0,122	-3,640
868,00	11,16	309,73	864,42	40,44	-45,33	13,31	1,586	1,131	-6,051
878,20	11,28	310,41	874,43	41,72	-46,85	13,72	0,525	0,353	2,000
918,00	10,70	315,76	913,50	46,89	-52,39	15,69	0,884	-0,437	4,033
999,00	8,88	319,88	993,32	57,06	-61,67	20,32	0,722	-0,674	1,526
1 079,80	9,07	320,34	1 073,13	66,73	-69,75	25,07	0,075	0,071	0,171
1 120,40	11,38	312,16	1 113,08	71,88	-74,76	27,27	2,012	1,707	-6,044
1 160,80	11,86	307,61	1 152,65	77,09	-81,01	28,94	0,768	0,356	-3,379
1 201,00	11,69	307,74	1 192,01	82,10	-87,50	30,32	0,128	-0,127	0,097
1 241,40	12,44	309,36	1 231,51	87,37	-94,10	31,87	0,611	0,557	1,203
1 281,80	12,06	309,62	1 270,99	92,82	-100,72	33,58	0,285	-0,282	0,193
1 322,40	10,85	310,25	1 310,79	97,99	-106,90	35,25	0,899	-0,894	0,466
1 335,60	10,17	309,34	1 323,76	99,53	-108,75	35,74	1,591	-1,545	-2,068
1 388,70	9,03	293,00	1 376,13	104,14	-116,21	36,30	1,662	-0,644	-9,232
1 468,00	15,99	273,27	1 453,53	107,20	-132,87	31,19	3,054	2,633	-7,464
1 509,10	17,10	272,45	1 492,93	107,78	-144,56	26,22	0,828	0,810	-0,599
1 547,40	17,12	261,15	1 529,54	107,15	-155,76	20,41	2,600	0,016	-8,851
1 588,80	18,12	249,89	1 569,01	104,00	-167,83	11,96	2,570	0,725	-8,159

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54,90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54,90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
Design:	F-1 C	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1 627,20	19,44	243,03	1 605,37	99,05	-179,14	2,29	2,008	1,031	-5,359
1 669,40	20,97	234,67	1 644,98	91,49	-191,56	-10,21	2,320	1,088	-5,943
1 710,10	20,97	231,97	1 682,99	82,79	-203,24	-23,37	0,712	0,000	-1,990
1 751,30	20,98	232,49	1 721,46	73,76	-214,89	-36,82	0,136	0,007	0,379
1 791,00	21,01	231,04	1 758,52	64,96	-226,07	-49,83	0,393	0,023	-1,096
1 833,50	21,06	229,16	1 798,19	55,17	-237,77	-63,96	0,478	0,035	-1,327
1 873,20	20,99	230,48	1 835,25	45,98	-248,65	-77,18	0,362	-0,053	0,997
1 913,90	20,99	232,25	1 873,25	36,88	-260,04	-90,56	0,467	0,000	1,305
1 952,50	21,02	234,84	1 909,28	28,66	-271,16	-103,04	0,722	0,023	2,013
2 032,90	21,59	231,88	1 984,19	11,23	-294,59	-129,43	0,454	0,213	-1,104
2 073,60	21,61	231,81	2 022,03	1,97	-306,37	-143,13	0,024	0,015	-0,052
2 115,30	20,99	231,93	2 060,88	-7,38	-318,29	-156,98	0,447	-0,446	0,086
2 155,10	17,84	234,16	2 098,42	-15,35	-328,84	-168,97	2,439	-2,374	1,681
2 195,50	14,86	235,78	2 137,18	-21,89	-338,14	-179,11	2,238	-2,213	1,203
2 235,00	11,48	240,64	2 175,64	-26,66	-345,76	-186,90	2,699	-2,567	3,691
2 275,50	7,80	249,10	2 215,56	-29,62	-351,84	-192,37	2,914	-2,726	6,267
2 316,10	5,19	266,14	2 255,90	-30,73	-356,25	-195,41	2,377	-1,929	12,591
2 356,10	3,96	307,43	2 295,78	-30,01	-359,15	-196,14	2,567	-0,922	30,967
2 397,10	4,71	2,44	2 336,68	-27,47	-360,21	-194,39	2,968	0,549	40,251
2 437,60	7,62	20,98	2 376,94	-23,30	-359,17	-190,22	2,585	2,156	13,733
2 477,80	11,51	27,03	2 416,58	-17,23	-356,40	-183,56	2,994	2,903	4,515
2 501,10	13,26	29,06	2 439,34	-12,83	-354,04	-178,57	2,322	2,253	2,614
2 548,10	16,86	33,41	2 484,72	-2,42	-347,67	-166,39	2,407	2,298	2,777
2 588,80	20,69	35,44	2 523,24	8,37	-340,25	-153,38	2,863	2,823	1,496
2 629,10	23,86	36,02	2 560,53	20,76	-331,32	-138,25	2,365	2,360	0,432
2 668,70	27,33	36,53	2 596,24	34,55	-321,20	-121,32	2,634	2,629	0,386
2 710,30	30,66	36,83	2 632,62	50,72	-309,15	-101,39	2,404	2,401	0,216
2 750,70	33,68	37,05	2 666,81	67,91	-296,23	-80,15	2,244	2,243	0,163
2 790,80	37,44	37,39	2 699,43	86,47	-282,12	-57,13	2,817	2,813	0,254
2 830,90	41,47	37,47	2 730,39	106,70	-266,63	-32,00	3,015	3,015	0,060
2 870,80	44,49	38,12	2 759,57	128,19	-249,96	-5,20	2,295	2,271	0,489
2 911,10	48,30	38,95	2 787,36	151,01	-231,78	23,48	2,871	2,836	0,618
2 951,30	51,15	39,44	2 813,35	174,78	-212,40	53,56	2,145	2,127	0,366
2 991,70	54,13	39,39	2 837,86	199,58	-192,01	85,03	2,213	2,213	-0,037
3 032,10	57,66	40,25	2 860,51	225,27	-170,59	117,76	2,674	2,621	0,639
3 071,30	60,97	40,46	2 880,51	250,95	-148,76	150,69	2,537	2,533	0,161
3 100,80	62,87	40,45	2 894,40	270,76	-131,88	176,10	1,932	1,932	-0,010
3 141,10	66,74	40,30	2 911,55	298,53	-108,26	211,71	2,883	2,881	-0,112
3 181,80	70,77	40,50	2 926,29	327,41	-83,68	248,75	2,974	2,971	0,147
3 221,70	75,83	40,95	2 937,76	356,37	-58,75	286,01	3,818	3,805	0,338
3 262,00	79,57	40,62	2 946,34	386,17	-33,03	324,40	2,794	2,784	-0,246
3 302,20	82,18	40,33	2 952,71	416,36	-7,27	363,15	1,959	1,948	-0,216
3 342,40	84,06	39,77	2 957,53	446,91	18,41	402,17	1,463	1,403	-0,418

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54,90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54,90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
Design:	F-1 C	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3 382,90	83,37	39,94	2 961,96	477,82	44,20	441,57	0,526	-0,511	0,126
3 423,40	81,81	39,41	2 967,19	508,73	69,84	480,89	1,219	-1,156	-0,393
3 464,20	78,33	40,11	2 974,22	539,62	95,54	520,23	2,609	-2,559	0,515
3 504,00	77,54	40,24	2 982,54	569,36	120,65	558,27	0,603	-0,595	0,098
3 544,40	77,09	40,50	2 991,41	599,39	146,18	596,77	0,384	-0,334	0,193
3 585,20	74,13	41,17	3 001,55	629,28	172,02	635,29	2,228	-2,176	0,493
3 625,50	69,54	40,96	3 014,11	658,15	197,17	672,58	3,420	-3,417	-0,156
3 666,00	65,98	41,60	3 029,44	686,31	221,89	709,06	2,673	-2,637	0,474
3 706,60	67,37	43,50	3 045,51	713,77	247,10	745,14	1,648	1,027	1,404
3 745,90	70,53	46,54	3 059,63	739,69	273,05	780,19	3,242	2,412	2,321
3 786,50	71,07	47,83	3 072,98	765,74	301,17	816,40	0,985	0,399	0,953
3 826,80	71,06	50,67	3 086,06	790,62	330,05	851,92	2,000	-0,007	2,114
3 866,80	69,52	52,20	3 099,55	814,10	359,49	886,46	1,581	-1,155	1,147
3 907,40	69,07	53,30	3 113,91	837,09	389,72	920,94	0,830	-0,333	0,813
3 947,80	69,09	55,53	3 128,33	859,04	420,40	954,73	1,547	0,015	1,656
3 987,90	69,09	56,45	3 142,65	880,00	451,45	987,80	0,643	0,000	0,688
4 028,00	69,15	56,90	3 156,94	900,58	482,76	1 020,67	0,318	0,045	0,337
4 069,20	69,11	57,78	3 171,62	921,35	515,17	1 054,22	0,599	-0,029	0,641
4 079,20	69,09	57,67	3 175,18	926,34	523,07	1 062,33	0,314	-0,060	-0,330
4 094,00	69,09	57,67	3 180,47	933,74	534,75	1 074,34	0,000	0,000	0,000

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
220,90	220,90	30"	30,000	36,000
1 348,40	1 336,37	20"	20,000	26,000
2 507,70	2 445,75	13 3/8"	13,375	17,500
3 047,00	2 868,34	9 5/8"	9,625	12,250
4 087,00	3 177,97	7"	7,000	8,500

7 Appendix 2: Operational experience

7.1 Appendix 2.1: Operational listing (optional)

WELLBORE ID: NO 15/9-F-1 B
INTERVAL: Preparations
START TIME: 25.Jan.2014 11:00
END TIME: 31.Jan.2014 07:45

Report date	Description
26.Jan.2014	Took over well F-1 B from production 11:00 hrs. R/U wireline pressure control equipment. Built cable head and prepared toolstring for run #1 (EVO plug).
27.Jan.2014	M/U toolstring run #1 (EVO plug). RIH, correlated and set plug at 3147 m. POOH and L/O toolstring. Prepared wireline run #2. Suspended operations on drillfloor and cantilever due to ice falling down.
28.Jan.2014	WOW - ice lumps falling to rigfloor/cantilever due to strong wind. Meanwhile: Evaluated and dicussed with onshore organization situation for pressure testing/ bled off on F-1 B. Pressure tested EVO plug to 220 bar/10 min. Bled of pressure via F-12. Performed housekeeping and maintenance while WOW.
29.Jan.2014	M/U wireline run #2 (tubing puncher) and RIH with same. Punched tubing. POOH and L/D toolstring. Reverse circulated well to kill mud. Circulated and conditioned mud.
30.Jan.2014	Circulated and conditioned kill mud. M/U and RIH with W/L run #3 (tubing cutter). Cut 7" tubing at 2948,5 m, POOH and L/D toolstring. R/D wireline equipment and inflow tested DHSV. N/D Xmas tree.
31.Jan.2014	N/D Xmas tree and spacer spool. Installed HP riser joint. Skidded cantilever. N/U BOP.

INTERVAL: Pull Completion
START TIME: 31.Jan.2014 07:45
END TIME: 03.Feb.2014 20:30

Report date	Description
01.Feb.2014	N/U BOP. Installed dart in BPV in tubing hanger. P/T BOP to 20/345 bar for 5/10 min. R/U Weatherford equipment. P/U and M/U THRT. Troubleshot TDS brake and BX elevator malfunction.
02.Feb.2014	Performed fault finding on TDS brake system. Removed brake disc and re-installed brake assembly on TDS. Prepared TDS for operation. P/U and M/U THRT, RIH with same to 21 m.
03.Feb.2014	Release TH and POOH with 7" completion on TH and THRT. R/U tubing handling equipment and Weatherford power tong. B/O and L/O TH. POOH and L/O 7" tubing from 2926 m to 1860 m, removed cable clamps and spooled control line/cable.

INTERVAL: Permanent P&A (DP) w/ RIG
START TIME: 03.Feb.2014 20:30
END TIME: 19.Feb.2014 21:00

Report date	Description
04.Feb.2014	POOH with 7" tubing and completion sub-assemblies from 1860 m to surface. Cleared and cleaned rigfloor. R/U 2 7/8" handling equipment and Weatherford power tong to pick up cement stinger.
05.Feb.2014	Changed out torque encoder on TDS. P/U and RIH with 9 5/8" Safelock plug to 587 m. Replaced TDS service loop.
06.Feb.2014	Replaced TDS service loop and tested TDS. Set 9 5/8" Safelock plug at 619 m and pressure tested same from below and above. Retrieved bowl protector. Retrieved MS-T seal assembly. RIH and engaged DrilQuip adjustable sub setting tool. Released tension from 10 3/4" surface tieback. Troubleshot stuck tool, released tieback adapter and POOH tieback with stuck tool inside. R/U to retrieve 10 3/4" tieback.
07.Feb.2014	Pulled casing hanger and 10 3/4" Tieback. Removed Diverter. Shut-in production. Pulled Overshot and removed BOP Tensioners. Nippled down BOP. Disconnected Surface riser and pulled same. Rigged down Cylinders, Tensioner ring and Centralizer. Removed hatches and installed sheaves for Guidewires.
08.Feb.2014	Removed hatches and installed sheaves for Guidewires. Ran Guidewires and Minireel. Removed dummy Hotstab and inserted Hotstab. Waited for boat handling. Disconnected Production Riser. Installed Spider and pulled PR and L/D same to deck.
09.Feb.2014	Worked on TDS brake disc. Tested same. Waited on Weather. Meanwhile tested BOP and P/U and tested Drilling Riser Connector. L/D same. Tested Kelly hose and IBOPs.
10.Feb.2014	Waited on Weather. Meanwhile tested manifolds. P/U Drilling Riser Connector. Performed maintenance and cleaned and cleared in drilling areas. Re-function tested Connector.
11.Feb.2014	Waited on Weather. Ran Drilling Riser joints, Tensioning joint and BOP joint. Removed Riser Spider. Ran down with Risers until just above Guide Funnel. Lowered Connector down inside Guide Funnel.

Report date	Description
12.Feb.2014	Parked Guide Funnel aside of WH. Landed Drilling Riser Connector and sat down 20 T. Installed Hot stab. Pressured up by Minireel to lock Connector. Performed OP test. Installed hatches and build scaffolding. Tested riser and Connector to 100 bar. Installed Centralizers and Tensioning system. Moved BOP and installed Diverter. Finalized nippling of BOP. Performed Connector test to 232 bar. P/U Safe Lock retrieving tool.
13.Feb.2014	P/U and RIH with Safe Lock pulling tool. Stopped at 170 m and displaced riser to 1,37 OBM. Continued RIH to above plug. Established circulation and latched onto plug. Released same. POOH and L/D plug and pulling tool. P/U and M/U cement stinger Muleshoe. RIH same on 350 m 2 7/8" tubing from deck. Continued RIH on 5 1/2" DP to 2735 m.
14.Feb.2014	Continued RIH with cement stinger from 1726 to 3097 m. Washed down and tagged EVO plug at 3148 m. Pulled up and conditioned mud. Placed a 300 m cement plug. Pulled out of cement and circulated btm up. POOH with cement stinger to 168 m.
15.Feb.2014	Continued RIH with 8 1/2" Clean-out assy from 168 to 2712 m. Washed down and drilled cement to confirmed hard cement at 2878 m. Confimed same by setting down 10 T. Circulated btm up and POOH to 1457 m. Circulated to take down MW from 1,38 to 1,28 SG. Continued POOH with 8 1/2" Clean-out assy to 534 m.
16.Feb.2014	Continued RIH with 9 5/8" Cutter assy from 815 to 1457 m. Cutted casing. POOH with casing Cutter assy. P/U Retrieval tool for Lock Down assy. RIH and retrieved Lock Down assy on third attempt. Re-dressed tool and RIH to pull 10 3/4" Seal assy. Pulled same. M/U Spear assy and RIH to 119 m. Slip and cut drill line.
17.Feb.2014	Finalized slip and cut. Postioned Spear at 144 m and engaged same. Pulled casing free. Circulated to take up mud weight from 1,28 to 1,40 SG. Pulled casing to RKB on HWDP. Rigged up casing handling equipment and pulled 10 3/4" x 9 5/8" casing to 721 m.
18.Feb.2014	Continued POOH with 9 5/8" casing from 721 to surface. Rigged down casing handling equipment. Installed Wearbushing. RIH with 13 3/8" casing Cutter and cut at 1399 m. POOH with same. Pulled WB. RIH with Seal assy RT to 174 m.
19.Feb.2014	Continued to retrieve 13 3/8" Seal assy. POOH with same. RIH with 13 3/8" Spear and engaged same. POOH with 13 3/8" casing Spear. Continued to POOH with 13 3/8" casing joints and L/D same to deck. Rigged down casing handling equipment. RIH with 20" Nominal Bore Protector to 132 m.

INTERVAL: Prepare Sidetrack (DP) w/ RIG
START TIME: 19.Feb.2014 21:00
END TIME: 21.Feb.2014 14:30

Report date	Description
20.Feb.2014	Flushed WH area. Installed 20" nominal bore protector in profile as per procedure. P/U and M/U cement stinger. RIH to 1457 m. Circulated and conditioned. Set Balanced cement plug from 1399 m to 1298 m. POOH to top of cement and circulated btm's up and string clean. POOH and LD CMT stinger. WOC, M/U 17 1/2" drilling assembly to 62 m.
21.Feb.2014	WOC. M/U 17 1/2" drilling assembly. RIH with 17 1/2" drilling assembly on 5 1/2" DP to 1200 m. Prepared to P/U and M/U casing. Clean out soft cement in 20" casing from 1293 m to 1302 m. Circulated btm's up and clean. M/U 13 3/8" casing and set same back in derrick. WOW. M/U 13 3/8" casing and set same back in derrick. Performed choke drill with crew.

WELLBORE ID: NO 15/9-F-1 C
INTERVAL: 17 1/2"
START TIME: 21.Feb.2014 14:30
END TIME: 03.Mar.2014 18:15

Report date	Description
22.Feb.2014	Drilled 17 1/2" hole from 1364 m to 1482 m. Circulated bottoms up. Perform check trip to 10 m inside 20" shoe. Drilled 17 1/2" hole to 1502 m.
23.Feb.2014	Drilled 17 1/2" hole from 1502 m to 1540 m. WOW circulated well clean. POOH with 17 1/2" BHA to inside 20" shoe. Performed general service on drilling equipment while WOW. RIH with 17 1/2" BHA to 1540 m. Drilled 17 1/2" hole to 1660 m.
24.Feb.2014	Drilled 17 1/2" hole from 1660 m to 1684 m. Waited one weather, not able to discharge cuttings to vessel due to bad weather. Circulated well clean. POOH with 17 1/2" BHA on 5 1/2" DP to inside 20" shoe. Performed general rig service and PM on drilling equipment.
25.Feb.2014	Waited one weather, not able to discharge cuttings to vessel due to bad weather. Performed general rig service and PM on drilling equipment. Transferred 50 ton cottings to Blue Guardian. Established circulation, down link battery on. RIH with 17 1/2" BHA to 1585 m.
26.Feb.2014	Waited one weather, not able to discharge cuttings to vessel due to bad weather. RIH with 17 1/2" BHA to 1684 m. Drilled 17 1/2" hole from 1684 m to 1864 m. Waited one weather, not able to discharge cuttings to vessel due to bad weather. Circulated well clean. Pumped out of hole to 1713 m. POOH with 17 1/2" BHA to 1319 m, inside 20" shoe. RIH with 17 1/2" BHA to 1864 m. Drilled 17 1/2" hole from 1864 m to 2080 m.
27.Feb.2014	Drilled 17 1/2" hole to 2088 m. Waited one weather, not able to discharge cuttings to vessel due to bad weather. Circulated well clean. Drilled 17 1/2" hole to 2514 m. Started to circulate hole clean.
28.Feb.2014	Circulate hole clean at 2514 m, TD. POOH with 17 1/2" BHA to 2088 m. Unplanned release of stand into derrick, retrieved same. POOH with 17 1/2" BHA from 2088 m, R/B and L/D BHA according to plan.
01.Mar.2014	Retrieved nominal seat protector. R/U 13 3/8" casing equipment. P/U 13 3/8" shoe track. RIH with 13 3/8" casing stand to 1848 m.
02.Mar.2014	RIH with 13 3/8" casing to 2362 m. R/D 13 3/8" casing equipment. RIH 13 3/8" casing on landing string, M/U cement head. Landed 13 3/8" casing at 2507,8 m. Performed cement job on 13 3/8" casing. Sat and pressure tested seal assy. POOH and L/D CSG running tool.
03.Mar.2014	Pressure tested 13 3/8" casing to 270 bar. Installed wear bushing in well head. Pressure tested BOP. Changed out IBOP and saver sub on TDS. Pressure tested TDS and STDP manifold. M/U 12 1/4" drilling assembly to 51 m.

INTERVAL: 12 1/4"
START TIME: 03.Mar.2014 18:15
END TIME: 11.Mar.2014 04:30

Report date	Description
04.Mar.2014	M/U 12 1/4" drilling assembly. RIH with 12 1/4" BHA to TOC. Performed choke drill. Drilled out plug and cement to 2475 m. WOW. Displaced to 1,28 SG OBM. Drilled out shoe track and rat hole to2514 m. Drilled 3 m new formation to 2517 m. Circulated well clean and condition mud. Performed FIT to 1,55 SG EMW. Drilled 12 1/4" hole to 2646 m.
05.Mar.2014	Drilled 12 1/4" hole from 2646 m to 3056 m. Circulated bottoms up and hole clean.
06.Mar.2014	Circulated hole clean at section TD 3056 m MD. POOH with 12 1/4" BHA and L/O same. Retrieved wearbushing and jetted wellhead area. R/U casing handling equipment. P/U and M/U 9 5/8" shoe track.

Report date	Description
07.Mar.2014	M/U 9 5/8" shoe track. RIH with 9 5/8" casing stands from derrick to 936 m. P/U and RIH with 9 5/8" casing joints from deck from 936 m to 2337 m.
08.Mar.2014	RIH with 9 5/8" casing from 2337 m to 2410 m. Changed handling equipment to 10 3/4". M/U X-over joint and RIH with 10 3/4" casing stands from 2410 m to 2902 m. Installed casing hanger and RIH on HWDP landing string. Landed casing at 3047 m. Circulated prior to cement job. Performed cement job.
09.Mar.2014	Pressure tested 10 3/4" x 9 5/8" casing to 345 bar/10 min. WOC prior to bullheading B-annulus to brine. Bullheaded B-annulus to 1,10 sg brine. Set 10 3/4" seal assembly and pressure tested same to 20/345 bar for 5/10 min. POOH with PADPRT and L/O same. Repaired annular BOP and performed maintenance on drilling equipment.
10.Mar.2014	Repaired annular BOP and performed maintenance on drilling equipment. Pressure tested BOP. Installed wearbushing. M/U 8 1/2" RSS BHA.

INTERVAL: 8 1/2"
START TIME: 11.Mar.2014 04:30
END TIME: 15.Mar.2014 16:30

Report date	Description
11.Mar.2014	P/U and M/U 8 1/2" RSS BHA, RIH to 1045 m. Performed choke drill with both crews. RIH with 8 1/2" BHA to 2914 m. Performed slip & cut. RIH, tagged TOC at 3015 m. Drilled out shoe track and rathole, drilled 3m new formation to 3059 m.
12.Mar.2014	Reamed rat hole below 9 5/8" shoe. Circulated and conditoned mud prior to FIT. Performed FIT to 1,60 sg EMW. Troubleshot downhole tools. Dropped gyro and POOH with 8 1/2" BHA. L/O AutoTrak RSS, M/U back-up steering head. RIH with 8 1/2" BHA to 140 m.
13.Mar.2014	RIH with 8 1/2" RSS BHA from 140 m to 3059 m. Drilled and oriented 8 1/2" from 3059 m to 3293 m.
14.Mar.2014	Drilled and geosteered 8 1/2" hole from 3290 m to 3806 m.
15.Mar.2014	Drilled and geosteered 8 1/2" hole from 3806 m to TD at 4094 m. Circulated hole clean. Performed re-logging and pipe stretch calibration. Took pressure points as per program.

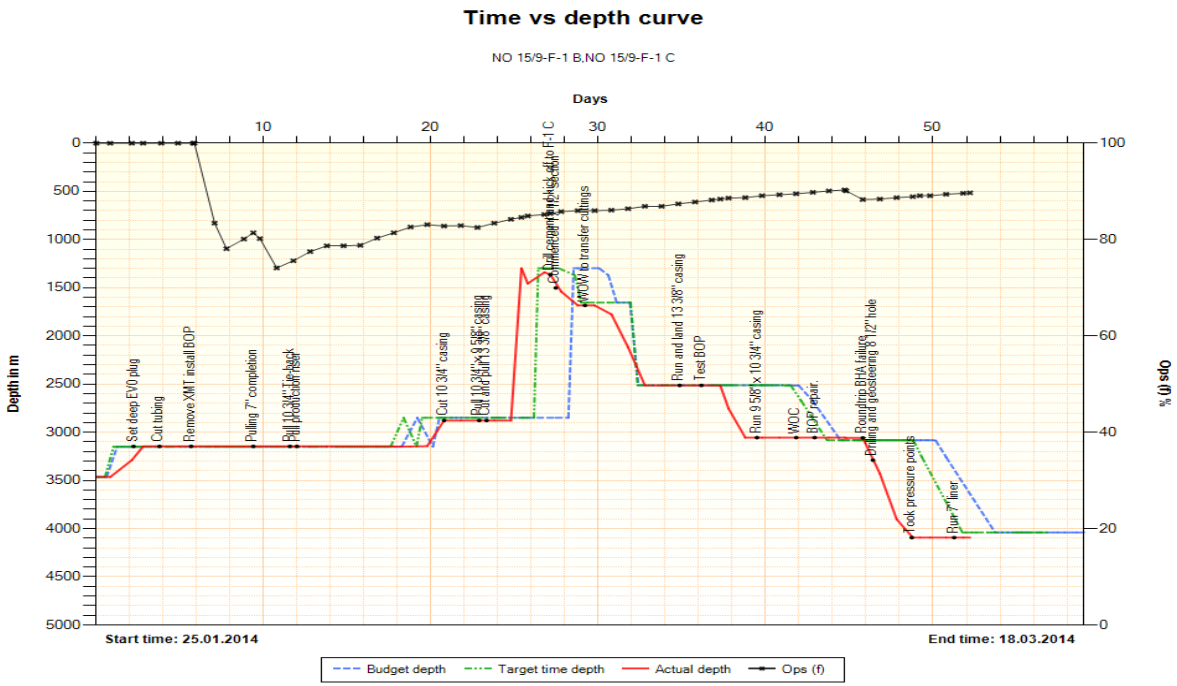
INTERVAL: Liner
START TIME: 15.Mar.2014 16:30
END TIME: 18.Mar.2014 16:15

Report date	Description

Report date	Description
16.Mar.2014	Obtained pressure points as per program. POOH with 8 1/2" RSS BHA and L/D same. R/U for running 7" liner. P/U and M/U 7" shoetrack. M/U and RIH with 7" liner to 110 m.
17.Mar.2014	M/U and RIH 7" liner from 110 m to 1149 m. Installed liner hanger. RIH with 7" liner on 5 1/2" landing string from 1149 m to 3040 m. Broke circulation and circ. at 9 5/8" shoe prior to enter open hole.
18.Mar.2014	RIH with 7" liner on 5 1/2" LS to 4036 m. Washed and worked liner to 4090 m. Decided to set liner at 4087 m. Installed cement head and positioned liner hanger on setting depth. Circulated hole clean prior to setting liner hanger. Set liner hanger.

8 Appendix 3: Figures and tables

8.1 Time/depth curve



8.2 Project planner

PROJECT NAME: F-1 B P&A
PROJECT NUMBER: T.O046V.DG.10072

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
25.Jan.2014 11:00	31.Jan.2014 07:45	126,0	5,2	104,9	4,4	64,5	141,1	5,9	0,0	Preparations [NO 15/9-F-1 B]	
25.Jan.2014 11:00	25.Jan.2014 13:00	0,0	0,0	0,0	0,0	0,0	2,0	0,1	0,0	1 ***Take over well F-1 B from production***	Maersk
25.Jan.2014 13:00	26.Jan.2014 10:00	3,7	0,2	3,1	0,1	6,0	21,0	1,0	0,0	4 RU wireline BOP w/pressure control	Schlum,Maersk,AWS
25.Jan.2014 13:00	25.Jan.2014 13:00	3,7	0,3	3,1	0,3	0,0	0,0	0,1	0,0	3 Line up AWV to choke manifold	Maersk
25.Jan.2014 13:00	25.Jan.2014 13:00	11,2	0,8	9,3	0,6	0,0	0,0	0,1	0,0	2 Skid rig over to F-1	Maersk
26.Jan.2014 10:00	26.Jan.2014 15:00	15,0	1,4	12,4	1,2	5,0	5,0	1,2	0,0	5 RIH and install deep set EV0 plug at 3150 m MD in 7in liner	Hall,Schlum,Maersk,AWS
26.Jan.2014 15:00	26.Jan.2014 21:30	5,6	1,6	4,7	1,4	3,0	6,5	1,4	0,0	6 POOH and L/D toolstring	Hall,Schlum,Maersk,AWS
26.Jan.2014 21:30	27.Jan.2014 16:30	0,9	1,7	0,8	1,4	0,5	19,0	2,2	0,0	7 Pressure test EV0 plug	Hall,Maersk
27.Jan.2014 16:30	28.Jan.2014 07:00	9,3	2,1	7,8	1,7	5,0	14,5	2,8	0,0	9 RIH and punch hole in tubing above packer	Hall,Schlum,Maersk,AWS
27.Jan.2014 16:30	27.Jan.2014 16:30	0,0	2,1	0,0	1,7	0,0	0,0	2,2	0,0	8 ***Perform drift run in 7" tubing and liner***	Maersk,AWS
28.Jan.2014 07:00	28.Jan.2014 13:00	3,7	2,2	3,1	1,8	4,0	6,0	3,1	0,0	10 POOH and L/D tool string	Schlum,Maersk,AWS
28.Jan.2014 13:00	29.Jan.2014 01:15	11,2	2,7	9,3	2,2	6,0	12,3	3,6	0,0	11 Reverse circulate annulus and tubing to 1.26-1.37 sg OBM kill mud and pump 10 m3 1.50 sg brine	Hall,Maersk
29.Jan.2014 01:15	29.Jan.2014 05:45	11,2	3,1	9,3	2,6	6,0	4,5	3,8	0,0	12 RIH and cut tubing above production packer	Hall,Schlum,Maersk,AOT,AWS
29.Jan.2014 05:45	29.Jan.2014 10:30	5,6	3,4	4,7	2,8	3,0	4,8	4,0	0,0	13 POOH and L/D tool string	Schlum,Maersk,AWS
29.Jan.2014 10:30	29.Jan.2014 16:45	1,9	3,5	1,6	2,9	6,0	6,3	4,2	0,0	14 R/D wireline.	Schlum,Maersk,AWS,AOT
29.Jan.2014 16:45	29.Jan.2014 19:00	3,7	3,6	3,1	3,0	2,0	2,3	4,3	0,0	15 Install BPV in TH and close DHSV	Hall,Schlum,Maersk,AOT,AWS
29.Jan.2014 19:00	30.Jan.2014 13:15	18,7	4,4	15,5	3,7	6,0	18,3	5,1	0,0	16 Remove X-mas tree and spacer spool	Vetco,Maersk
30.Jan.2014 13:15	31.Jan.2014 03:00	18,7	5,2	15,5	4,3	10,0	13,8	5,7	0,0	17 Install short HP riser and BOP	Vetco,Maersk,Ocean
31.Jan.2014 03:00	31.Jan.2014 07:45	1,9	5,3	1,6	4,4	2,0	4,8	5,9	0,0	18 Install dart in BPV. P/T BOP	Schlum,Maersk,AWS,AOT
31.Jan.2014 07:45	03.Feb.2014 20:30	59,8	2,5	49,7	2,1	32,0	84,8	3,5	37,3	Pull Completion [NO 15/9-F-1 B]	

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
31.Jan.2014 07:45	02.Feb.2014 00:45	3,7	5,4	3,1	4,5	2,0	41,0	7,6	34,8	19 RIH with THRT, release TH	Schlum,Vetco,Maersk
02.Feb.2014 00:45	03.Feb.2014 20:30	56,1	7,7	46,6	6,4	30,0	43,8	9,4	2,5	20 Retrieve TH and tubing and subassemblies down to tubing cut	Vetco,Maersk,Weath,BHA,BOT
03.Feb.2014 20:30	19.Feb.2014 21:00	513,7	21,4	491,9	20,5	277,0	385,2	16,0	43,8	Permanent P&A (DP) w/ RIG [NO 15/9-F-1 B]	
03.Feb.2014 20:30	04.Feb.2014 09:30	9,3	8,1	7,8	6,8	5,0	13,0	9,9	6,8	22 M/U 2 7/8" cement stinger with muleshoe	Hall,Maersk
03.Feb.2014 20:30	03.Feb.2014 20:30	0,0	8,1	0,0	6,8	0,0	0,0	9,9	0,0	21 ***If not 24hr weather window - M/U cement stinger***	Hall,Maersk
04.Feb.2014 09:30	05.Feb.2014 08:00	5,6	8,4	4,7	7,0	3,0	22,5	10,9	16,5	23 Install shallow Safe Lock (VMB) plug in 9 5/8" casing at ~605 m MD tag same	Hall,Maersk,AOT
05.Feb.2014 08:00	05.Feb.2014 11:00	7,5	8,7	6,2	7,2	4,0	3,0	11,0	0,0	24 POOH to 170m. Displace to SW. POOH and P/T plug from above	Hall,Maersk,AOT
05.Feb.2014 11:00	05.Feb.2014 12:45	1,9	8,8	1,6	7,3	1,0	1,8	11,1	0,0	25 Rig up Vetco equipment to retrieve MS-T seal	Vetco,Maersk
05.Feb.2014 12:45	05.Feb.2014 14:45	3,7	8,9	3,1	7,4	2,0	2,0	11,2	0,0	26 RIH with MS-T retrieval tool and recover seal assembly	Vetco,Maersk
05.Feb.2014 14:45	05.Feb.2014 18:45	7,5	9,2	6,2	7,7	4,0	4,0	11,3	2,5	27 RIH with Adjustable sub setting tool and remove tension in string	DQ,Vetco,Maersk
05.Feb.2014 18:45	05.Feb.2014 18:45	0,0	9,2	0,0	7,7	0,0	0,0	11,3	0,0	28 RIH with torque tool to disconnect tie back adaptor	DQ,Maersk
05.Feb.2014 18:45	06.Feb.2014 01:00	1,9	9,3	1,6	7,7	1,0	6,3	11,6	0,0	29 R/U 10-3/4" Casing handling equipemnt	Maersk,Weath
06.Feb.2014 01:00	06.Feb.2014 07:00	7,5	9,6	6,2	8,0	4,0	6,0	11,8	0,0	31 Pull Tie back string and lay out	Vetco,Maersk,Weath
06.Feb.2014 01:00	06.Feb.2014 01:00	0,0	9,6	0,0	8,0	0,0	0,0	11,6	0,0	30 P/U and RIH with Casing hanger RT	Maersk,Weath
06.Feb.2014 07:00	06.Feb.2014 11:00	7,5	9,9	6,2	8,3	4,0	4,0	12,0	0,0	32 Nipple down BOP	Maersk
06.Feb.2014 11:00	06.Feb.2014 11:45	5,6	10,2	4,7	8,5	3,0	0,8	12,0	0,0	33 Prepare for retrieving production riser	Vetco,Maersk
06.Feb.2014 11:45	06.Feb.2014 13:00	5,6	10,4	4,7	8,6	3,0	1,3	12,1	0,0	34 Retrieve HP surface riser joint	Vetco,Maersk
06.Feb.2014 13:00	06.Feb.2014 13:30	2,8	10,5	2,3	8,7	1,5	0,5	12,1	0,0	35 Rig down Multibowl valves	Vetco,Maersk
06.Feb.2014 13:30	06.Feb.2014 14:30	1,9	10,6	1,6	8,8	1,0	1,0	12,1	0,0	36 RIH with Multibowl RT c/w handling pup	Vetco,Maersk
06.Feb.2014 14:30	07.Feb.2014 04:30	9,3	11,0	7,8	9,1	5,0	14,0	12,7	0,0	37 RD cylinders, tensioner ring and centralizer	Vetco,Maersk
07.Feb.2014 04:30	07.Feb.2014 09:15	9,3	11,4	11,6	9,6	2,0	4,8	12,9	0,0	38 Unlock PR Tie Back Connector (3.75 hrs.WOW)	Vetco,Ocean,Maersk
07.Feb.2014 09:15	08.Feb.2014 00:00	28,0	12,5	23,8	10,6	15,0	14,8	13,5	0,0	39 Retrieve production riser HEAVY LIFT OPERATION - NO PRODUCTION (0.5 hrs WOW)	Vetco,Maersk,Ocean
08.Feb.2014 00:00	08.Feb.2014 13:30	0,0	12,5	13,5	11,2	15,0	13,5	14,1	0,0	41 While WOW: Install TDS brake disc	Maersk,Ocean
08.Feb.2014 00:00	08.Feb.2014 00:00	11,2	13,0	9,3	11,6	0,0	0,0	14,1	0,0	40 Simultaneously with TDS repair: RIH with riser guide funnel	Vetco,Maersk,Ocean

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
08.Feb.2014 13:30	10.Feb.2014 05:00	0,0	13,0	39,5	13,2	0,0	39,5	15,8	0,0	42 WOW and RU for running HPDR. Install spider, prepare fairings, guides, etc. Ensure equipment incl Tensioner Ring ready at Centr deck.	Vetco,Maersk,Ocean
10.Feb.2014 05:00	10.Feb.2014 19:30	37,4	14,6	31,1	14,5	20,0	14,5	16,4	0,0	43 Run in with HPDR to Tensioner joint. HEAVY LIFT OPERATION - NO PRODUCTION	Vetco,Maersk,Ocean
10.Feb.2014 19:30	11.Feb.2014 02:00	13,1	15,1	10,9	15,0	7,0	6,5	16,6	0,0	44 MU BOP jt w/Claxton NT-2 running tool. RIH, land and lock the Tie-back connector (TBC) using ROV. HEAVY LIFT OPERATION - NO PRODUCTION	Vetco,Maersk,Ocean
11.Feb.2014 02:00	11.Feb.2014 03:15	3,7	15,3	3,1	15,1	2,0	1,3	16,7	0,0	45 Confirm TBC is locked by applying 10 ton overpull on riser string.	Vetco,Maersk,Ocean
11.Feb.2014 03:15	11.Feb.2014 08:00	1,9	15,3	1,6	15,2	1,0	4,8	16,9	0,0	46 Fill the riser, and pressure test to 100 bar.	Hall,Vetco,Maersk
11.Feb.2014 08:00	11.Feb.2014 14:00	5,6	15,6	4,7	15,3	3,0	6,0	17,1	0,0	47 Install centralizer and Boat Collision Stool on centralizer deck.	Vetco,Maersk
11.Feb.2014 14:00	11.Feb.2014 16:00	11,2	16,0	9,3	15,7	6,0	2,0	17,2	0,0	48 Install tensioner ring and cylinders, and tensioning up system.	Vetco,Maersk
11.Feb.2014 16:00	12.Feb.2014 00:00	13,1	16,6	10,9	16,2	7,0	8,0	17,5	0,0	49 Install BOP and Diverter. Pressure test same.	Hall,Maersk
12.Feb.2014 00:00	12.Feb.2014 08:30	9,3	17,0	7,8	16,5	5,0	8,5	17,9	1,0	50 Retrieve the shallow 9 5/8" Safe Lock (VMB) plug	Vetco,Maersk
12.Feb.2014 08:30	13.Feb.2014 01:15	9,3	17,4	7,8	16,8	4,0	16,8	18,6	0,0	51 RIH w. 2 7/8" cement stinger to +/- 3147 m.	Maersk
13.Feb.2014 01:15	13.Feb.2014 01:45	1,9	17,4	1,6	16,9	1,0	0,5	18,6	0,0	52 Tag deep set 7" EV0 plug	Maersk
13.Feb.2014 01:45	13.Feb.2014 05:15	4,7	17,6	3,9	17,1	3,5	3,5	18,8	0,0	53 Circulate and condition mud	Maersk
13.Feb.2014 05:15	13.Feb.2014 08:30	3,7	17,8	3,1	17,2	2,0	3,3	18,9	0,0	54 Set cement plug from 3147 m MD to 2850 m MD	Hall,Maersk
13.Feb.2014 08:30	13.Feb.2014 11:00	5,6	18,0	4,7	17,4	3,0	2,5	19,0	0,0	55 POOH above TOC and circulate string clean	Hall,Maersk
13.Feb.2014 11:00	13.Feb.2014 22:30	9,3	18,4	7,8	17,7	5,0	11,5	19,5	1,3	56 POOH with cement stinger. L/D same.	Hall,Maersk
13.Feb.2014 22:30	13.Feb.2014 22:30	22,4	19,3	18,6	18,5	0,0	0,0	19,5	0,0	57 WOC	Hall,Maersk
13.Feb.2014 22:30	14.Feb.2014 13:00	22,4	20,3	18,6	19,3	12,0	14,5	20,1	4,3	58 MU & RIH with 8 1/2" clean-out assembly and drill until firm cement and tag to 10 mT	Hall,Maersk,BHA,BOT
14.Feb.2014 13:00	14.Feb.2014 15:00	9,3	20,7	7,8	19,6	2,0	2,0	20,2	0,0	59 POOH to planned cutting depth	Hall,Maersk,BHA
14.Feb.2014 15:00	14.Feb.2014 21:00		20,7	5,0	19,8	5,0	6,0	20,4	0,0	60 Circulate and cut down on MW to 1,28 SG. Continue POOH and L/D BHA.	Hall,Maersk,BHA

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
14.Feb.2014 21:00	15.Feb.2014 06:00	18,7	21,4	17,3	20,5	10,0	9,0	20,8	1,8	61 RIH and cut 10 3/4" casing at 1433 m MD (1.75 hrs. WOW)	Hall,Maersk,Weath,BOT
15.Feb.2014 06:00	15.Feb.2014 09:30	5,6	21,7	4,7	20,7	3,0	3,5	20,9	0,0	62 POOH and L/D BHA	Maersk,BHA,BOT
15.Feb.2014 09:30	15.Feb.2014 14:00	7,5	22,0	6,2	21,0	4,0	4,5	21,1	2,5	63 Retrieve 10 3/4" lock down assembly	DQ,Hall,Vetco,Maersk
15.Feb.2014 14:00	15.Feb.2014 19:30	0,0	22,0	0,0	21,0	3,0	5,5	21,4	1,3	64 Retrieve 10 3/4" seal assembly	DQ,Hall,Vetco,Maersk
15.Feb.2014 19:30	17.Feb.2014 08:30	46,7	23,9	38,8	22,6	28,0	37,0	22,9	5,8	65 RIH with spear, slip and cut, circulate and weight up to 1,40 SG. Retrieve 10 3/4" casing	Hall,Maersk,Weath,BOT
17.Feb.2014 08:30	17.Feb.2014 11:30	5,6	24,2	4,7	22,8	3,0	3,0	23,0	0,0	66 Install 13 3/8" wear bushing	Vetco,Maersk
17.Feb.2014 11:30	17.Feb.2014 19:00	28,0	25,3	23,3	23,8	15,0	7,5	23,3	0,0	67 RIH and cut 13 3/8" casing at ~1400m MD	Hall,Maersk,Weath,BOT
17.Feb.2014 19:00	17.Feb.2014 22:30	7,5	25,6	6,2	24,0	4,0	3,5	23,5	0,0	68 Retrieve 13 3/8" wear bushing	Vetco,Maersk
17.Feb.2014 22:30	18.Feb.2014 03:15	5,6	25,9	4,7	24,2	3,0	4,8	23,7	0,0	69 Retrieve 13 3/8" seal assembly	Vetco,Maersk
18.Feb.2014 03:15	19.Feb.2014 02:15	41,1	27,6	34,2	25,6	22,0	23,0	24,6	0,0	70 RIH with spear and retrieve 13 3/8" casing. Set NBP.	Hall,Vetco,Maersk,Weath,BOT
19.Feb.2014 02:15	19.Feb.2014 10:45	18,7	28,4	15,5	26,3	10,0	8,5	25,0	0,0	71 M/U cement stinger and RIH to +/- 1530 m MD and set a 130 m WBM hi-vis pill	Hall,Maersk
19.Feb.2014 10:45	19.Feb.2014 11:00	0,0	28,4	0,0	26,3	0,0	0,3	25,0	0,0	72 POOH to top of 13 3/8", circulate to verify top of hi-vis pill	Hall,Maersk
19.Feb.2014 11:00	19.Feb.2014 13:15	7,5	28,7	6,2	26,5	4,0	2,3	25,1	0,0	73 Set surface barrier / KOP from top of cut 13 3/8" casing and 50 m into 20" casing	Hall,Maersk
19.Feb.2014 13:15	19.Feb.2014 17:00	5,6	28,9	4,7	26,7	3,0	3,8	25,3	0,0	74 POOH above TOC and circulate out excess cmt.	Hall,Maersk
19.Feb.2014 17:00	19.Feb.2014 21:00	5,6	29,1	4,7	26,9	3,0	4,0	25,4	0,0	75 POOH and rack back cement stinger	Hall,Maersk
19.Feb.2014 21:00	21.Feb.2014 14:30	38,5	1,6	43,1	1,8	22,0	41,6	1,7	0,0	Prepare Sidetrack (DP) w/ RIG [NO 15/9-F-1 B]	
19.Feb.2014 21:00	20.Feb.2014 05:15	25,4	30,2	19,9	27,8	15,0	8,3	25,8	0,0	76 WOC.MU and RIH with 17 1/2" BHA to TOC	Hall,Maersk,BHA
20.Feb.2014 05:15	21.Feb.2014 14:30	13,1	30,8	23,2	28,7	7,0	33,3	27,1	0,0	77 Drill out 20" cement, K/O to F-1 C (12.25 hrs. WOW)	Hall,Maersk,BHA

PROJECT NAME: F-1 C Northwest Producer
PROJECT NUMBER: T.O046V.DG.10070

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
21.Feb.2014 14:30	03.Mar.2014 18:15	273,0	11,4	309,6	12,9	271,3	244,7	10,2	3,3	17 1/2" [NO 15/9-F-1 C]	
21.Feb.2014 14:30	21.Feb.2014 22:15	12,0	31,3	9,0	29,1	10,0	7,8	27,5	0,0	1 Drill 17 1/2" hole to 1482 mMD while racking 13 3/8" casing	
21.Feb.2014 22:15	22.Feb.2014 05:00	12,0	31,8	9,0	29,5	10,0	6,8	27,8	0,0	2 Drill 17 1/2" hole to 1540 mMD at controlled ROP of 12 m/hr due to WOW. Meanwhile racking 13 3/8" casing	
22.Feb.2014 05:00	22.Feb.2014 19:00	0,0	31,8	14,0	30,1	0,0	14,0	28,3	0,0	3 WOW to discharge cutting to cuttings vessel. Meanwhile racking 13 3/8" casing	
22.Feb.2014 19:00	23.Feb.2014 01:00	7,0	32,0	0,0	30,1	6,0	6,0	28,6	0,0	4 Drill 17 1/2" hole to 1684 m MD while racking 13 3/8" casing	
23.Feb.2014 01:00	23.Feb.2014 16:00	0,0	32,0	15,0	30,7	0,0	15,0	29,2	0,0	5 WOW to discharge cutting to cuttings vessel. Meanwhile racking 13 3/8" casing	
23.Feb.2014 16:00	25.Feb.2014 01:00	0,0	32,0	33,0	32,1	0,0	33,0	30,6	0,0	6 WOW to discharge cutting to cuttings vessel.	
25.Feb.2014 01:00	25.Feb.2014 10:00	12,0	32,5	10,0	32,5	90,0	9,0	31,0	0,0	7 Drill 17 1/2" hole to 1864 m MD	Hall,Schlum,Maersk,BHA
25.Feb.2014 10:00	25.Feb.2014 16:00	0,0	32,5	6,0	32,7	0,0	6,0	31,2	0,0	8 WOW to discharge cuttings to cutting vessel.	Hall,Schlum,Maersk,BHA
25.Feb.2014 16:00	26.Feb.2014 00:45	33,0	33,9	33,0	34,1	33,0	8,8	31,6	0,0	9 Drill 17 1/2" hole to 2088 m MD	Hall,Schlum,Maersk,BHA
26.Feb.2014 00:45	26.Feb.2014 05:00	0,0	33,9	4,3	34,3	0,0	4,3	31,8	0,0	10 WOW to discharge cuttings to cutting vessel.	Hall,Schlum,Maersk,BHA
26.Feb.2014 05:00	26.Feb.2014 23:45	68,0	36,8	67,0	37,1	30,0	18,8	32,5	0,0	11 Drill 17 1/2" hole to TD (30 mTVD above Ty Fm)	Hall,Schlum,Maersk,BHA
26.Feb.2014 23:45	27.Feb.2014 04:00	5,0	37,0	4,0	37,2	4,0	4,3	32,7	0,0	13 Circulate well clean	Hall,Schlum,Maersk,BHA
26.Feb.2014 23:45	26.Feb.2014 23:45	0,0	37,0	0,0	37,2	0,0	0,0	32,7	0,0	12 WOW to discharge cuttings to cuttings vessel	Hall,Schlum,Maersk,BHA
27.Feb.2014 04:00	27.Feb.2014 06:00	2,0	37,0	2,0	37,3	2,0	2,0	32,8	0,0	14 Weigh up mud from 1.40 sg to 1.41 sg	Hall,Schlum,Maersk,BHA
27.Feb.2014 06:00	27.Feb.2014 08:45	3,0	37,2	2,0	37,4	1,0	2,8	32,9	0,0	15 POOH to 2088 m MD	Hall,Schlum,Maersk,BHA
27.Feb.2014 08:45	27.Feb.2014 12:00		37,2	0,0	37,4	0,0	3,3	33,0	3,3	16 Retrieve released stand of DP in derrick	Hall,Schlum,Maersk,BHA
27.Feb.2014 12:00	27.Feb.2014 18:15	7,0	37,5	6,0	37,7	6,0	6,3	33,3	0,0	17 POOH until BHA at rotary	
27.Feb.2014 18:15	27.Feb.2014 23:45	4,0	37,6	3,0	37,8	3,0	5,5	33,5	0,0	18 Lay down BHA	
27.Feb.2014 23:45	28.Feb.2014 05:45	5,0	37,8	4,0	37,9	4,0	6,0	33,8	0,0	19 RIH, retrieve nominal seat protector, wash WH and riser BOP, POOH	Hall,Schlum,Maersk,BHA
28.Feb.2014 05:45	28.Feb.2014 11:45	5,0	38,0	3,0	38,1	2,0	6,0	34,0	0,0	20 RU casing running equipment	Hall,Schlum,Maersk,Weath,BHA

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
28.Feb.2014 11:45	28.Feb.2014 13:00		38,0	2,0	38,2	2,0	1,3	34,1	0,0	21 P/U shoe track. Baker lock joint # C4 and shoe track.	Hall,Schlum,Maersk,BHA,Weath
28.Feb.2014 13:00	01.Mar.2014 02:45	27,0	39,2	25,0	39,2	18,0	13,8	34,7	0,0	22 Run 13 3/8" casing to 2370 mMD.	
01.Mar.2014 02:45	01.Mar.2014 05:45	5,0	39,4	3,0	39,3	2,0	3,0	34,8	0,0	23 RD casing handling equipment and RU 5 1/2" handling equipment	
01.Mar.2014 05:45	01.Mar.2014 08:00	3,0	39,5	3,0	39,4	1,0	2,3	34,9	0,0	24 Run and land 13 3/8" casing	
01.Mar.2014 08:00	01.Mar.2014 09:15	1,0	39,5	1,0	39,5	1,0	1,3	34,9	0,0	25 Pick up cement head.	
01.Mar.2014 09:15	01.Mar.2014 16:00	9,0	39,9	8,0	39,8	8,0	6,8	35,2	0,0	26 Circulate, pump and displace cement.	Hall,Schlum,Maersk,Weath,BHA
01.Mar.2014 16:00	01.Mar.2014 21:00	0,5	39,9	0,5	39,8	1,0	5,0	35,4	0,0	27 Set and test seal assy.	Schlum,Maersk,Weath
01.Mar.2014 21:00	02.Mar.2014 01:30	3,0	40,1	1,0	39,9	1,5	4,5	35,6	0,0	28 Release running tool and POOH. LD RT and CMT head.	
02.Mar.2014 01:30	02.Mar.2014 03:45	2,0	40,1	2,3	40,0	2,0	2,3	35,7	0,0	29 RIH set wear bushing.	Schlum,Maersk,Weath
02.Mar.2014 03:45	02.Mar.2014 09:00	3,0	40,3	2,5	40,1	4,5	5,3	35,9	0,0	30 Pressure test BOP	
02.Mar.2014 09:00	02.Mar.2014 13:00	0,0	40,3	0,0	40,1	5,0	4,0	36,1	0,0	31 Planned maintenance. Change out IBOP/saver sub on TDS	
02.Mar.2014 13:00	02.Mar.2014 15:00	6,0	40,5	5,0	40,3	0,3	2,0	36,2	0,0	32 Pressure test IBOPs & TDS.	Hall,Schlum,Vetco,Maersk
02.Mar.2014 15:00	02.Mar.2014 17:45	6,0	40,8	5,0	40,5	0,0	2,8	36,3	0,0	33 POOH with WBRT/ BOP test tool.	Hall,Schlum,Maersk
02.Mar.2014 17:45	03.Mar.2014 08:00	18,0	41,5	15,0	41,1	12,0	14,3	36,9	0,0	34 MU and RIH with 12 1/4" BHA to TOC. Choke drill	Hall,Schlum,Maersk
03.Mar.2014 08:00	03.Mar.2014 09:00	8,0	41,9	4,0	41,3	4,0	1,0	36,9	0,0	35 Drill cement and 3 m of new formation. Dilute mud to 1,28 sg OBM while drilling out shoe track.	Hall,Maersk,BHA
03.Mar.2014 09:00	03.Mar.2014 11:00	0,0	41,9	2,0	41,4	2,0	2,0	37,0	0,0	36 WOW Not able to operate crane.	Hall,Maersk,BHA
03.Mar.2014 11:00	03.Mar.2014 15:30	4,0	42,0	4,0	41,5	4,0	4,5	37,2	0,0	37 Drill out shoe track and 3 m new formation. Displace well to 1,28 SG OBM.	Hall,Maersk,BHA
03.Mar.2014 15:30	03.Mar.2014 18:15	2,5	42,1	2,0	41,6	2,0	2,8	37,3	0,0	38 Circulate B/U, condition mud and perform FIT	Hall,Maersk,BHA
03.Mar.2014 18:15	11.Mar.2014 04:30	196,5	8,2	176,1	7,3	158,0	178,8	7,4	3,3	12 1/4" [NO 15/9-F-1 C]	
03.Mar.2014 18:15	04.Mar.2014 01:45	20,0	43,0	15,0	42,2	13,0	7,5	37,6	0,0	39 Drill 12 1/4" hole through Ty Fm. to Top Shetland Gp. ~2674 m MD	Hall,Maersk,BHA
04.Mar.2014 01:45	04.Mar.2014 22:30	40,0	44,6	37,0	43,8	36,0	20,8	38,5	0,0	40 Drill 12 1/4" hole to section TD approx. 3083 m MD (ROP=15 m/hr)	Hall,Maersk,BHA
04.Mar.2014 22:30	05.Mar.2014 14:45	18,0	45,4	16,0	44,5	15,0	16,3	39,2	1,5	42 Circulate well clean. Drop Gyro, POOH and R/B BHA.	Scient,Hall,Maersk,BHA

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
04.Mar.2014 22:30	04.Mar.2014 22:30	2,0	45,5	1,0	44,5	0,0	0,0	38,5	0,0	41 Drop gyro, POOH	Scient,Hall,Maersk,BHA
05.Mar.2014 14:45	05.Mar.2014 18:15	5,0	45,7	4,0	44,7	3,0	3,5	39,3	0,0	43 RIH, retrieve nominal seat protector, wash WH, POOH.	Hall,Maersk
05.Mar.2014 18:15	05.Mar.2014 22:45	3,0	45,8	2,0	44,7	2,0	4,5	39,5	0,0	44 R/U 9 5/8" casing handling equipment.	Hall,Maersk,Weath
05.Mar.2014 22:45	07.Mar.2014 00:45	17,5	46,5	15,0	45,4	13,0	26,0	40,6	0,0	45 Run 9 5/8" casing	
07.Mar.2014 00:45	07.Mar.2014 07:00	2,0	46,6	2,0	45,5	1,0	6,3	40,8	1,3	46 Change from 9 5/8" handling equipment to 10 3/4"	Maersk,Weath
07.Mar.2014 07:00	07.Mar.2014 13:45	6,0	46,9	4,0	45,6	4,0	6,8	41,1	0,5	47 Run 10 3/4" casing to (~2930 m).	Maersk,Weath
07.Mar.2014 13:45	07.Mar.2014 16:00	3,0	47,0	2,0	45,7	2,0	2,3	41,2	0,0	48 RD casing handling equipment and RU 5 1/2" handling equipment	Maersk,Weath
07.Mar.2014 16:00	07.Mar.2014 19:30	1,0	47,0	1,0	45,7	1,0	3,5	41,4	0,0	49 Run and land casing in MS700 wellhead	Maersk
07.Mar.2014 19:30	08.Mar.2014 01:00	11,4	47,5	10,0	46,2	6,0	5,5	41,6	0,0	51 Cement casing in place (TOC ~300 m above shoe) and pressure test	Hall,Maersk
07.Mar.2014 19:30	07.Mar.2014 19:30	1,0	47,5	1,0	46,2	0,0	0,0	41,6	0,0	50 Pick up cement head	Maersk
08.Mar.2014 01:00	08.Mar.2014 01:00	7,1	47,8	6,9	46,5	0,0	0,0	41,8	0,0	52 Rig down casing running equipment	Weath
08.Mar.2014 01:00	08.Mar.2014 07:00		47,8	0,0	46,5	6,0	6,0	41,8	0,0	53 WOC to bullhead 9 5/8" x 13 3/8" annulus to brine	Weath
08.Mar.2014 07:00	08.Mar.2014 14:15	12,0	48,3	11,0	47,0	3,0	7,3	42,1	0,0	55 Bullhead 9 5/8" x 13 3/8" annulus to brine	
08.Mar.2014 07:00	08.Mar.2014 07:00	6,0	48,6	5,0	47,2	0,0	0,0	41,8	0,0	54 Perform injection test pumping brine	
08.Mar.2014 14:15	08.Mar.2014 16:00	5,0	48,8	4,7	47,4	1,0	1,8	42,2	0,0	56 Set 10 3/4" seal assembly in MS700 wellhead	Maersk
08.Mar.2014 16:00	08.Mar.2014 16:30	3,6	48,9	7,1	47,7	3,0	0,5	42,2	0,0	57 Pressure test seal assembly, perform one BOP test.	
08.Mar.2014 16:30	08.Mar.2014 19:30		48,9		47,7	2,0	3,0	42,4	0,0	58 POOH with PADPRT, L/D same.	Maersk
08.Mar.2014 19:30	09.Mar.2014 09:30		48,9	0,0	47,7	12,0	14,0	42,9	0,0	59 Repair annular, perform general maintenance	Maersk
09.Mar.2014 09:30	09.Mar.2014 09:30	6,0	49,2	5,0	47,9	0,0	0,0	43,4	0,0	60 Install wear bushing.	Hall,Maersk
09.Mar.2014 09:30	09.Mar.2014 19:45	5,0	49,4	4,0	48,0	6,0	10,3	43,4	0,0	61 Pressure test BOP	Maersk
09.Mar.2014 19:45	09.Mar.2014 21:45	1,0	49,4		48,0	1,5	2,0	43,4	0,0	62 Pull wear bushing running tool.	Maersk
09.Mar.2014 21:45	09.Mar.2014 21:45	2,9	49,6	2,4	48,1	0,0	0,0	43,7	0,0	63 L/D cement head and rig down casing running equipment	Hall,Maersk,BHA
09.Mar.2014 21:45	10.Mar.2014 04:30	7,0	49,9	6,0	48,4	10,0	6,8	43,7	0,0	65 M/U 8 1/2" RSS BHA.	Hall,Maersk,BHA
09.Mar.2014 21:45	09.Mar.2014 21:45	0,0	49,9	0,0	48,4	0,0	0,0	43,7	0,0	64 Rig up wireline. Run CBL/USIT log with wireline, R/D wireline	Hall,Schlum,Maersk,AWS

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
10.Mar.2014 04:30	10.Mar.2014 07:30		49,9	1,0	48,4	1,5	3,0	43,9	0,0	66 RIH with 8 1/2" BHA to 1000 m. Performing shallow test	Hall,Maersk,BHA
10.Mar.2014 07:30	10.Mar.2014 11:30		49,9	2,0	48,5	4,0	4,0	44,0	0,0	67 RIH with 8 1/2" BHA from 1000 m to TOC. Perform choke drill.	
10.Mar.2014 11:30	10.Mar.2014 15:45		49,9	2,0	48,6	3,0	4,3	44,2	0,0	68 Perform slip/cut.	
10.Mar.2014 15:45	11.Mar.2014 00:00	7,0	50,1	6,0	48,8	4,0	8,3	44,5	0,0	69 Drill out 9 5/8" shoe track and rat hole	Hall,Maersk,BHA
11.Mar.2014 00:00	11.Mar.2014 04:30	4,0	50,3	3,0	49,0	5,0	4,5	44,7	0,0	70 Drill 3 m new formation and perform FIT to 1,60 sg EMW.	Hall,Maersk,BHA
11.Mar.2014 04:30	15.Mar.2014 16:30	123,0	5,1	106,0	4,4	103,0	108,1	4,5	23,3	8 1/2" [NO 15/9-F-1 C]	
11.Mar.2014 04:30	12.Mar.2014 06:30		50,3	0,0	49,0	16,0	26,0	45,8	21,8	71 Drop Gyro,POOH, chang RSS, RIH	Hall,Maersk,BHA
12.Mar.2014 06:30	12.Mar.2014 21:00	15,0	50,9	12,0	49,5	13,0	14,5	46,4	1,5	72 Drill & geosteer 8 1/2" hole to top Hugin at ~3259 mMD (ROP=15 m/hr)	Hall,Maersk,BHA
12.Mar.2014 21:00	14.Mar.2014 16:00	70,0	53,9	58,0	51,9	48,0	43,0	48,2	0,0	73 Drill & geosteer 8 1/2" hole reservoir to TD at ~4039 mMD with restricted ROP to obtain LWD data	Hall,Maersk,BHA
14.Mar.2014 16:00	15.Mar.2014 04:45	10,0	54,3	5,0	52,1	8,0	12,8	48,7	0,0	74 Circulate well clean at TD. Take torque readings. Take pressure points.	Hall,Maersk,BHA
15.Mar.2014 04:45	15.Mar.2014 05:45	10,0	54,7	9,0	52,5	6,0	1,0	48,8	0,0	75 Start POOH to 9 5/8" shoe. Take torque readings.	Hall,Maersk,BHA
15.Mar.2014 05:45	15.Mar.2014 11:15	11,0	55,1	6,0	52,7	6,0	5,5	49,0	0,0	76 POOH to surface	Hall,Maersk,BHA
15.Mar.2014 11:15	15.Mar.2014 16:30	7,0	55,4	16,0	53,4	6,0	5,3	49,2	0,0	77 L/D 8 1/2" BHA	Maersk,BHA
15.Mar.2014 16:30	18.Mar.2014 16:15	90,0	3,8	86,0	3,6	48,0	71,8	3,0	0,8	Liner [NO 15/9-F-1 C]	
15.Mar.2014 16:30	15.Mar.2014 20:00	6,0	55,7	5,6	53,6	3,0	3,5	49,4	0,0	78 RU 7" casing handling equipment.	Maersk,Weath
15.Mar.2014 20:00	16.Mar.2014 12:45	40,0	57,4	39,0	55,2	12,0	16,8	50,1	0,8	79 MU and RIH w/7" liner. (appr. 1100 m)	Maersk,Weath
16.Mar.2014 12:45	16.Mar.2014 16:15	6,0	57,6	5,3	55,5	4,0	3,5	50,2	0,0	80 MU liner hanger/RT. RD casing handling equipment, RU DP handling equipment.	Maersk,Weath
16.Mar.2014 16:15	17.Mar.2014 17:30	10,0	58,0	9,0	55,8	9,0	25,2	51,3	0,0	81 RIH with 7" liner on DP	Hall,Maersk,Weath
17.Mar.2014 17:30	18.Mar.2014 05:30	12,0	58,5	11,8	56,3	10,0	12,0	51,8	0,0	82 RIH Inst cmt head. Set liner hanger. Circulate and cement 7" liner.	Hall,Maersk,BHA,Weath
18.Mar.2014 05:30	18.Mar.2014 08:15	5,0	58,7	4,6	56,5	4,0	2,8	51,9	0,0	83 Set packer. Circulate out excess cement.	Hall,Maersk,BHA,Weath
18.Mar.2014 08:15	18.Mar.2014 16:15	11,0	59,2	10,7	57,0	6,0	8,0	52,2	0,0	84 P/T well, continue POOH with liner running tool, L/D same	Hall,Maersk,Weath

8.3 Borehole stability figure with MW and final pressure/fracture gradients

Ref. FWR F-1 C G&G

8.4 Bit record

WELL: NO 15/9-F-1 C

Run no	Bit size	Bit no	BHA no	Bit type	IADC code	Bit manufacturer
1	17 1/2"	1	1	HP633F		UNKNOWN
5	12 1/4"	2	5	QT406	M333	Hughes Christensen
8	8 1/2"	3	8	TD606X	M223	Hughes Christensen
9	8 1/2"	3RR1	9	TD606X	M223	Hughes Christensen

					Nozzles (n/32")				
Run no	Bit size	Bit no	BHA no	Serial no	no x n	no x n	no x n	no x n	Flow area in2
1	17 1/2"	1	1	7034208	3 x 12	4 x 14	2 x 16	x	1,3260
5	12 1/4"	2	5	7146692	5 x 16	1 x 18	x	x	1,2310
8	8 1/2"	3	8	7148041	4 x 14	2 x 15	x	x	0,9470
9	8 1/2"	3RR1	9	7148041	4 x 14	2 x 15	x	x	0,9470

Run no	Bit size	Pump rate l/min	Pump press bar	Depth in mMD	Depth out mMD	Form drld m	Total drld m	Drld hrs	Circ hrs	ROP m/hr
1	17 1/2"	4237,5	277,5	1364	2514	1150	1150	40,0	87,8	28,8
5	12 1/4"	3750,0	245,5	2514	3056	542	582	23,1	41,0	23,5
8	8 1/2"	2373,0	209,0	3056	3059	3	3	0,1	14,0	30,0
9	8 1/2"	2163,7	197,0	3059	4094	1035	1035	44,9	65,8	23,1

Run no	Bit size	Min WOB tonne	Max WOB tonne	Min RPM	Max RPM	Torque Min Nm	Torque Max Nm	Con drag Min 1000 1000 daN	Con drag Max 1000 1000 daN
1	17 1/2"	1	21,50	123,50	135,50	11000	22500	143	158
5	12 1/4"	2,50	15	164,50	208,50	13000	22500	168,50	178
8	8 1/2"	1	5	66	90	12010	19150		
9	8 1/2"	1	12,3	97,7	157,3	11416,7	23140	159	164

Run no	Bit size	I	O	DC	L	B	G	OC	RP
1	17 1/2"	0	1	WT	S	E	I	NO	TD
5	12 1/4"	1	1	WT	A	X	I	NO	TD
8	8 1/2"	0	0	NO	A	X	I	NO	DTF
9	8 1/2"	1	1	WT	A	X	I	NO	TD

Run no	Bit size	Remarks
1	17 1/2"	Drill 17 1/2" hole to TD
5	12 1/4"	12 1/4" OTKII + CoPilot BHA
8	8 1/2"	8 1/2" G3 + LihoTrak and TesTrak
9	8 1/2"	8 1/2" G3 + LihoTrak and TesTrak

8.5 Bha list

WELLBORE: NO 15/9-F-1 B
BHA NO: 2
RUN TYPE: Cement run
DESCRIPTION: Cement stinger for deepsat cement plug
RUN NAME: 1

String component	OD in	ID in	Length m	Acc length m
MULE SHOE STINGER	3,250	2,250	3,16	3,16
CEMENT STINGER,			348,25	351,41
XO			0,39	351,80
X/O			1,23	353,03
XO			0,85	353,88

BHA NO: 3
RUN TYPE: Clean-out run
DESCRIPTION: 8 1/2" Clean-out assy
RUN NAME: 1

String component	OD in	ID in	Length m	Acc length m
BIT			0,25	0,25
Bit sub			0,56	0,81
MILL, STRING			1,96	2,77
FLOAT SUB			0,92	3,69
XO			0,27	3,96
PUP			3,21	7,17
HWDP			203,23	210,40

BHA NO: 4
RUN TYPE: Cut casing run
DESCRIPTION: 9 5/8" casing cutter assy
RUN NAME: 1

String component	OD in	ID in	Length m	Acc length m
BIT	8,500		1,52	1,52
XO			0,32	1,84
CUTTER			1,90	3,74
X/O			0,84	4,58
STABILIZER	8,500		2,30	6,88
FLOAT SUB			0,87	7,75
PUP			3,21	10,96
HWDP			203,23	214,19

BHA NO: 6
RUN TYPE: Pull Casing run

DESCRIPTION: Spear assy for pulling 10 3/4" x 9 578" casing.
RUN NAME: 1

String component	OD in	ID in	Length m	Acc length m
TAPER MILL			1,51	1,51
SPEAR PACK-OFF			0,94	2,45
SPEAR			1,35	3,80
BUMPER SUB			2,76	6,56
XO			1,23	7,79
HANDLING JOINT			3,21	11,00

BHA NO: 9
RUN TYPE: Cut casing run
DESCRIPTION: 13 3/8" casing cut at 1399 m
RUN NAME: 2

String component	OD in	ID in	Length m	Acc length m
BIT	8,500		1,42	1,42
CUTTER			3,74	5,16
STABILIZER	8,500		2,37	7,53
FLOAT SUB			0,91	8,44
XO			1,24	9,68
PUP			3,20	12,88
HWDP			189,68	202,56

BHA NO: 11
RUN TYPE: Wellhead run
DESCRIPTION: Run for retrieving 13 3/8" Seal assy
RUN NAME: 4

String component	OD in	ID in	Length m	Acc length m
JET SUB			3,84	3,84
HWDP			40,64	44,48
PUP			3,00	47,48
PULLING TOOL			2,67	50,15
PUP			1,96	52,11
HWDP			121,97	174,08

BHA NO: 12
RUN TYPE: Pull Casing run
DESCRIPTION: 13 3/8" Spear assy
RUN NAME: 3

String component	OD in	ID in	Length m	Acc length m
MILL			0,37	0,37
XO			1,00	1,37
SPEAR			1,05	2,42
XO			0,44	2,86
SPEAR			1,34	4,20
BUMPER SUB			2,65	6,85
XO			1,18	8,03
PUP			3,23	11,26

BHA NO: 14
RUN TYPE: Wellhead run
DESCRIPTION: Run for setting 20" Nominal Bore Protector
RUN NAME: 5

String component	OD in	ID in	Length m	Acc length m
JET SUB			3,84	3,84
HWDP			40,64	44,48
PUP			2,47	46,95
PULLING TOOL			1,73	48,68
PUP			2,51	51,19
HWDP			121,97	173,16

BHA NO: 15
RUN TYPE: Cement run
DESCRIPTION: Cement stinger for Kick-off plug in 20" casing
RUN NAME: 2

String component	OD in	ID in	Length m	Acc length m
MULE SHOE			4,28	4,28
XO			1,14	5,42

WELLBORE: NO 15/9-F-1 C
BHA NO: 1
RUN TYPE: Drilling run
DESCRIPTION: Drill 17 1/2" hole to TD
RUN NAME: 1

String component	OD in	ID in	Length m	Acc length m
BIT	17,500	3,000	0,50	0,50
AUTOTRAK		3,000	2,52	3,02
FLEX SUB W/ STAB	17,375	3,250	3,67	6,69
ONTRAK	11,750	3,250	6,95	13,64
MOD STAB	16,875	3,000	1,77	15,41
BCPM		3,250	3,66	19,07
STOP SUB		3,125	1,08	20,15
STAB STRING	16,000	3,000	2,28	22,43
FLOAT SUB		2,875	0,98	23,41
DRILL COLLAR, FLEX NON MAG		2,812	9,34	32,75
STAB STRING	12,000	2,875	2,23	34,98
DRILL COLLAR, FLEX NON MAG		2,875	27,47	62,45
JAR		3,000	9,69	72,14
DRILL COLLAR, FLEX NON MAG		2,875	26,77	98,91
ACCELERATOR,		3,000	9,70	108,61
XO		3,000	0,79	109,40
HW DRILL PIPE		3,875	121,94	231,34

BHA NO: 5
RUN TYPE: Drilling run
DESCRIPTION: 12 1/4" OTKII + CoPilot BHA
RUN NAME: 2

String component	OD in	ID in	Length m	Acc length m
BIT	12,250		0,34	0,34
AUTOTRAK		2,480	2,51	2,85
CoPilot		2,875	2,15	5,00
XTRA-A	12,125	6,780	9,23	14,23
FLEX SUB W/ STAB	12,125	3,125	3,71	17,94
ONTRAK	11,750	3,000	6,95	24,89
BCPM		3,000	3,68	28,57
STOP SUB		3,000	1,09	29,66
STAB STRING	11,750	2,875	2,20	31,86
FLOAT SUB		2,875	0,80	32,66
DRILL COLLAR, FLEX NON MAG		2,875	18,79	51,45
JAR + PONY DC		3,000	9,69	61,14
DRILL COLLAR, FLEX NON MAG		2,875	8,68	69,82
XO		3,000	0,79	70,61
HW DRILL PIPE		3,000	121,96	192,57

BHA NO: 8
RUN TYPE: Drilling run
DESCRIPTION: 8 1/2" G3 + LihoTrak and TesTrak
RUN NAME: 3

String component	OD in	ID in	Length m	Acc length m
BIT	8,500		0,32	0,32
AUTOTRAK		2,000	2,17	2,49
ZoneTrak G		3,000	1,24	3,73
FLEX SUB W/ STAB	8,375	3,000	2,51	6,24
CoPilot		3,000	2,21	8,45
ONTRAK		3,000	6,70	15,15
BCPM		2,250	4,90	20,05
FLEX SUB W/ STAB	8,375	2,250	2,48	22,53
ORD	8,375	2,250	2,84	25,37
CCN		2,250	2,76	28,13
TESTRAK		2,250	7,50	35,63
STOP SUB		2,250	0,77	36,40
STAB STRING	8,250	2,875	1,67	38,07
FLOAT SUB		3,000	0,91	38,98
HW DRILL PIPE		3,125	39,91	78,89
JAR + PONY DC		2,750	9,90	88,79
HW DRILL PIPE		3,125	40,17	128,96
XO		2,875	0,97	129,93
Drill Pipe		4,778	9,00	138,93

BHA NO: 9
RUN TYPE: Drilling run
DESCRIPTION: 8 1/2" G3 + LihoTrak and TesTrak
RUN NAME: 4

String component	OD in	ID in	Length m	Acc length m
BIT	8,500		0,32	0,32
AUTOTRAK		2,000	2,18	2,50
ZoneTrak G		3,000	1,24	3,74
FLEX SUB W/ STAB	8,375	3,000	2,51	6,25
CoPilot		3,000	2,21	8,46
ONTRAK		3,000	6,70	15,16
BCPM		2,250	4,90	20,06
FLEX SUB W/ STAB	8,375	2,250	2,48	22,54
ORD	8,375	2,250	2,84	25,38
CCN		2,250	2,76	28,14
TESTRAK		2,250	7,50	35,64
STOP SUB		2,250	0,77	36,41
STAB STRING	8,250	2,875	1,67	38,08
FLOAT SUB		3,000	0,91	38,99
HW DRILL PIPE		3,125	39,91	78,90
JAR + PONY DC		2,750	9,90	88,80
HW DRILL PIPE		3,125	40,17	128,97
XO		2,875	0,97	129,94
Drill Pipe		4,778	9,00	138,94

8.6 Drilling fluids

Drilling Fluids			
Hole section	Base fluid	Type	Density (SG)
17 1/2"	OBM	Enviromul	1,40-1,41
12 1/4"	OBM	Enviromul	1,28
8 1/2"	OBM	Enviromul	1,28-1,34

8.7 Cementing operations

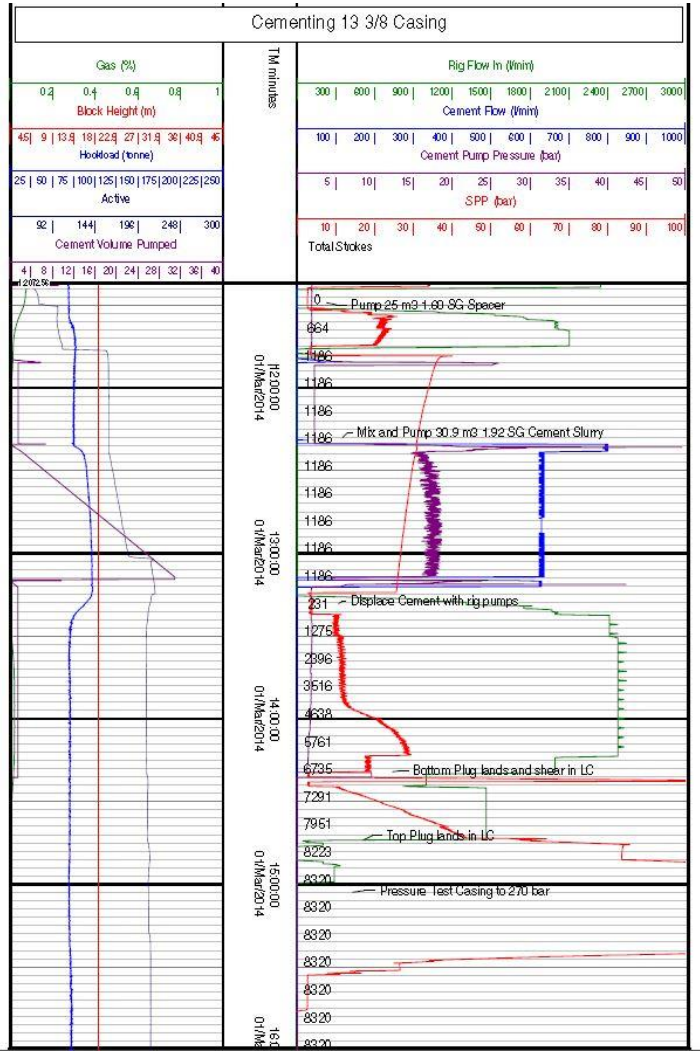
8.7.1 Casing/Liner cementing

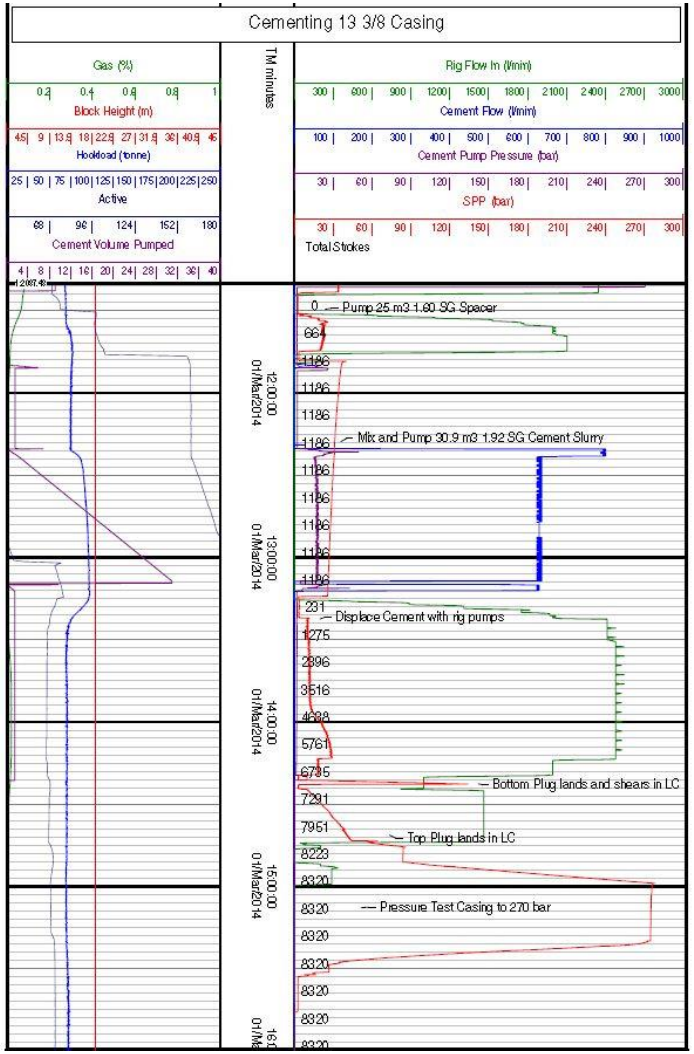
WELLBORE ID: NO 15/9-F-1 C
CASING SIZE: 13 3/8"
STAGE CEMENTING: No
REPORT DATE: 02.Mar.2014
THEORETHICAL TOC: 2207 mMD
EVALUATED TOC: mMD
EVALUATED BY BOND LOG:
LINER ROTATION PLANNED: N
LINER ROTATION ACHIEVED: Y
REMARKS:

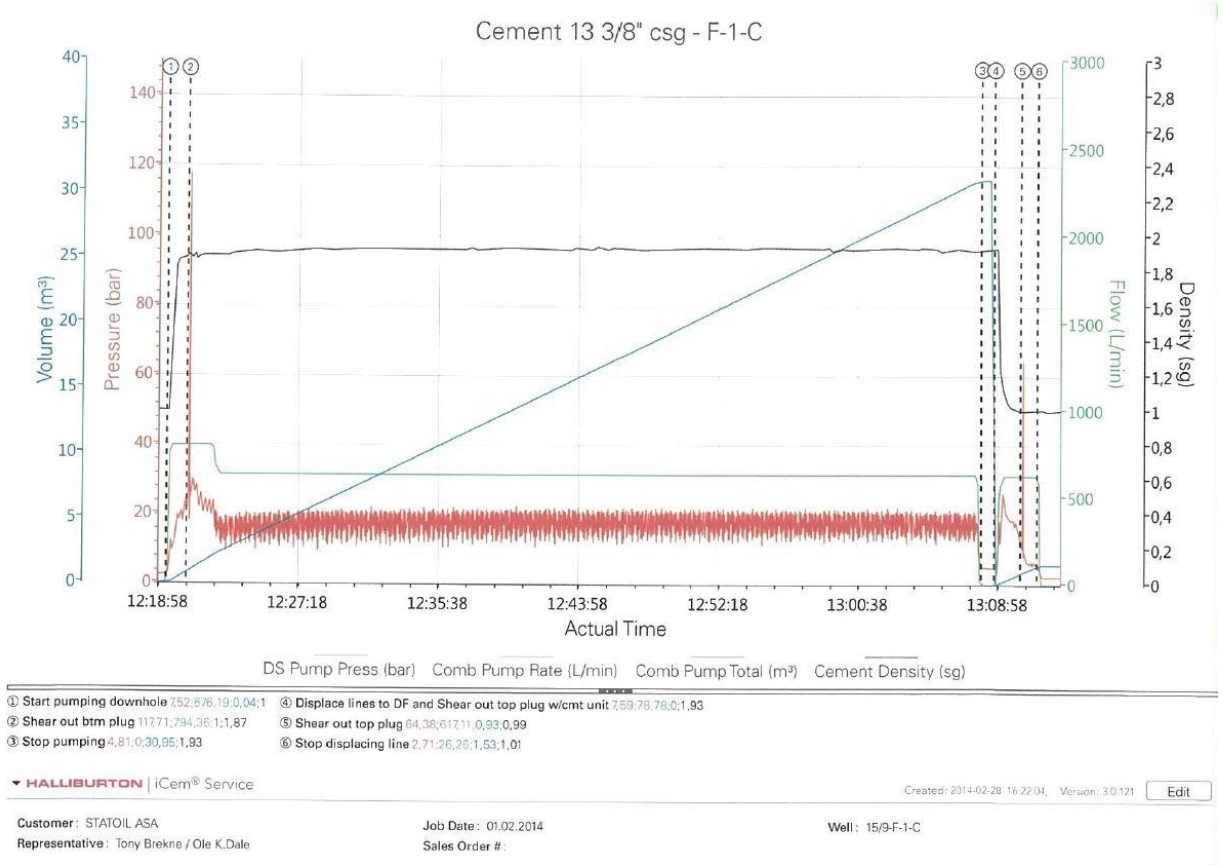
Objective:
Main objective was to cement 13 3/8" casing 300 m above shoe.

Execution:
Cement job was mixed "on the fly" using CMS (chemical injection) pumps. Spacer was prepared in Pit 12 + 14. During weighing up cement at startup we had problems with manual weight scale (mudcup) readings. Found out after a few weight measurements that we needed to move weight scale to workbench. Most likely as ventilation is blowing from roof.
Btm / Top plug teoretical shear vol: 860 Ltr
Btm plug actual shear vol: 1000 ltr
Top plug actual shear vol: 930 ltr

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Spacer before	Spacer WSOBM	1,60	25,0	2100	23	F	BARITE	755,80	kg/m3	4,23	
							Freshwater	779,10	l/m3	1,00	
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	
Lead	Cement Slurry Non Standard	1,92	30,9	636	18	F	EZ-FLO added to cmt onshore	0,10	%		
							Freshwater	39,94	l/100kg	1,00	
							HALAD-99LE+	3,00	l/100kg	1,02	
							HR-5L	0,70	l/100kg	1,16	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	100,00	%	3,20	


Cementing 13 3/8 Casing
Gas (%)
Block Height (m)
Hookload (tonne)
Active
Cement Volume Pumped
TM minutes
Rig Flow In (l/min)
Cement Flow (l/min)
Cement Pump Pressure (bar)
SPP (bar)
Total Strokes
Pump 25 m3 1.60 SG Spacer
Mix and Pump 30.9 m3 1.92 SG Cement Slurry
Displace Cement with rig pumps
Bottom Plug lands and shear in LC
Top Plug lands in LC
Pressure Test Casing to 270 bar


Cementing 13 3/8 Casing
Gas (%)
Block Height (m)
Hookload (tonne)
Active
Cement Volume Pumped
TM minutes
Rig Flow In (l/min)
Cement Flow (l/min)
Cement Pump Pressure (bar)
SPP (bar)
Total Strokes
Pump 25 m3 1.60 SG Spacer
Mix and Pump 30.9 m3 1.92 SG Cement Slurry
Displace Cement with rig pumps
Bottom Plug lands and shears in LC
Top Plug lands in LC
Pressure Test Casing to 270 bar



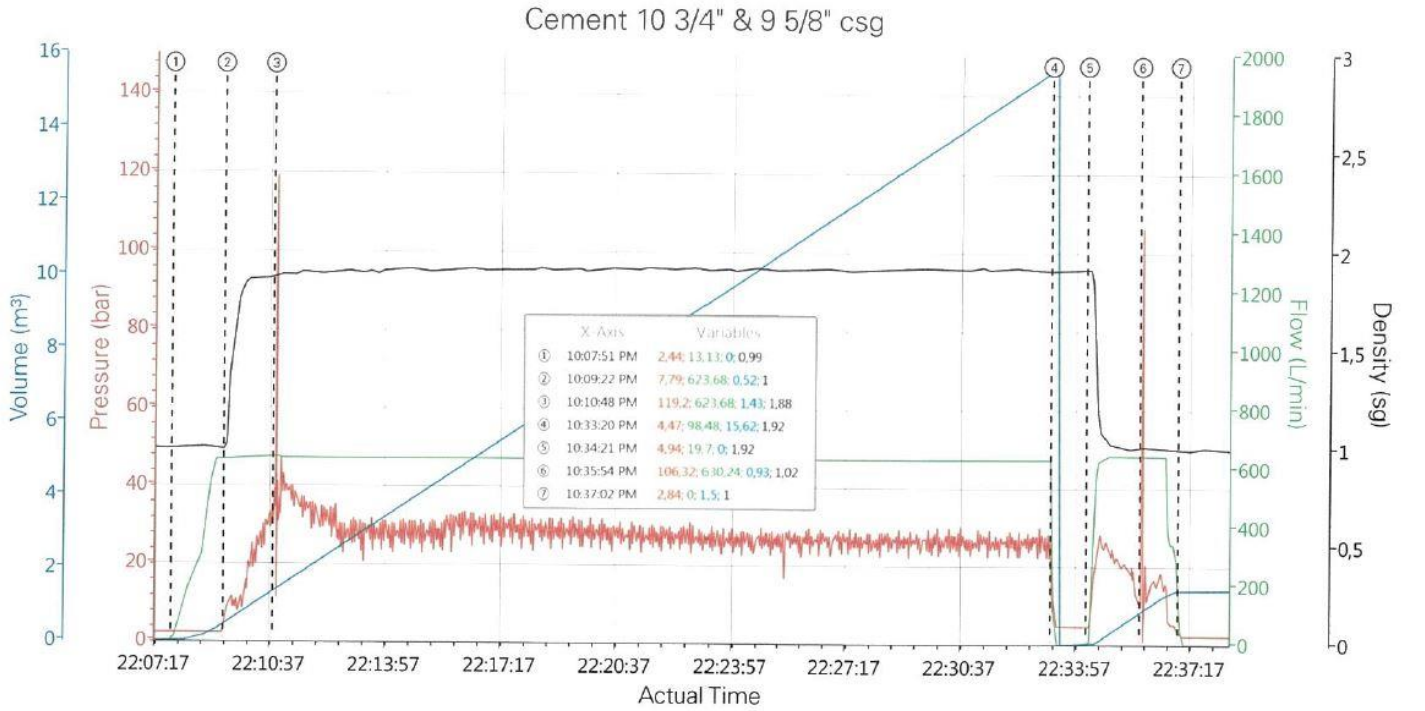
Evaluation:
Job was mixed at designed density of 1,92 sg and pumped at 636 lpm. Pump rates were lower than simulated, but we also kept lower rates than simulations. Top plug was bumped and casing tested on bump. No backflow when checking floats in casing shoe.

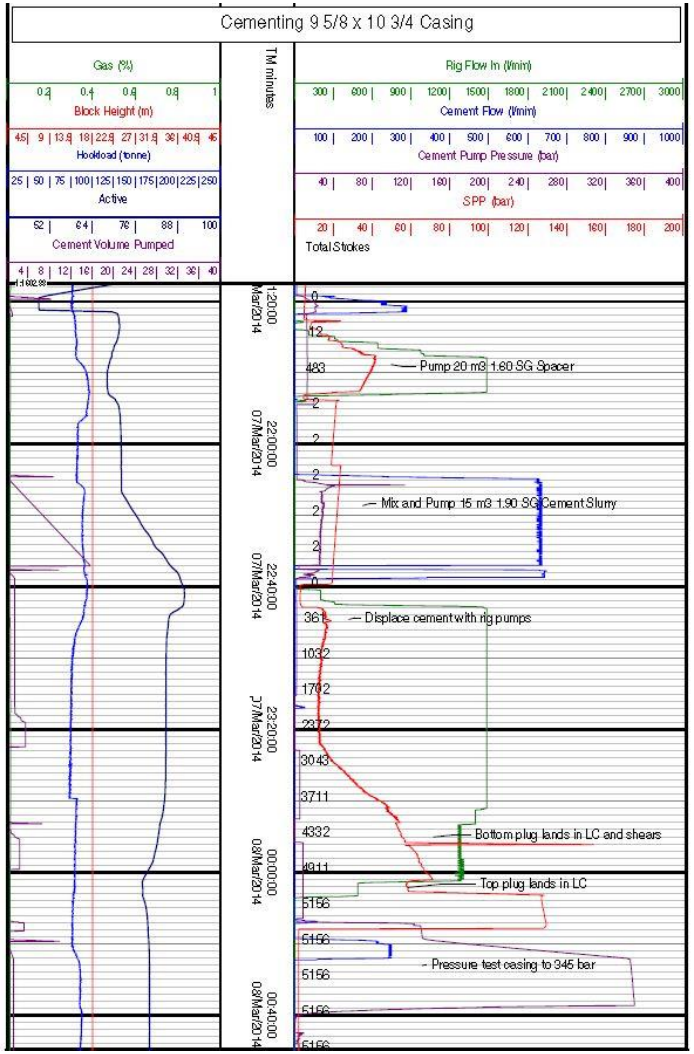
CASING SIZE: 9 5/8"
STAGE CEMENTING: No
REPORT DATE: 06.Mar.2014
THEORETHICAL TOC: 2747 mMD
EVALUATED TOC: mMD
EVALUATED BY BOND LOG:
LINER ROTATION PLANNED: N
LINER ROTATION ACHIEVED: Y
REMARKS:

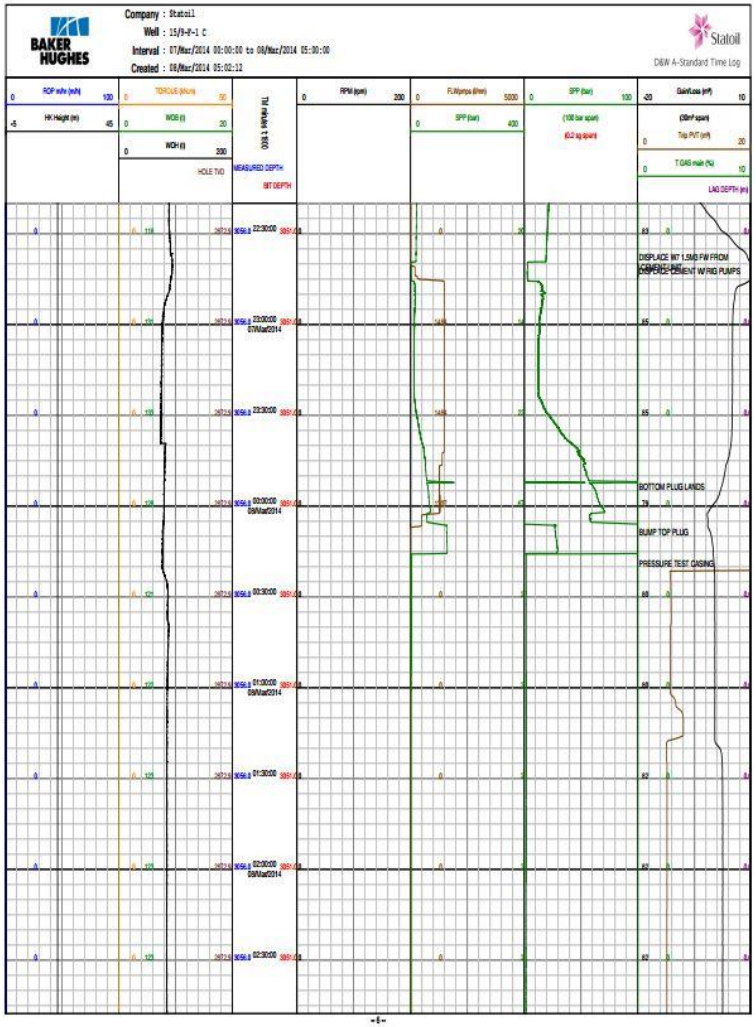
Objective:
Cement 10 3/4" x 9 5/8" casing 300m above shoe.

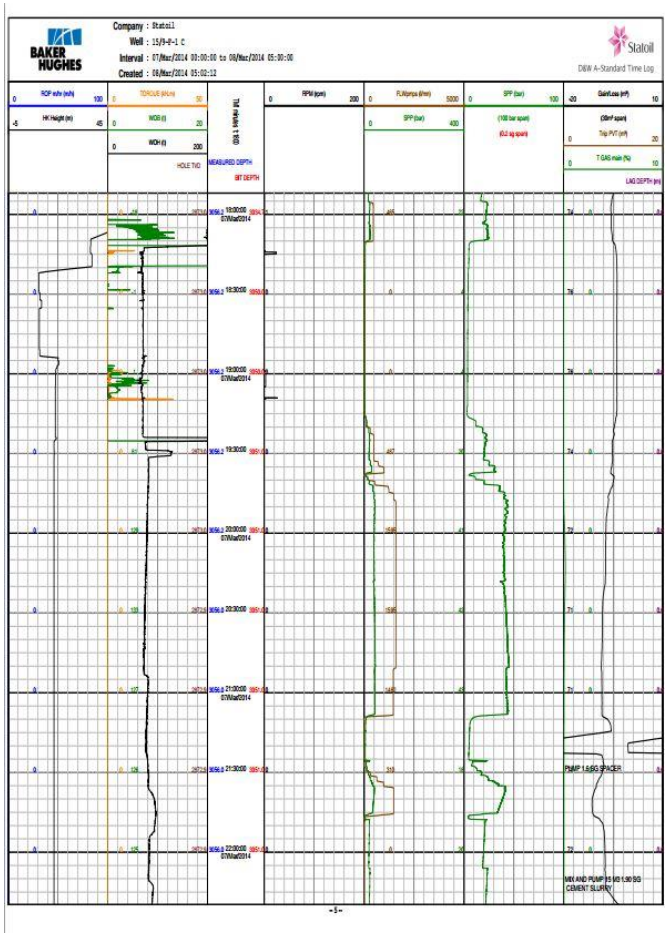
Execution:
Job was mixed "on the fly" using CMS (chemical injection pumps). Spacer was mixed in Pit 12. Observed a total loss of 8 m3 during displacement. Shear of both plugs at plugset was as expected. Btm plug shear pressure at landing collar: 153 bar.
FCP: 58 bar @ 500 lpm.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Spacer before	Spacer WSOBM	1,60	20,0	1500	38	F	BARITE	755,80	kg/m3	4,23	
							Freshwater	779,10	l/m3	1,00	
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	
Tail	Cement Slurry Non Standard	1,90	15,0	636	28	F	CFR-8L	1,50	l/100kg		
							EZ-FLO added to cmt onshore	0,10	%		
							Freshwater	46,51	l/100kg	1,00	
							GASCON	4,50	l/100kg	1,10	
							HALAD 400L	3,50	l/100kg		
							HR-5L	1,40	l/100kg	1,16	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	100,00	%	3,21	
							SSA-1	35,00	%	2,63	


Cement 10 3/4" & 9 5/8" csg
Volume (m³)
Pressure (bar)
Flow (L/min)
Density (sg)
Actual Time
22:07:17
22:10:37
22:13:57
22:17:17
22:20:37
22:23:57
22:27:17
22:30:37
22:33:57
22:37:17
X-Axis Variables
① 10:07:51 PM 2,44; 13,13; 0; 0,99
② 10:09:22 PM 7,79; 623,68; 0,52; 1
③ 10:10:48 PM 119,2; 623,68; 1,43; 1,88
④ 10:33:20 PM 4,47; 98,48; 15,62; 1,92
⑤ 10:34:21 PM 4,94; 19,7; 0; 1,92
⑥ 10:35:54 PM 106,32; 630,24; 0,93; 1,02
⑦ 10:37:02 PM 2,84; 0; 1,5; 1
DS Pump Press (bar) 0.2678 Comb Pump Rate (L/min) 0 Comb Pump Total (m³) 0 Cement Density (sg) 0.989
① Start pumping downhole(Forgot to open valve) 2,44;13,13;0;0,99
② Start pumping downhole (Opened valve after 500ltr pumped) 7,79;623,68;0,52;1
③ Shear out bottom plug (After 920 ltr) 119,2;623,68;1,43;1,88
④ Stop pumping downhole 4,47;98,48;15,62;1,92
⑤ Drop dart for top plug and displace lines w/1500 ltr DW 4,94;19,7;0;1,92
⑥ Shear out top plug (After 930 ltr) 106,32;630,24;0,93;1,02
⑦ Stop displacing lines 2,84;0;1,5;1
HALLIBURTON | iCem® Service
Created: 2014-03-07 09:36:26, Version: 3.0.121
Edit
Customer: STATOIL ASA
Job Date: 07.03.2014
Well: 15/9-F-1-C
Representative: Tony Brekne / Ole K. Dale / Thomas Mæland
Sales Order #:


Cementing 9 5/8 x 10 3/4 Casing
Gas (%)
Block Height (m)
Hookload (tonne)
Active
Cement Volume Pumped
TM minutes
Rig Flow In (l/min)
Cement Flow (l/min)
Cement Pump Pressure (bar)
SPP (bar)
Total Strokes
Pump 20 m3 1.60 SG Spacer
Mix and Pump 15 m3 1.90 SG Cement Slurry
Displace cement with rig pumps
Bottom plug lands in LC and shears
Top plug lands in LC
Pressure test casing to 345 bar


BAKER HUGHES
Statoil
D&W A-Standard Time Log
DISPLACE WT 1.5MB FW FROM CEMENT LINE
DISPLACE CEMENT W/RIG PUMPS
BOTTOM PLUG LANDS
BUMP TOP PLUG
PRESSURE TEST CASING



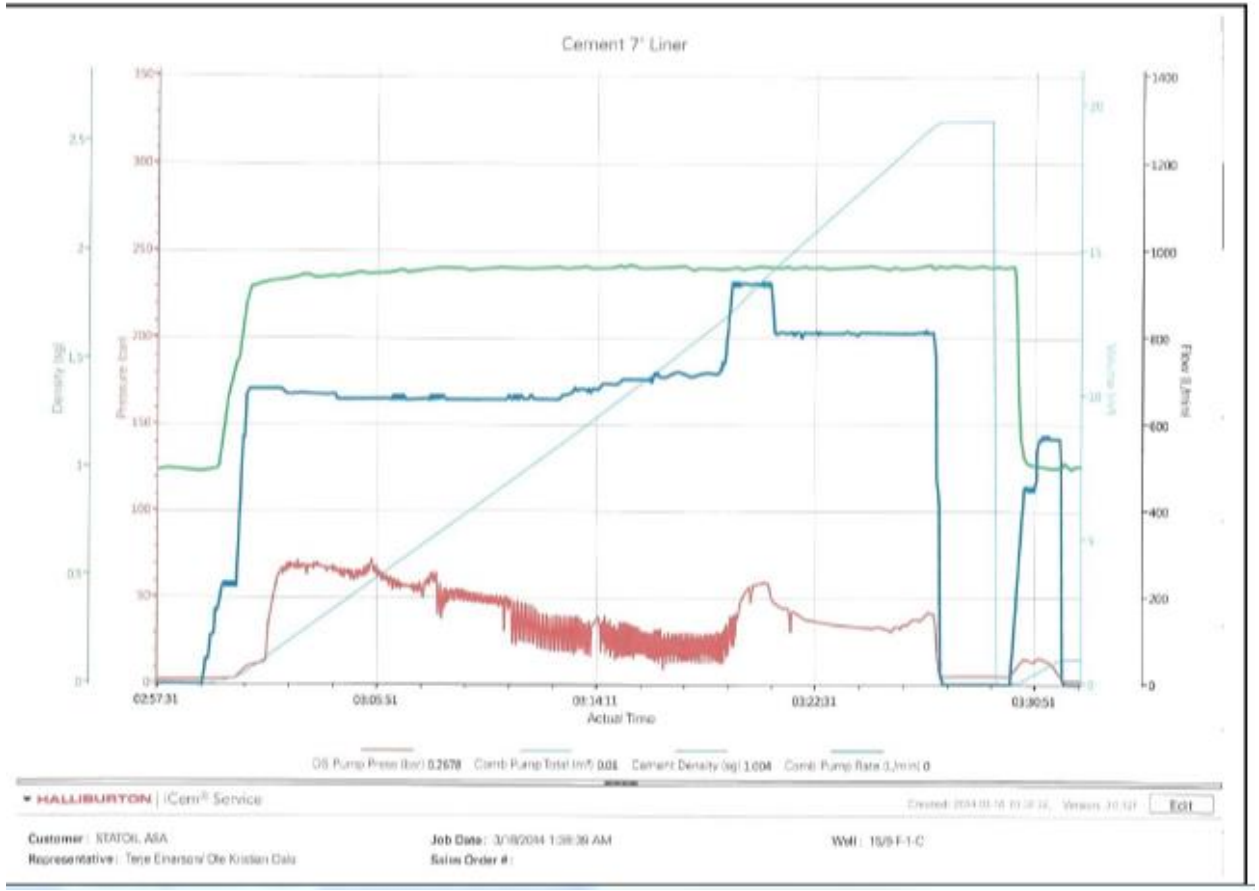
Evaluation:
Observed total loss of 8 m3. Casing was tested to 345 bar when bumping top plug.

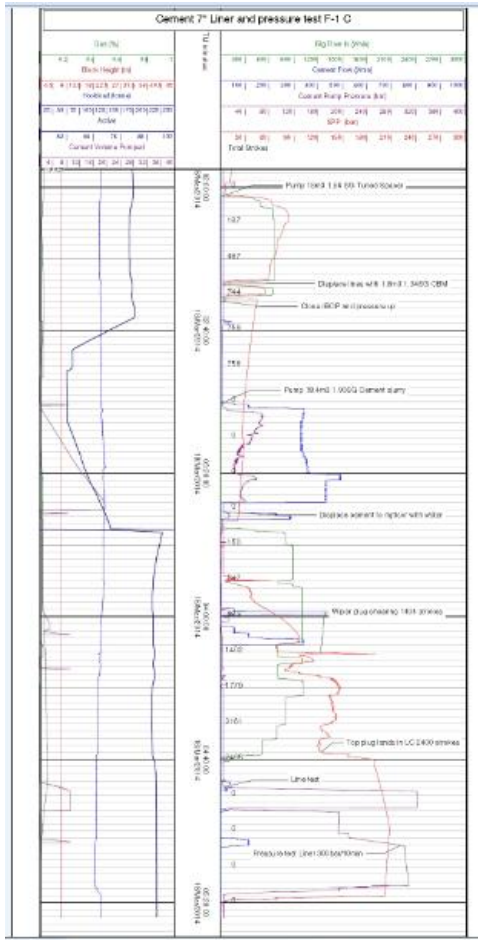
CASING SIZE: 7"
STAGE CEMENTING: No
REPORT DATE: 19.Mar.2014
THEORETHICAL TOC: 2940 mMD
EVALUATED TOC: mMD
EVALUATED BY BOND LOG:
LINER ROTATION PLANNED: Y
LINER ROTATION ACHIEVED: Y
REMARKS:

Objective:
Being TOC to TOL to isolate liner lap. Provide structual support to the liner and zonal isolation across the interval.
TOC is 2940

Execution:
Job was mixed with Automatic density controlled unit and CMS (auto chemical injection). We use CMS system for the chemicals. This system mix drillwater and chemicals, on fly. No need to prepare mixwater before the job.
Spacer was premixed in pit and pumped with rig pumps.
No losses or problems during cement operation.
Liner rotation during cement job.
Had no problems with cement unit or bulk transfer under the cement job.
Volume of actual wiperplug shear and pump as planned. Used 95,8% pump efficiency while displacing with rig pump.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Spacer before	Tuned Spacer E+	1,54	17,0	800	55	F	BARITE	703,20	kg/m3	4,20	
							DRILL WATER	791,40	l/m3	1,00	
							MUSOL E	12,00	l/m3	0,90	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	1,06	
							TUNED SPACER E+	40,40	kg/m3	2,30	
Lead	Cement Slurry GTTTZ	1,90	19,8	650	68	F	CFR-8L	2,00	l/100kg		
							Freshwater	45,93	l/100kg	1,00	
							GASCON	4,50	l/100kg	1,39	
							HALAD 400L	3,50	l/100kg		
							HR-5L	1,60	l/100kg	1,16	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	100,00	%	3,22	
							SSA-1	35,00	%	2,63	





Evaluation:
Job objectives met. Pump rate and density stayed inside requirements. Pressure test on liner was good, and we bleed back the same volume.

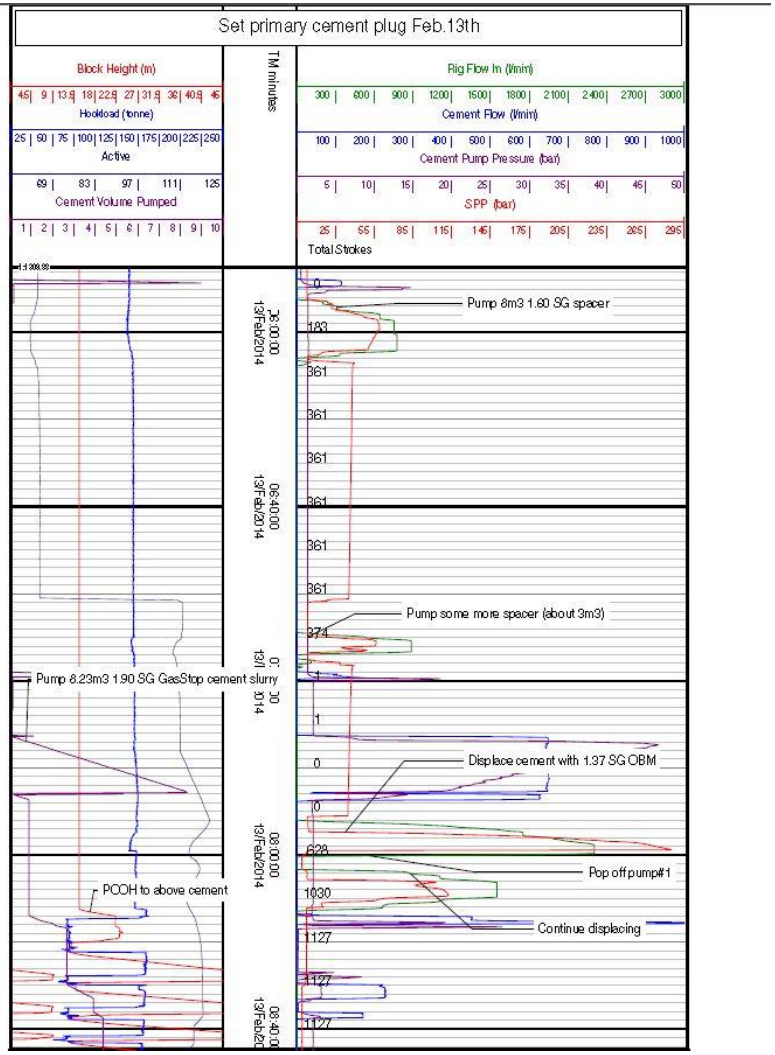
8.7.2 Plugging

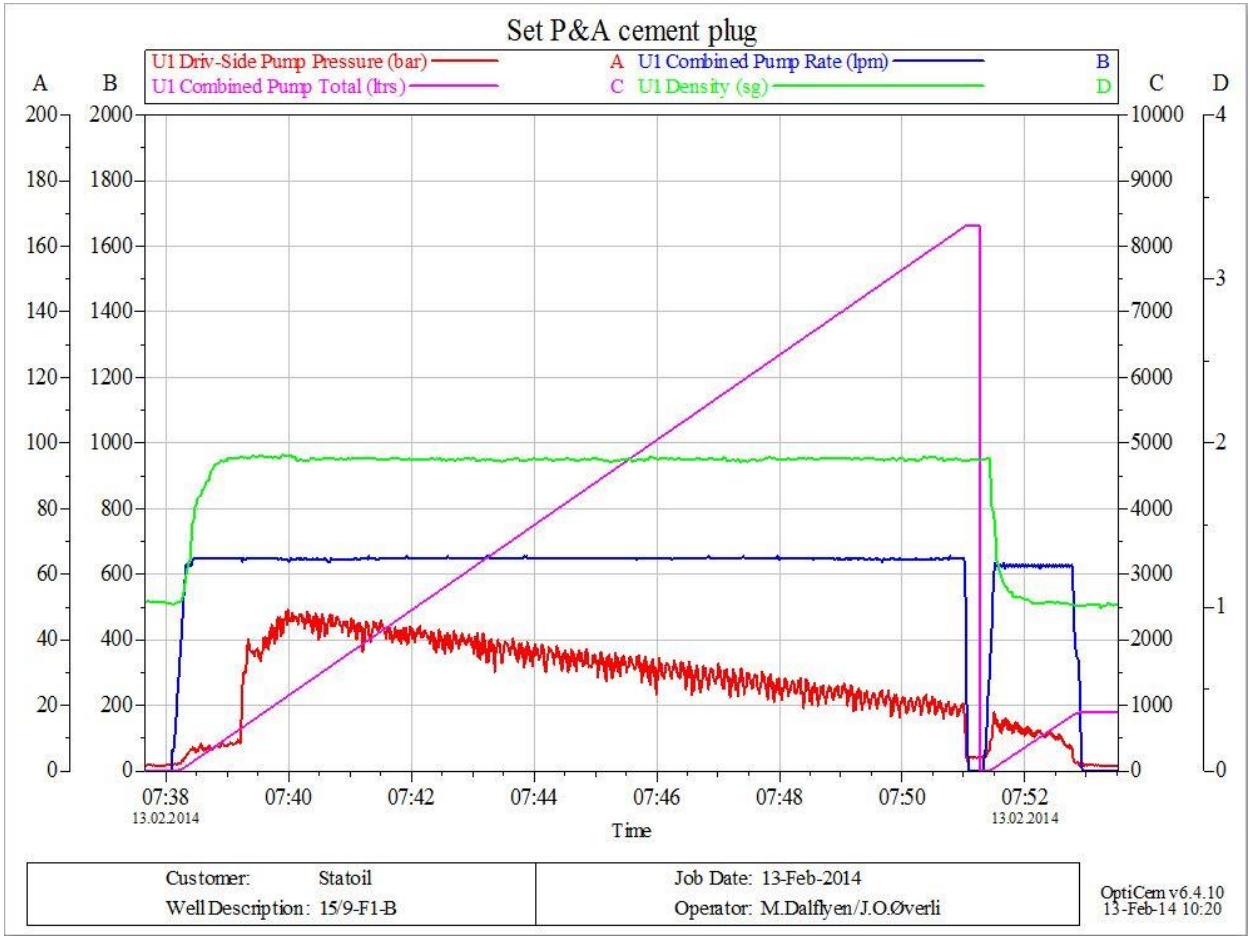
WELLBORE ID: NO 15/9-F-1 B
REPORT DATE: 14.Feb.2014
PLUG NO: 1
TOP DEPTH: 2850
BOTTOM DEPTH: 3147
CASING SIZE: 7"
HOLE DIAMETER: NA
CEMENTING TYPE: CASING
REMARKS:

Objective:
Plug back the well and set a KOP below 20 " later.
We have set a banced plug on top of EVO plug. Estimatet TOC is 2850 mtr.

Execution:
A balance plug was set on top of a EVO plug.
Slurry was mixed on the fly with CMS.
Spacer was prepared in pit 12 and pumped with the rig pumps. No losses prior or during the cement operation.
No problems during mixing cement.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Spacer before	Tuned Spacer E+	1,60	9,2	900	45		BARITE	755,80	kg/m3		
							Freshwater	779,10	l/m3	1,00	
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	
Slurry	Cement Slurry GTT90	1,90	8,3	636	45		CFR-8L	1,00	l/100kg		
							Freshwater	37,63	l/100kg	1,00	
							GASCON	3,50	l/100kg	1,10	
							HALAD 400L	3,00	l/100kg		
							HR-5L	1,20	l/100kg	1,16	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	100,00	kg/100kg	3,22	
Spacer after	Tuned Spacer E+	1,60	1,9	900	45		BARITE	755,80	kg/m3		
							Freshwater	779,10	l/m3	1,00	
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	


Set primary cement plug Feb.13th
Block Height (m)
Hookload (tonne)
Active
Cement Volume Pumped
TM minutes
Rig Flow In (l/min)
Cement Flow (l/min)
Cement Pump Pressure (bar)
SPP (bar)
Total Strokes
Pump 8m3 1.60 SG spacer
Pump some more spacer (about 3m3)
Pump 8.23m3 1.90 SG GasStop cement slurry
Displace cement with 1.37 SG OBM
Pop off pump#1
Continue displacing
POOH to above cement



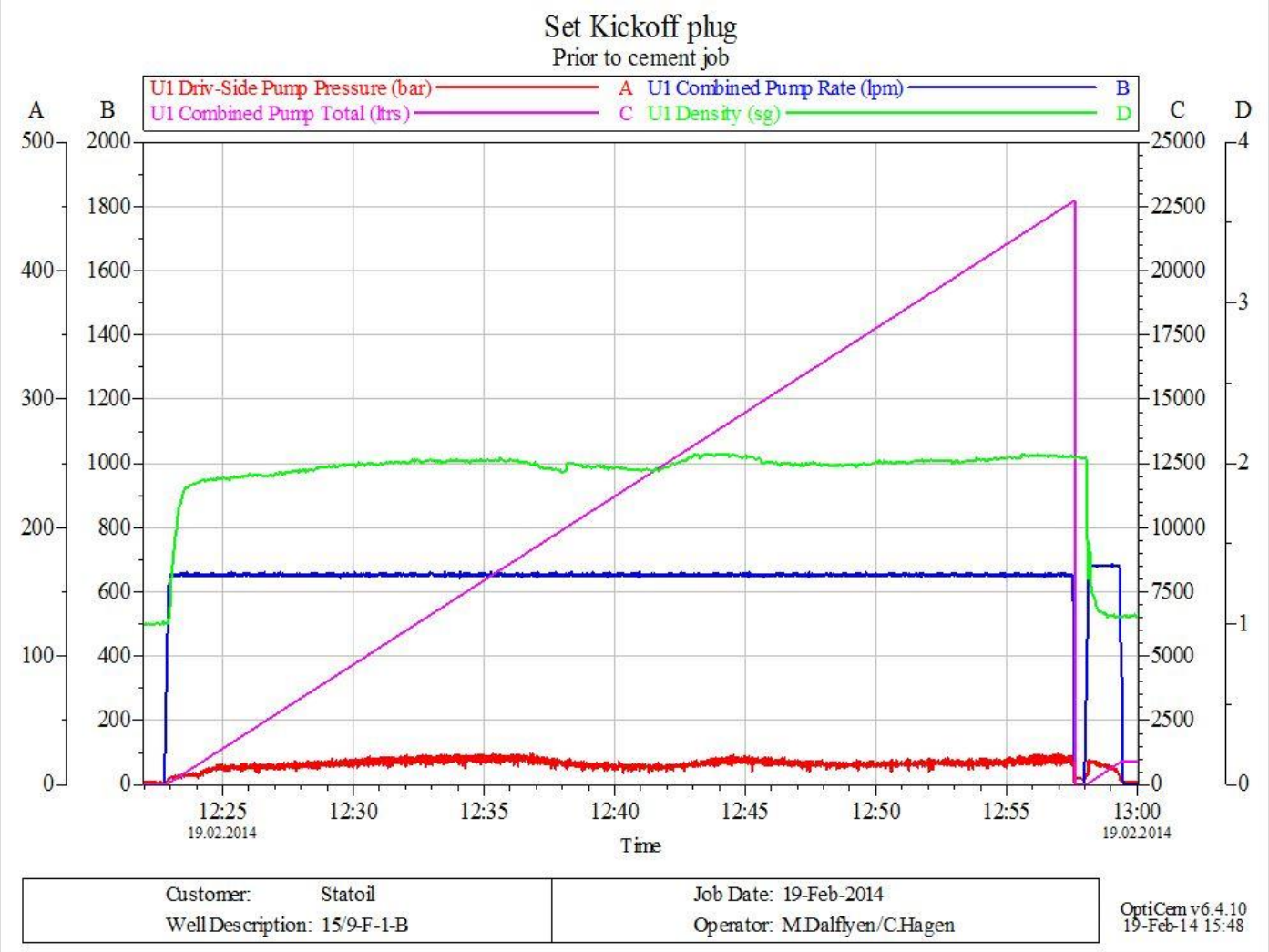
Evaluation:
Job objectives met. Pump rate and density steady inside requirements. Planned to tag cmt tonight.

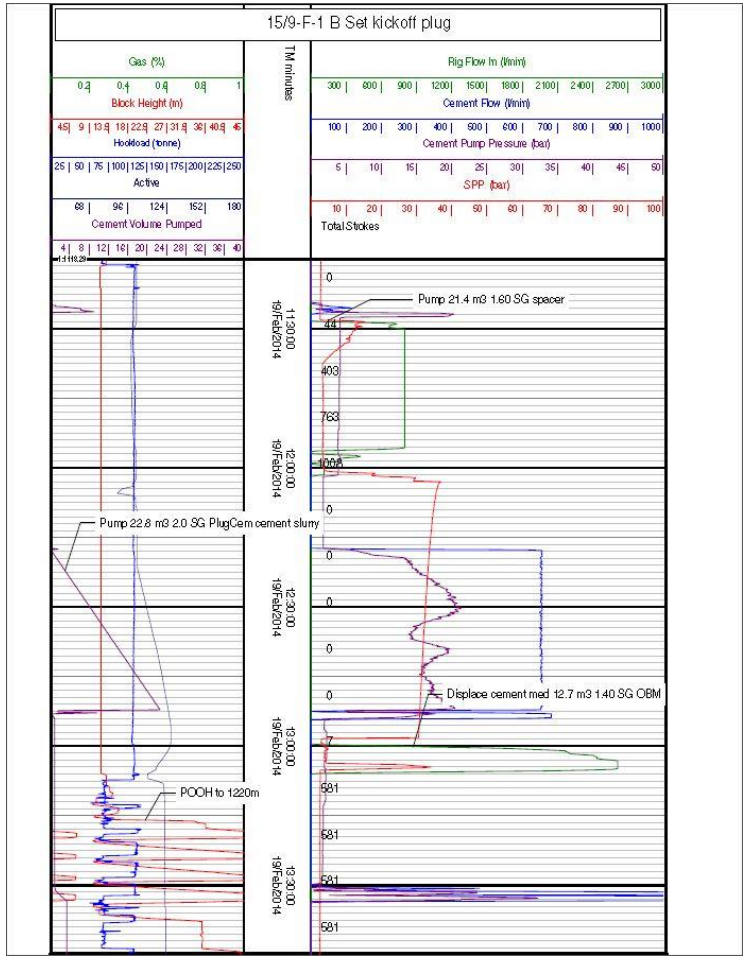
REPORT DATE: 20.Feb.2014
PLUG NO: 1
TOP DEPTH: 1298
BOTTOM DEPTH: 1399
CASING SIZE: 20"
HOLE DIAMETER: 17.5" x 26"
CEMENTING TYPE: OPEN HOLE/CSG
REMARKS: Kickoff plug

Objective:
Successfully set KOP below 20" csg choe and perform side track of well. Plug is set from 1399 mtr in 17 1/2" OH, 26" rat hole from 1355 and 50 mtr inside 20" CSG from 1348 mtr. TOC 1298 mtr.

Execution:
Job was mixed with Automatic density controlled unit and CMS (auto chemical injection). We use CMS system for the chemicals. This system mix drillwater and chemicals, on fly. No need to prepare mixwater before the job.
Spacer was premixed in pit and pumped with rig pumps.
No losses or problems during cement operation.
Liner rotation during cement job.
Had no problems with cement unit or bulk transfer under the cement job. Had some problems with cmt manual weight and calibration of densiometer halfway through the job as shown on plot.
Used 95,6% pump efficiency while displacing with rig pump.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Spacer before	Tuned Spacer E+	1,60	20,0	800	15		BARITE	755,80	kg/m3	4,23	
							DRILL WATER	779,10	l/m3		
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	
Slurry	Cement Slurry KOPMP	2,00	22,8	650	35		CFR-8L	1,00	l/100kg		
							Freshwater	36,06	l/100kg	1,00	
							HR-4L	0,70	l/100kg	1,18	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	100,00	kg/100kg	3,22	
Spacer after	Tuned Spacer E+	1,60	1,4	2200	35		BARITE	755,80	kg/m3	4,23	
							DRILL WATER	779,10	l/m3		
							MUSOL E	12,00	l/m3	0,92	
							NF-6, DEFOAMER	3,00	l/m3	0,94	
							SEM-8, WATER WETTING AGENT	12,00	l/m3	0,99	
							TUNED SPACER E+	40,00	kg/m3	2,65	


Set Kickoff plug
Prior to cement job
U1 Driv-Side Pump Pressure (bar) A
U1 Combined Pump Rate (lpm) B
U1 Combined Pump Total (ltrs) C
U1 Density (sg) D
A
B
C
D
500
400
300
200
100
0
2000
1800
1600
1400
1200
1000
800
600
400
200
0
25000
22500
20000
17500
15000
12500
10000
7500
5000
2500
0
4
3
2
1
0
12:25
19.02.2014
12:30
12:35
12:40
12:45
12:50
12:55
13:00
19.02.2014
Time
Customer: Statoil
Well Description: 15/9-F-1-B
Job Date: 19-Feb-2014
Operator: M.Dalflyen/C.Hagen
OptiCem v6.4.10
19-Feb-14 15:48



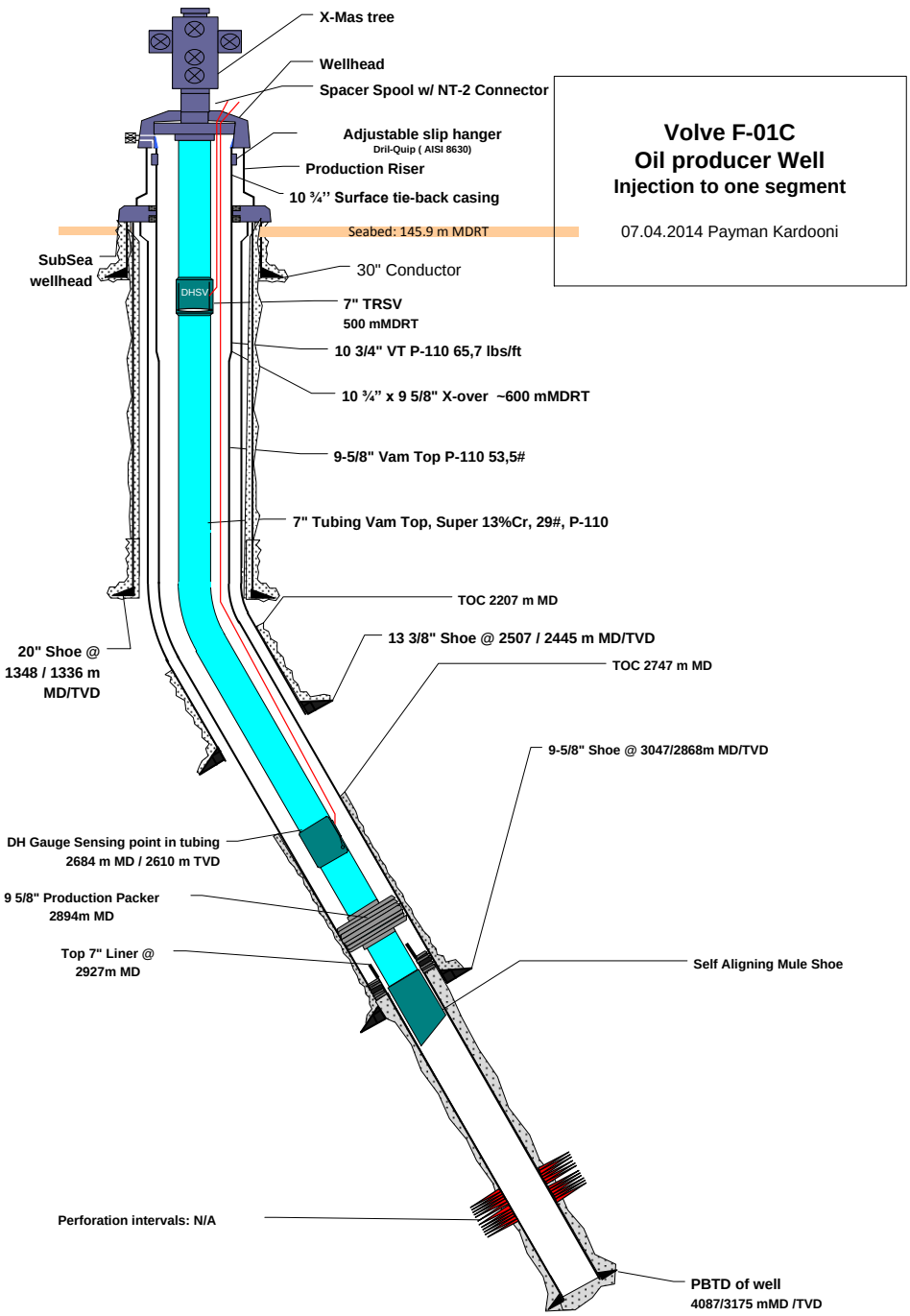
Evaluation:
Tagged TOC @ 1293 - 7 MT after 17 hrs WOC. Still a bit soft cmt at that point.

8.7.3 Mechanical plug

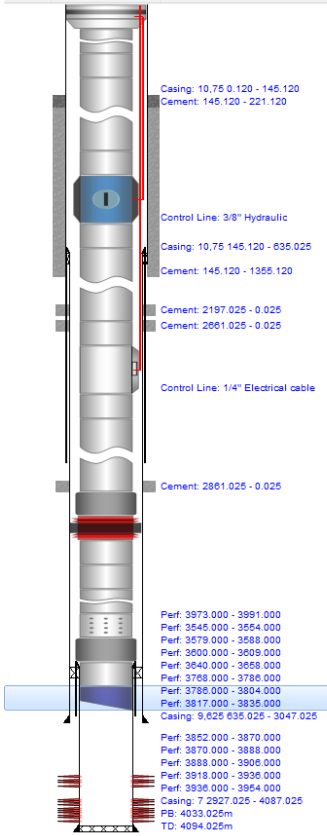
WELLBORE ID: NO 15/9-F-1 B

Report date	Depth mMD	Diam. in	Plug type	Tagged depth mMD	Pressure test bar	Setting time	Pulling time	Remarks
13.feb.2014	617,0	9 5/8"	Peak V Series Mechanical barrier plug			12.feb.2014	12.feb.2014	

8.8 Well schematic

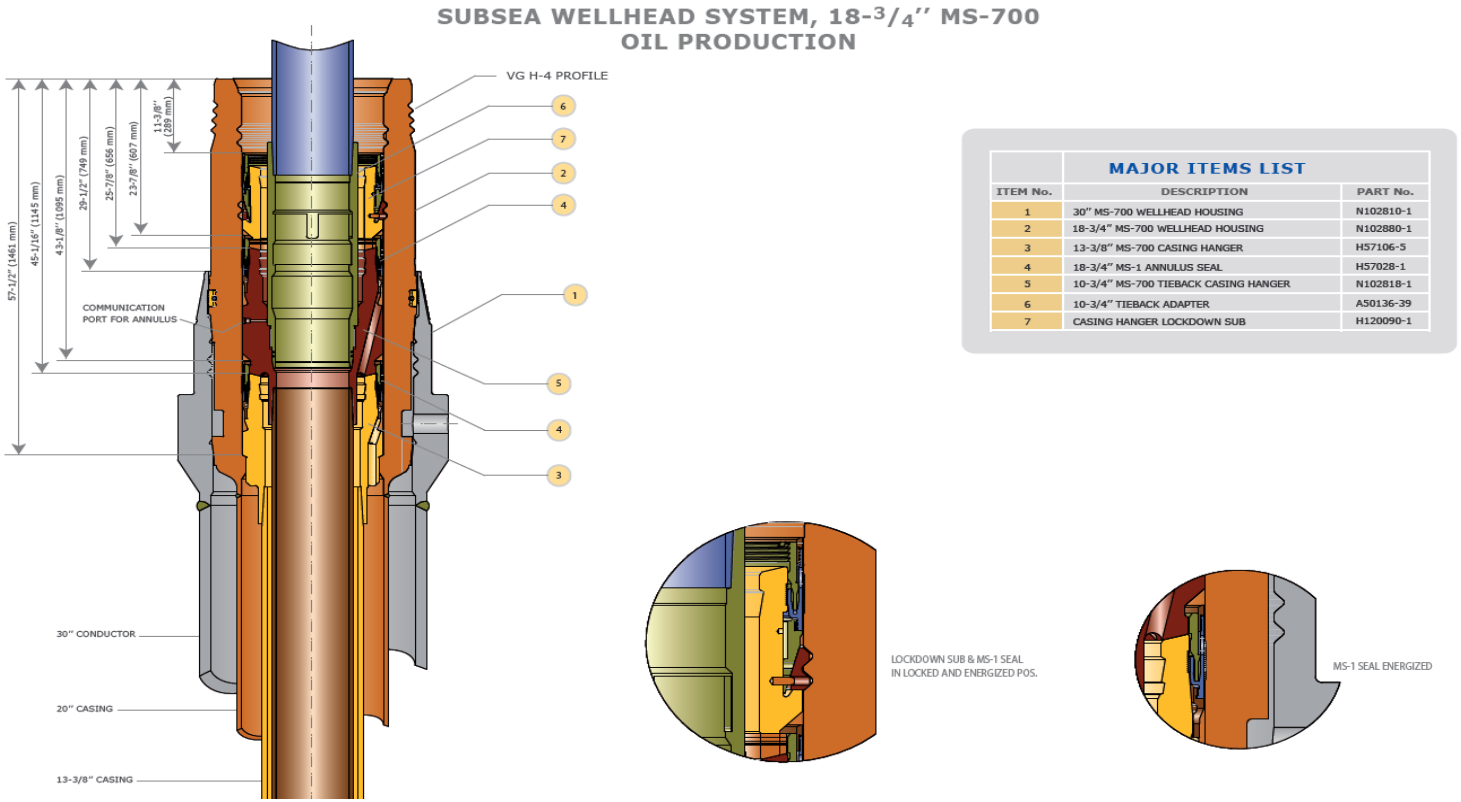


8.9 Completion schematic



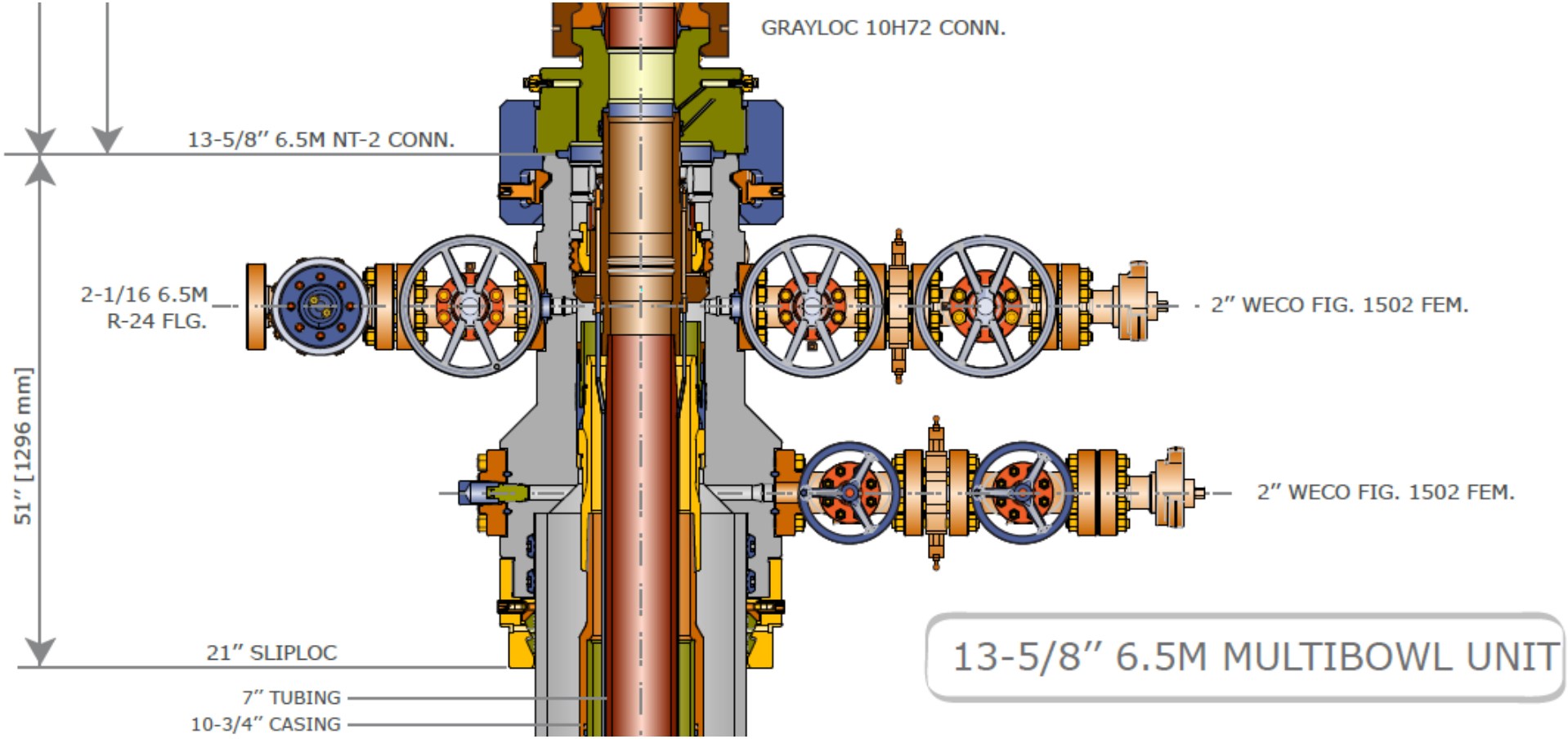
Assen	Symbol	Symbol Extra Info	Forr	MD [RKB] Top [m]	Length [m]	Min ID [inch]	Drift ID [inch]	Max OD [inch]	Description	MD [RKB] Bottom [m]	Comments
				20.560	1.444	6.084	6.084	13.500	13 5/8" x 7" VetcoGray Tubing Hanger Volve	22.004	Hang off point 20.560 m
				22.004	12.056	6.184	6.102	7.699	7" 29# Vam Top Tubing	34.060	
				34.060	6.053	6.184	6.059	7.680	7" 29# Vam Top Pup Joint P x P	40.113	Pupjoint manufactured from special drift base pipe
				40.113	443.136	6.184	6.102	7.699	7" 29# Vam Top Tubing	483.249	
				483.249	2.864	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	486.113	Pupjoint manufactured from special drift base pipe
				486.113	2.714	5.812	5.810	9.500	Baker Mod: ONYX-7.5" TRSV Sz.: 5.812"	488.827	
				488.827	1.968	6.184	6.059	7.670	7" 29# Vam Top Pup Joint B x P	490.795	Pupjoint manufactured from special drift base pipe
				490.795	2190.158	6.184	6.102	7.699	7" 29# Vam Top Tubing	2680.953	
				2680.953	3.000	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	2683.953	Pupjoint manufactured from special drift base pipe
				2683.953	2.390	5.900	5.869	8.311	7" Single Gauge Mandrel for single NPQG-BA Tubing P/T Gauge	2686.343	
				2686.343	2.010	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	2688.353	Pupjoint manufactured from special drift base pipe
				2688.353	202.328	6.184	6.102	7.699	7" 29# Vam Top Tubing	2890.681	Centralized tubing
				2890.681	2.858	6.094	5.969	7.700	7" 29# Vam Top Collar	2893.539	
				2893.539	1.538	6.000	5.969	8.310	Granite Packer 9 5/8" x 7" (Size 831 x 600)	2895.077	90-80-90 Nitrile PE
				2895.077	1.969	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	2897.046	Pupjoint manufactured from special drift base pipe
				2897.046	3.006	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	2900.052	Pupjoint manufactured from special drift base pipe
				2900.052	23.846	6.184	6.102	7.699	7" 29# Vam Top Tubing	2923.898	
				2923.898	2.345	6.184	6.089	7.690	7" 29# Vam Top Perforated Joint	2926.243	
				2926.243	0.150	6.184	6.059	7.750	7" 29# Collar Vam Top HC BxB	2926.393	
				2926.393	3.120	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	2929.513	
				2929.513	1.385	6.184	6.059	7.050	7" 29# 13 CR Muleshoe	2930.898	

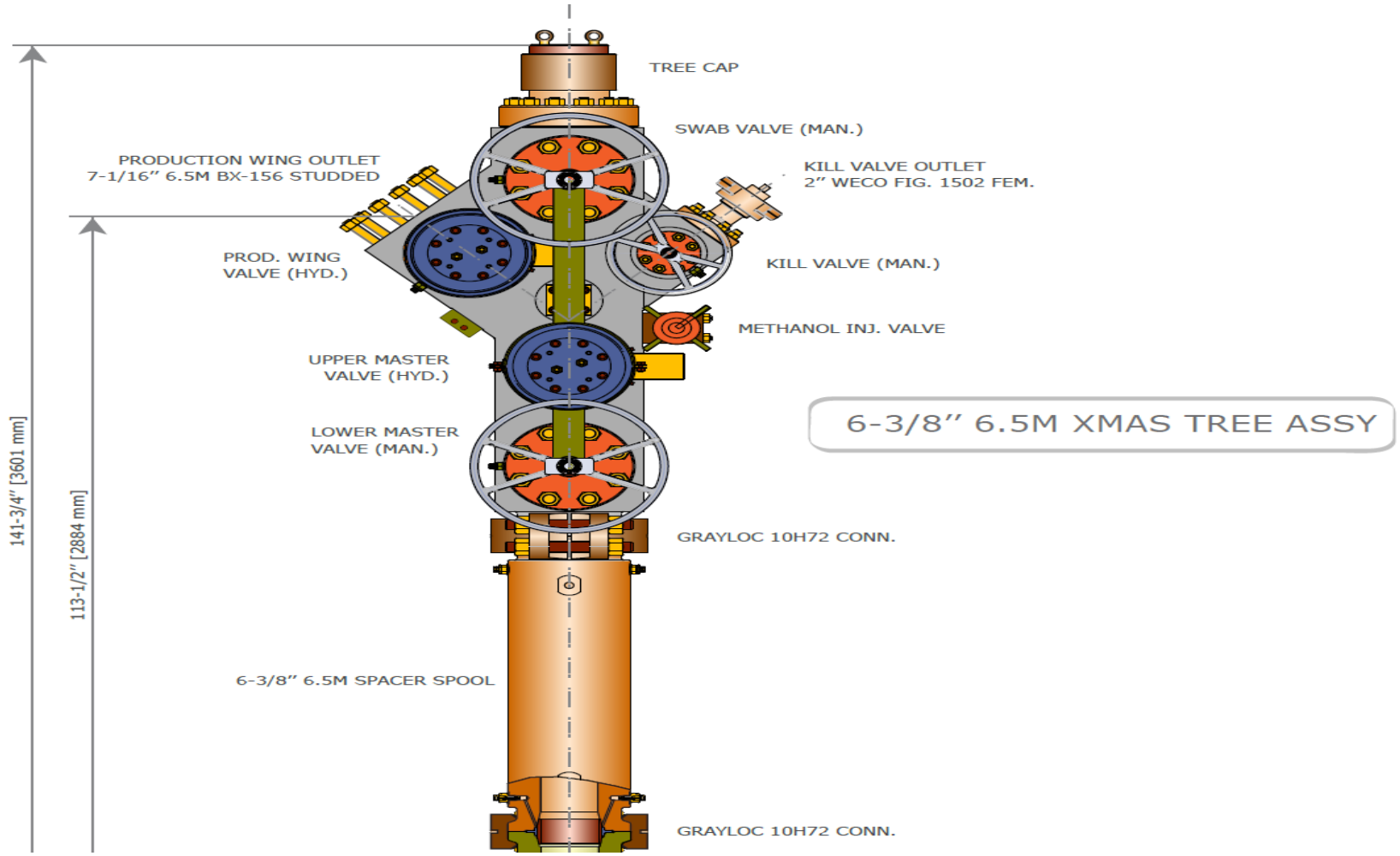
8.10 Wellhead / TSR / X-tree schematic



MAJOR ITEMS LIST

ITEM No.	DESCRIPTION	PART No.
1	30" MS-700 WELLHEAD HOUSING	N102810-1
2	18-3/4" MS-700 WELLHEAD HOUSING	N102880-1
3	13-3/8" MS-700 CASING HANGER	H57106-5
4	18-3/4" MS-1 ANNULUS SEAL	H57028-1
5	10-3/4" MS-700 TIEBACK CASING HANGER	N102818-1
6	10-3/4" TIEBACK ADAPTER	A50136-39
7	CASING HANGER LOCKDOWN SUB	H120090-1


GRAYLOC 10H72 CONN.
13-5/8" 6.5M NT-2 CONN.
2-1/16 6.5M
R-24 FLG.
2" WECO FIG. 1502 FEM.
2" WECO FIG. 1502 FEM.
51" [1296 mm]
21" SLIPLOC
7" TUBING
10-3/4" CASING
13-5/8" 6.5M MULTIBOWL UNIT


TREE CAP
SWAB VALVE (MAN.)
PRODUCTION WING OUTLET
7-1/16" 6.5M BX-156 STUDDED
KILL VALVE OUTLET
2" WECO FIG. 1502 FEM.
PROD. WING
VALVE (HYD.)
KILL VALVE (MAN.)
METHANOL INJ. VALVE
UPPER MASTER
VALVE (HYD.)
LOWER MASTER
VALVE (MAN.)
6-3/8" 6.5M XMAS TREE ASSY
GRAYLOC 10H72 CONN.
6-3/8" 6.5M SPACER SPOOL
GRAYLOC 10H72 CONN.
141-3/4" [3601 mm]
113-1/2" [2884 mm]

9 Appendix 4: Contractors list

Service	Contract no	Contractor
Plug	SAP4600013125	Archer Oil Tools
Drilling Fluids	SAP4600007720	Halliburton
Electric Wireline Logging	SAP4600014591	Schlumberger
Subsea WH/X-mas tree	SAP4600013629	Vetco Gray
ROV Systems	SAP4600016957	Oceaneering
Fishing service	SAP4600008016	Baker Oil Tools
Directional survey	SAP4600015977	Scientific Drilling International
Mud Logging	SAP4600017565	Baker Hughes Norge AS
Directional Drilling	SAP4600017565	Baker Hughes Norge AS
Drilling Fluids	SAP4600015237	Schlumberger
Wireline	SAP460001246x	Aker Well Service
Measurement While Drilling	SAP4600017565	Baker Hughes Norge AS
Rig Operations	SAP4500847188	Maersk Drilling
Subsea WH/X-mas tree	SAP4600013011	Dril-Quip
Surf. WH/X-mas tree	SAP4600003983	Vetco Gray
ROV Systems	SAP4600006702	Oceaneering
Cementing	SAP4600007720	Halliburton
Casing	SAP4600011449	Weatherford Norge AS
Compl/sand contr. equip.	SAP4600009869	Baker Hughes Norge AS